The maximum current that a 10W, 80 kΩ resistor can safely conduct is:

Select one:

A. 40 kA

B. 10 mA

C. 160 A

D. 10 μA

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A load draws 10 A when connected to a 240V source. How long does it take to consume 180 kJ?

Select one:

- a. 25s
- b. 7.5 minutes
- c. 75s
- d. 500ms

A resistor absorbs an instantaneous power of $30\cos^2 t$ mW when connected to a voltage source v=30Cos t V. Find R

Select one:

- a. 30 kΩ
- b. 7.5Ω
- c. 75kΩ
- d. 330mΩ

The prefix Tera stands for:

Select one:

a. 10^{-6}

b. 10^{12}

c. 10^{-12}

d. 10^{9}

The maximum current that a 8W, 50 kΩ resistor can safely conduct is:

Select one:

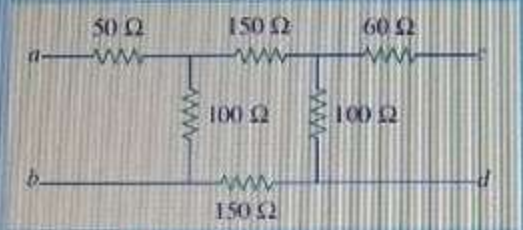
- a. 40 kA
- b. 10 μA
- c. 10 mA
- d. 160 A

The parallel internal resistance of ideal current source is:

Select one:

- a. Infinite
- b. finite
- c. None of the answers
- d. zero

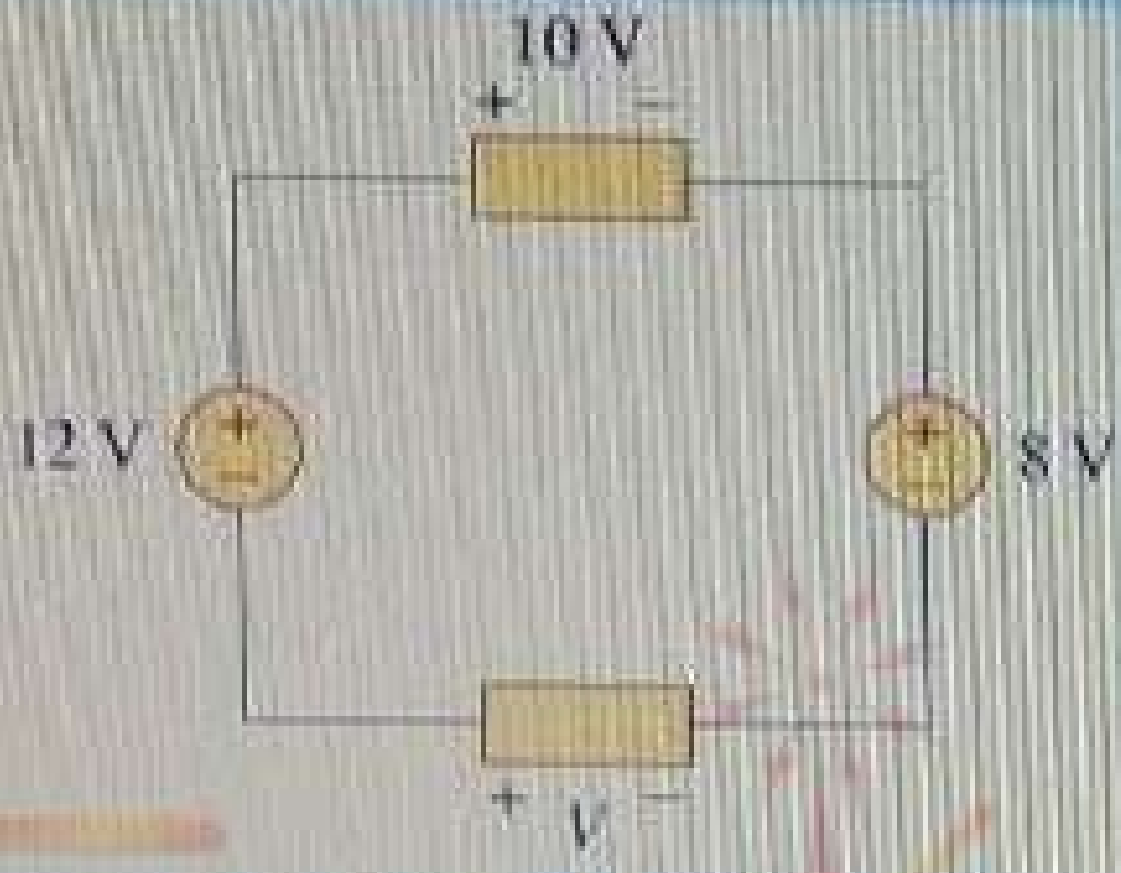
Refer to the figure below; the value of the equivalent resistance at terminals ab is:



Select one:

- a. 160 Ω
- b. 260 Ω
- c. 130 Ω
- d. 175 Ω

In the circuit below V is



Select one:

- a. 6 V
- b. 18 V
- c. -18 V
- d. -6 V

The conductance of a 10 kΩ resistor in milli siemens is:

Select one:

A. 100mS

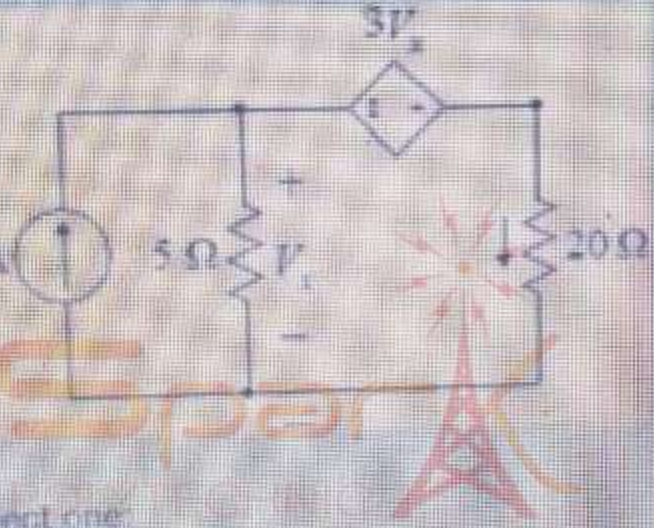
B. 0.1mS

C. 10 mS

D. 0.01mS

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In the circuit shown, The delivered power by the 4 A current source is:



Select one:

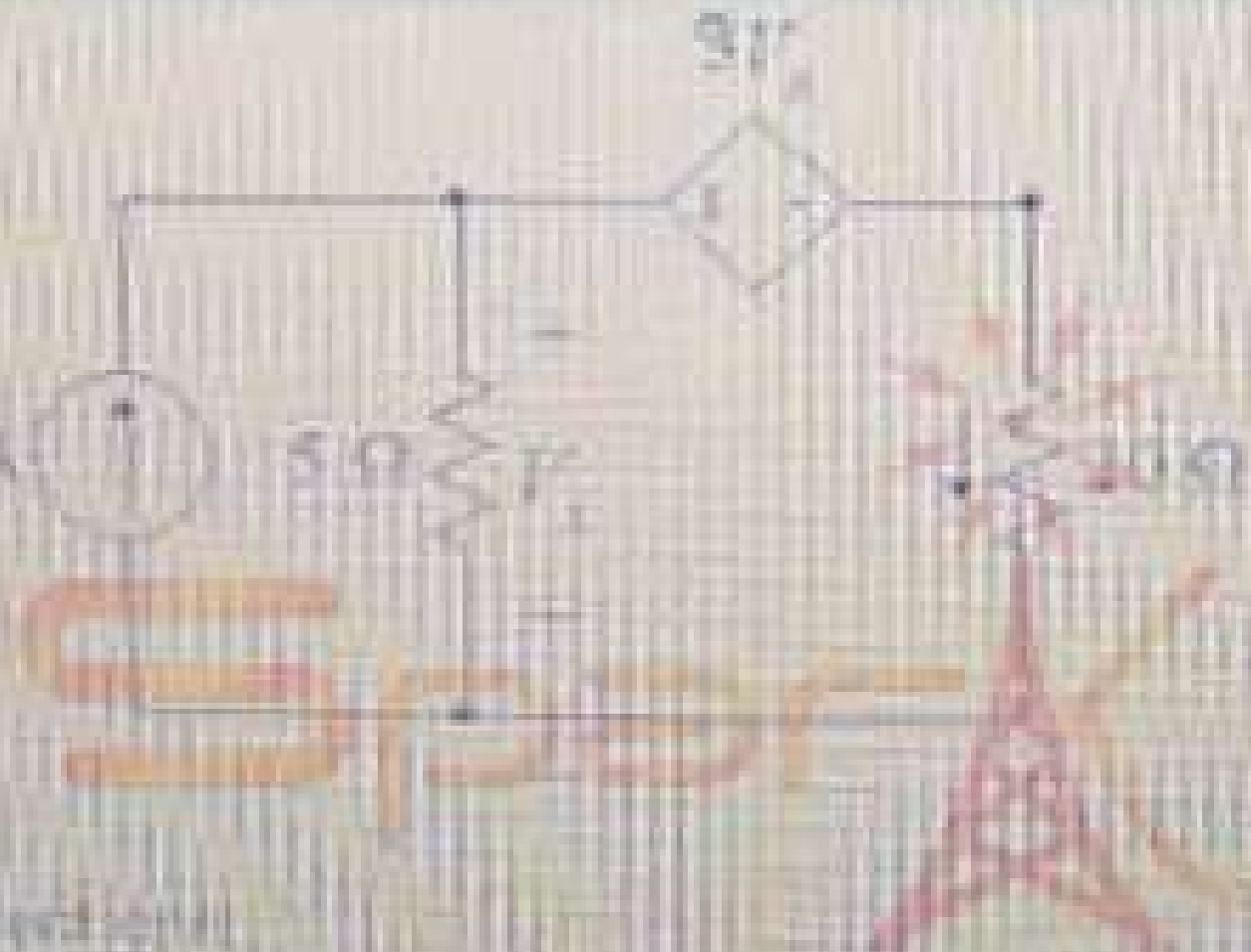
A. 40 W

B. 16 W

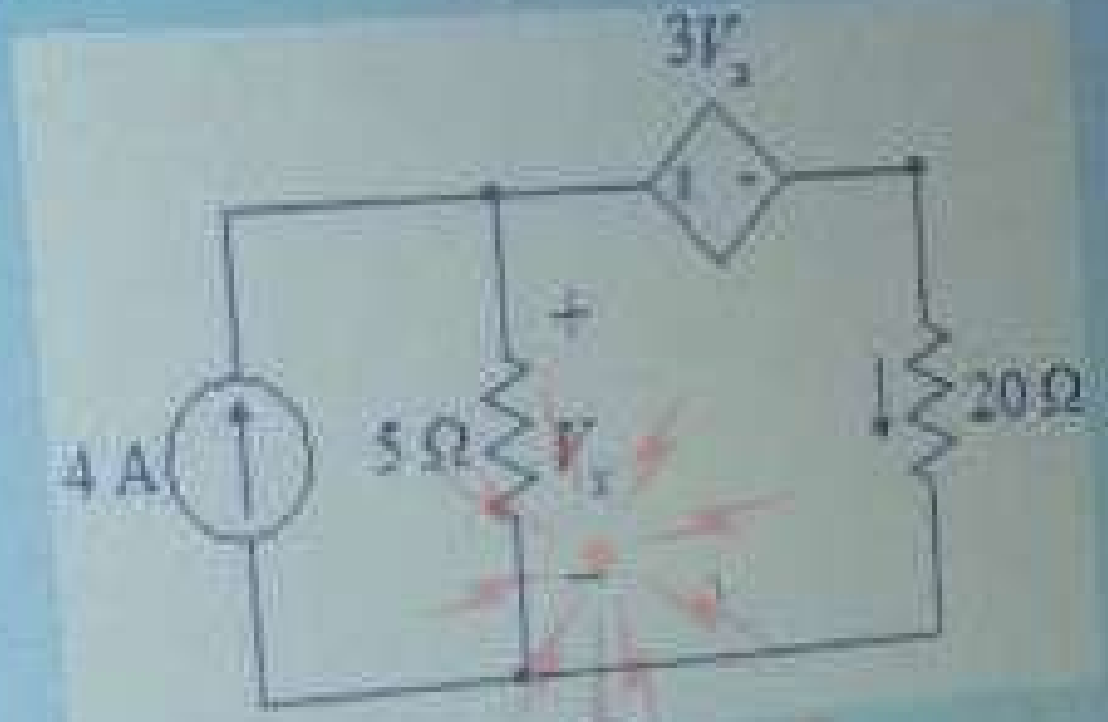
C. 20 W

D. 6 W

In the circuit shown, The delivered power by the 4 A current source is:



In the circuit shown, The current in 20 Ω resistor is:

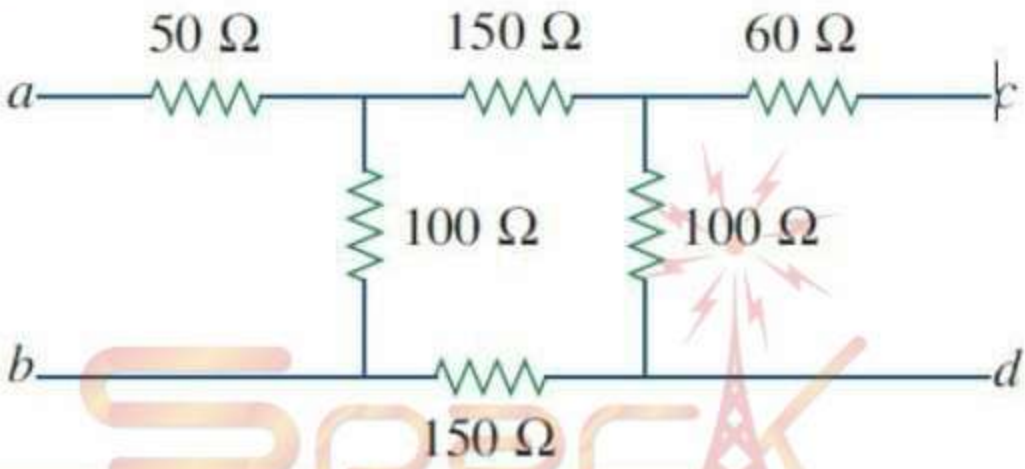


Select one:

- A. 2.8A
- B. 2A
- C. 1A
- D. 200 mA

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Refer to figure ; the value of the equivalent resistance at terminals ab is:



Select one:

- A. 160 Ω
- B. 130 Ω
- C. 175Ω
- D. 260 Ω

Protected with free version of Watermarkly. Full version doesn't put this mark.

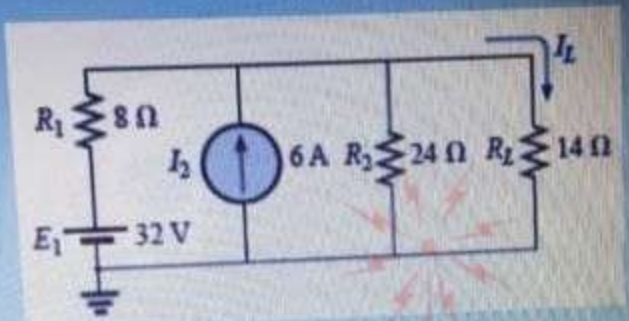
The current through a branch in a linear network is 2 A when the input source voltage is 10 V. If the voltage is reduced to 1 V and the polarity is reversed, the current through the branch is:

Select one:

- A. 2 A
- B. -2 A
- C. 0.2 A
- D. -0.2 A

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For the given circuit, Calculate the current I_L:



Select one:

1

None of all

2

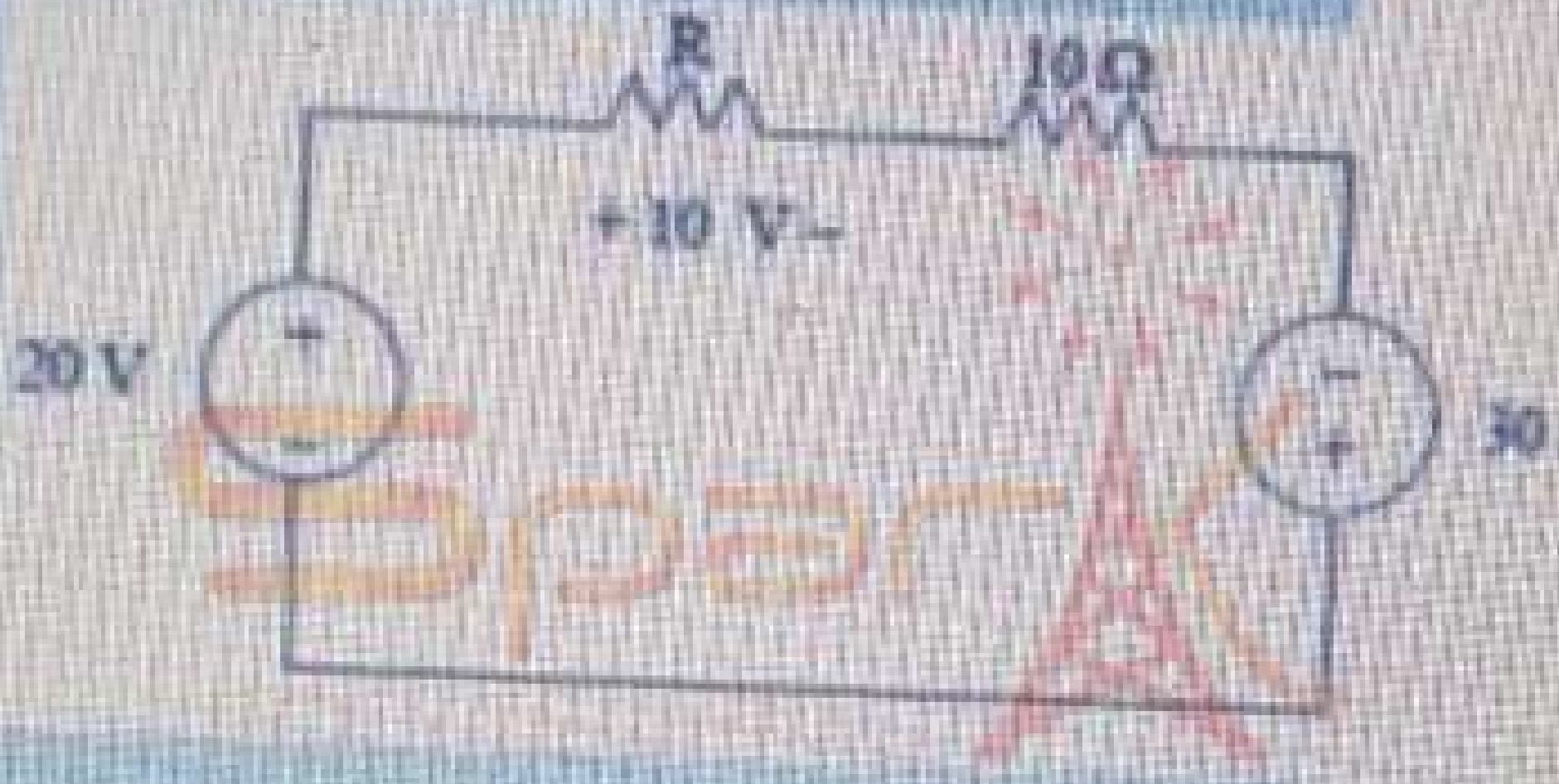
3

7 A

4

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Find R for the circuit in Figure.



Answer:

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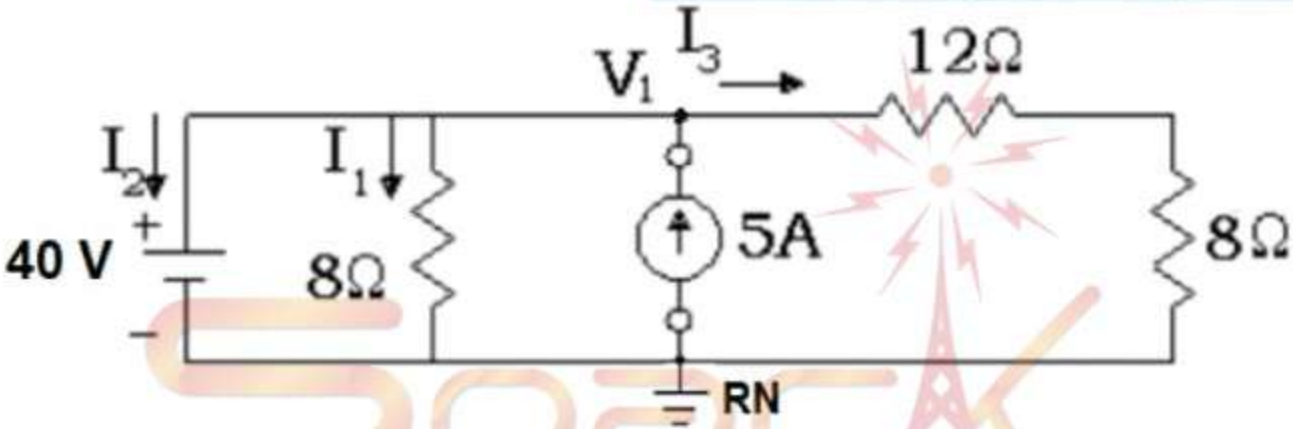


The maximum current that a 10W, 80 kΩ resistor can safely conduct is:

Select one:

- A. 40 kA
- B. 10 mA
- C. 160 A
- D. 10 μA

:Refer to the figure, the voltage V1 is



إختر أحد الخيارات:

a. 40 V

b. 20 V

c. 100 V

d. 50 V

Protected with free version of Watermarkly. Full version doesn't put this mark.

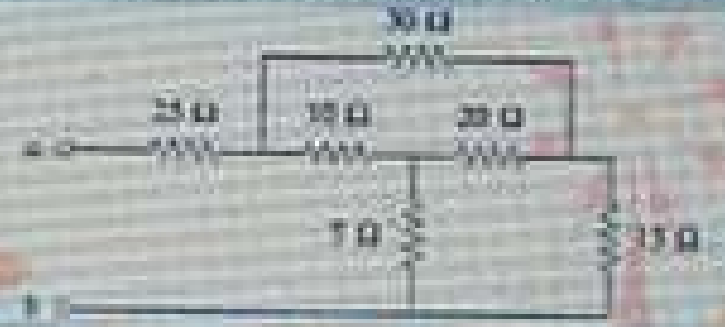
The current through a branch in a linear network is 5 A when the input source voltage is 10 V. If the voltage is reduced to 1 V and the polarity is reversed, the current through the branch is:

Select one:

- A. -0.5 A
- B. 0.2 A
- C. 2 A
- D. -2 A

Spark POLYTECHNIC

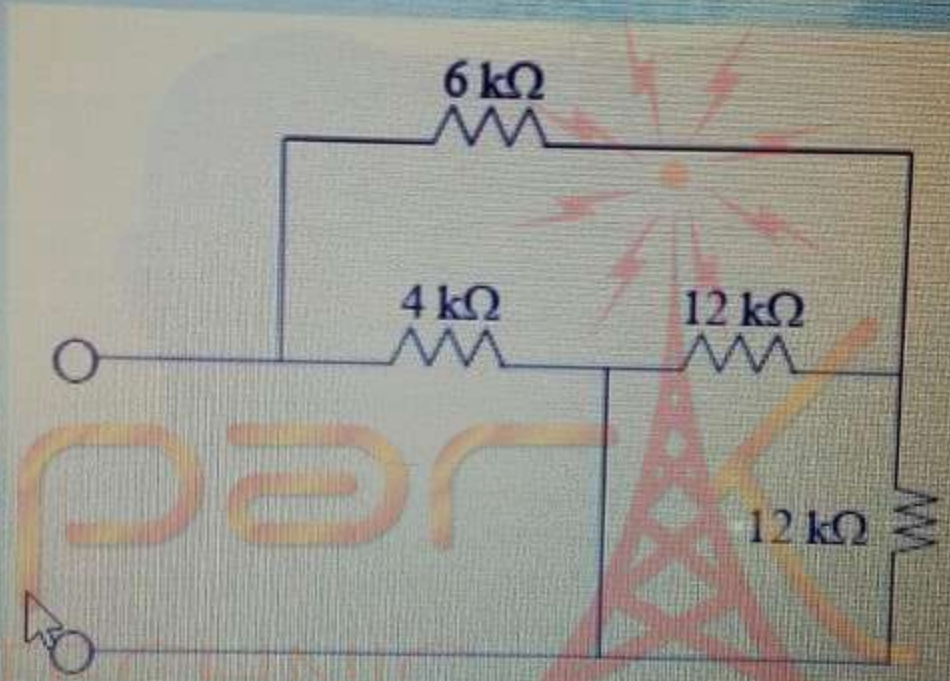
Refer to figure shown; the value of the equivalent resistance between a-b is:



Select one:

- A. 62.5 Ω
- B. 45.5 Ω
- C. 36.25 Ω
- D. 75 Ω

Evaluate R_{eq} in kΩ for the circuit shown in Figure:



Answer:

Protected with free version of Watermarkly. Full version doesn't put this mark.

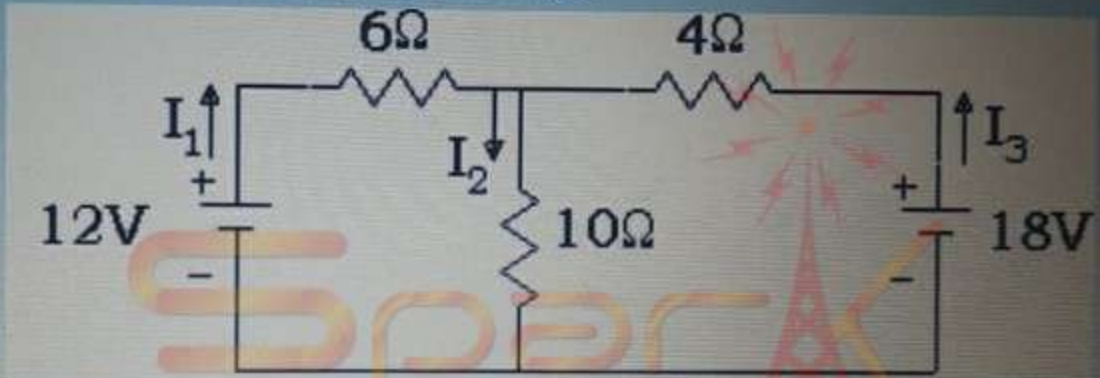
The current through a branch in a linear network is 6 A when the input source voltage is 10 V. If the voltage is reduced to 1 V and the polarity is reversed, the current through the branch is:

Select one:

- A. 0.2 A
- B. -0.6 A
- C. 2 A
- D. -2 A

Protected with free version of Watermarkly. Full version doesn't put this mark.

For the circuit shown in the figure , the current I_1 is:



Select one:

- A. 1.62A
- B. 1.42A
- C. -0.1A
- D. -1.62A

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The conductance of a 20 Ω resistor in milli siemens is:

Select one:

- A. 0.01mS
- B. 10 mS
- C. 50mS
- D. 0.1mS

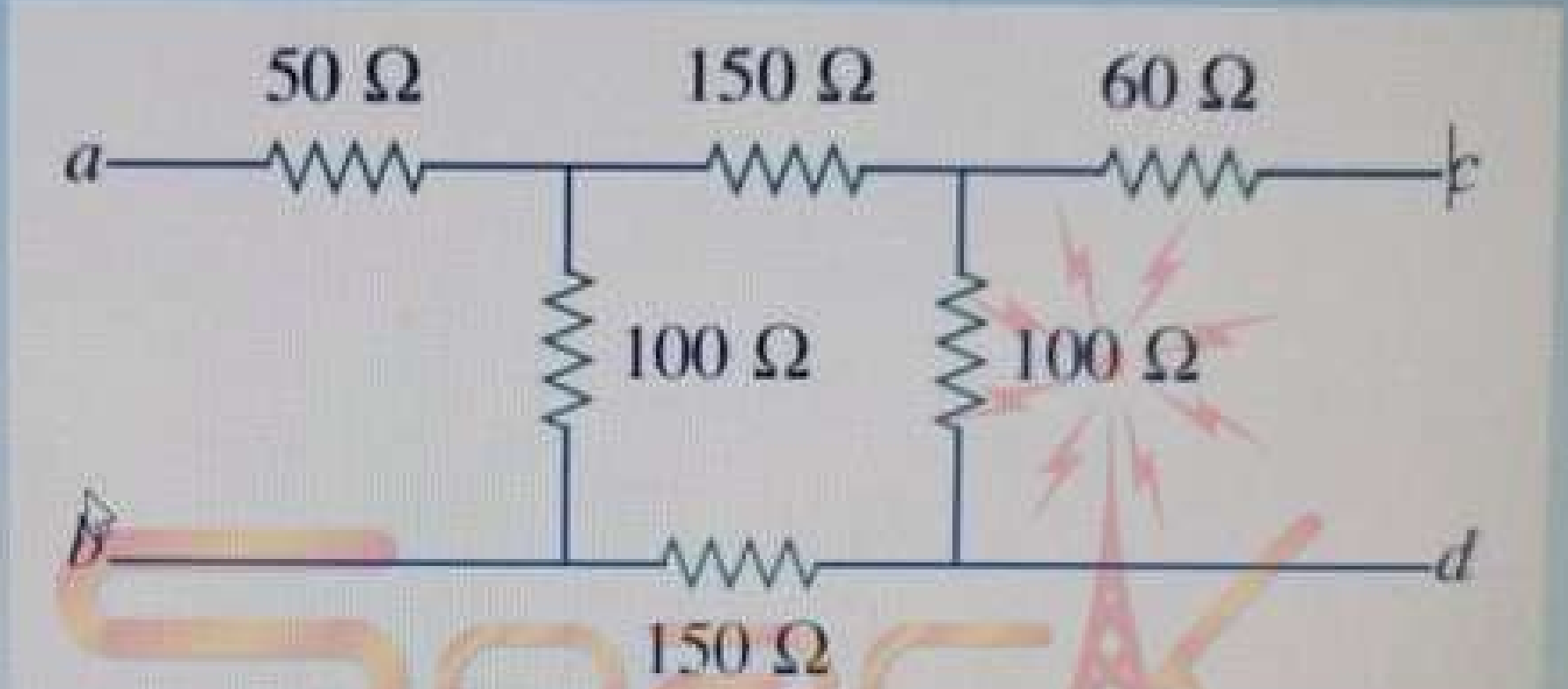
Refer to the figure shown, the power dissipated in 3k Ω is:



Select one:

- A. 17.5W
- B. 675mW
- C. 160mW
- D. 607 kW

: Refer to figure ; the value of the equivalent resistance at terminals cd



اختر احد الخيارات

A. 175 Ω

B. 160Ω

C. 140Ω

D. 260 Ω

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A Norton equivalent circuit with IN=3A and RN = 10Ω is equivalent to Thevenin circuits with:

إختر أحد الخيارات:

a. Vth = 15 V, Rth = 5 Ω

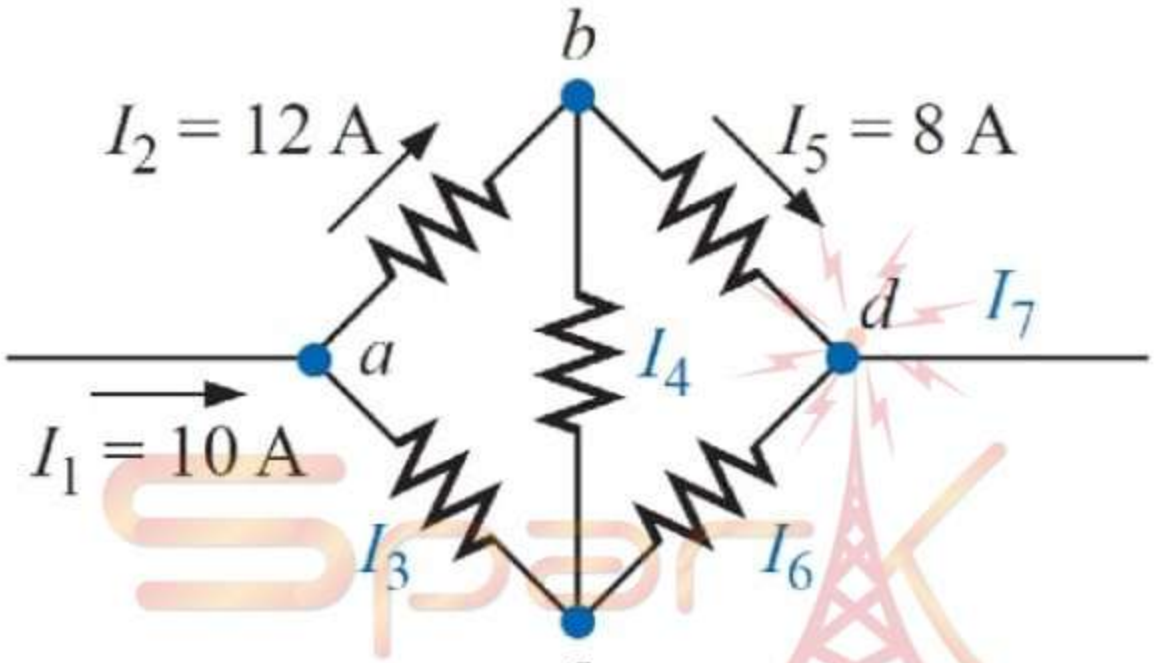
b. Vth = 3 V, Rth = 4 Ω

c. Vth =30 V, Rth = 10 Ω

d. Vth = 3 V, Rth = 10 Ω

Spark POLYTECHNIC
Proceed with free version of Watermarkly. Full version doesn't put this mark.

Find the magnitude of the current I6 for the given network:



POLYTECHNIC

إختر أحد الخيارات:

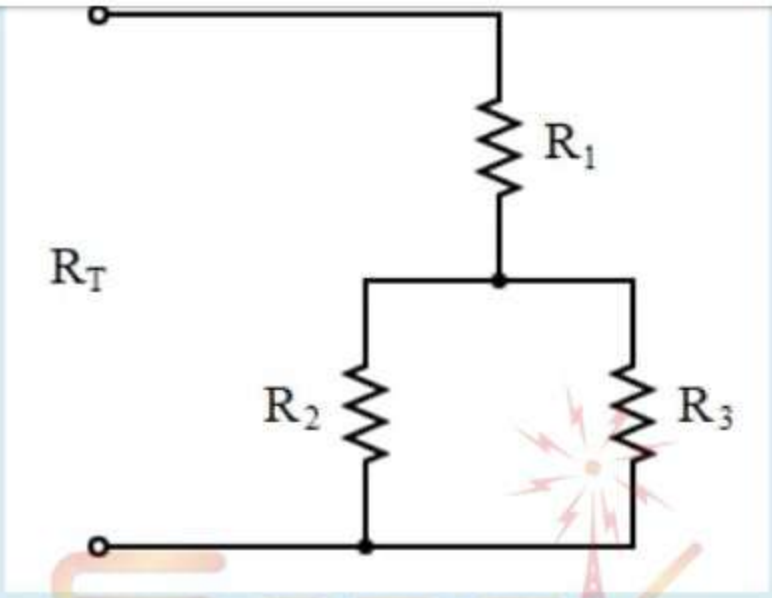
a. 2 A

b. 10 A

c. None of all

d. 4 A

Protected with free version of Watermarkly. Full version doesn't put this mark.



إختر أحد الخيارات:

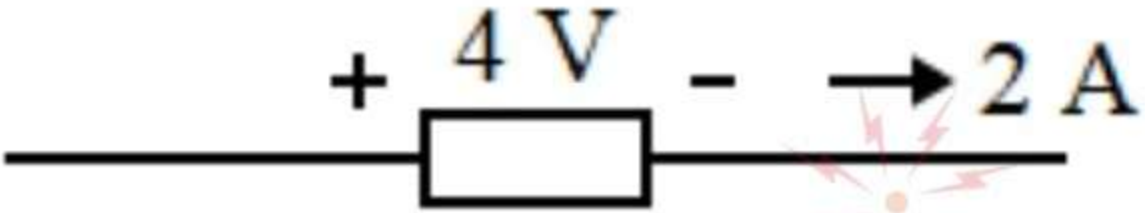
a. 11.43 Ω

b. 6.66 Ω

c. 5.33 Ω

d. None of all

.Find the power generated by the device



إختر أحد الخيارات:

a. 8 W

b. None of all

c. 4 W

d. - 8 W

Spar POLYTECHNIC
Protected with free version of Watermarkly. Full version doesn't put this mark.

A charge entering a terminal is given by $q = .5t^2$ mC. Calculate the current at t = 0.5 S

إختر أحد الخيارات:

a. None of all

b. 5 mA

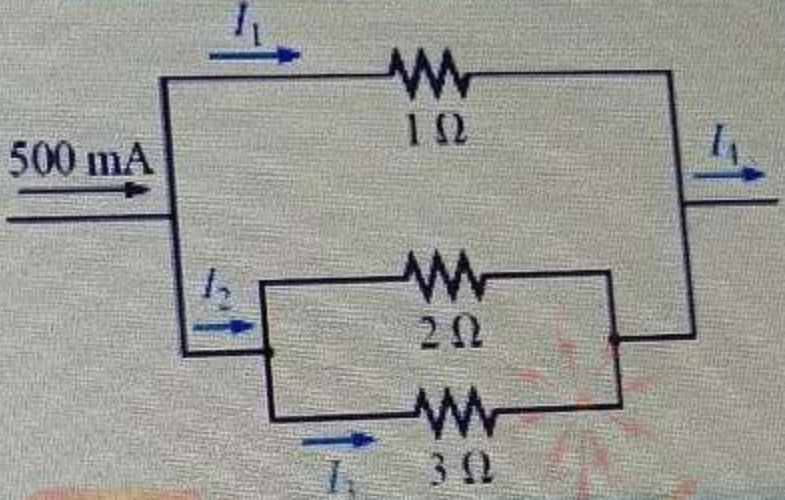
c. 2.5 mA

d. 5 A

Spark POLYTECHNIC

Protected with free version of Watermarkly. Full version doesn't put this mark.

In the given network, find I_2 :

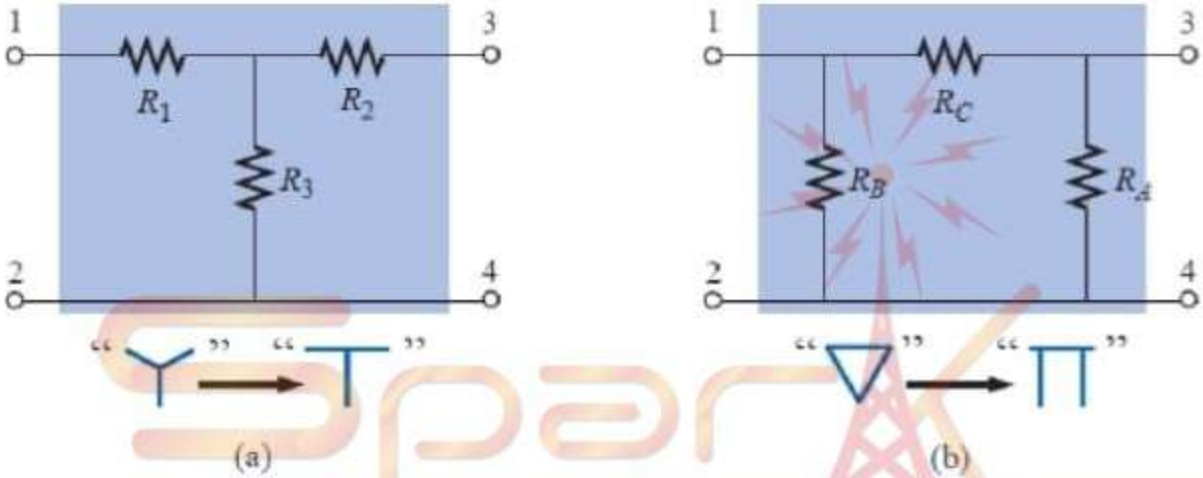


Select one:

- a. None of all
- b. 91 mA
- c. 273 mA
- d. 227 mA

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If the networks a and b are equivalent, find
. R1 if Ra = 10 Ω ; Rb = 20 Ω ; Rc = 5 Ω



POLYTECHNIC

إختر أحد الخيارات:

a. 5.71 Ω

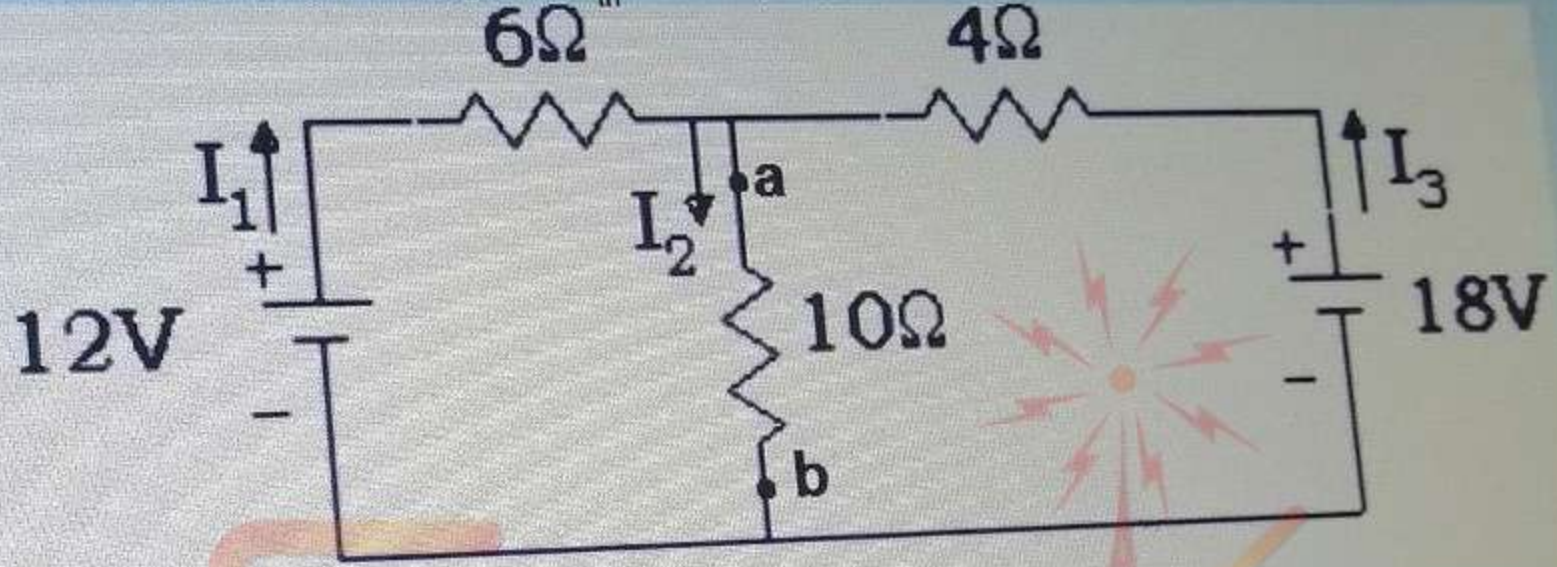
b. None of all

c. 1.43 Ω

d. 2.86 Ω

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For the circuit shown in the figure , V_{th} with respect to terminals a-b is:

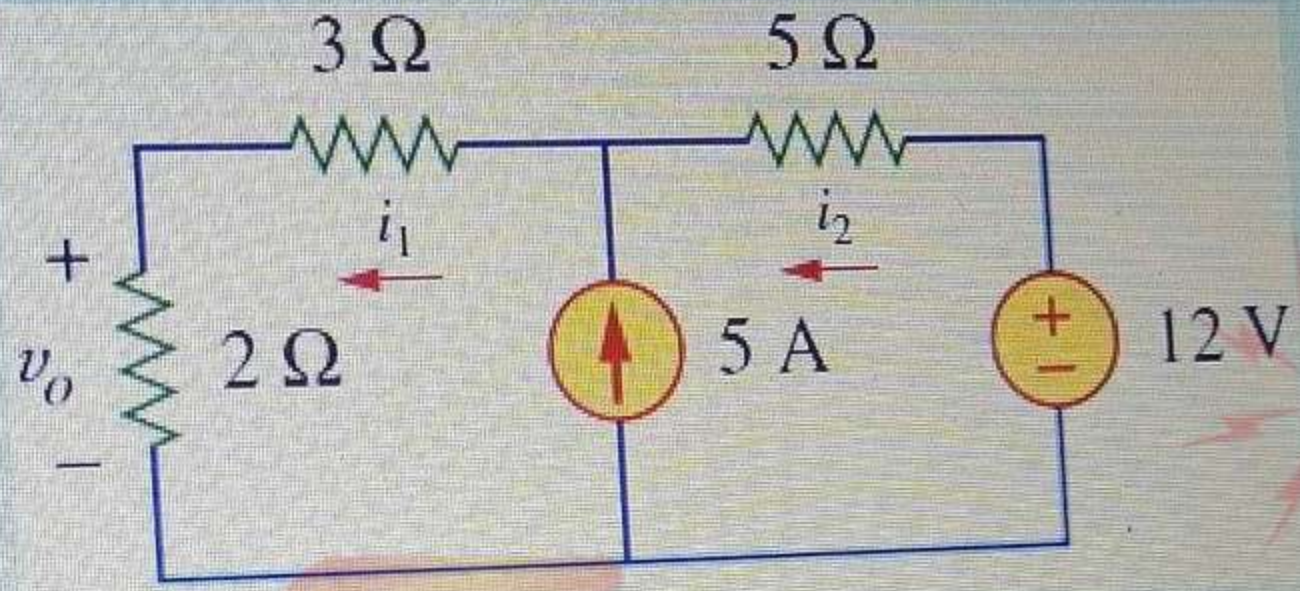


Select one:

- a. 20.4 V
- b. -30 V
- c. -12.6 V
- d. 15.6 V

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Using the superposition theorem, find i_1 due to 5A current source in the circuit of the figure:



Select one:

- a. 2 A
- b. 5 A
- c. 4 A
- d. 2.5 A

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A charge entering a terminal is given by $q = 5t^2$ mC. Calculate the current at t = 0.5 S.

Select one:

- a. 2.5 mA
- b. 5 A
- c. None of all
- d. 5 mA

Spark POLYTECHNIC

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The potential difference between two points in an electric circuit is 24 V. If 0.4 J of energy were dissipated in a period of 5 mS, what would the current be between the two points:

إختر أحد الخيارات:

a. 5 mA

b. 5 A

c. None of all

d. 3.33 A

Spark POLYTECHNIC

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A resistor absorbs an instantaneous power of $30\cos^2 t$ mW when connected to a voltage source v=10Cos t V. Find *R*.

Select one:

- A. 33 Ω
- B. 330 Ω
- C. 3.33 kΩ
- D. 330 mΩ

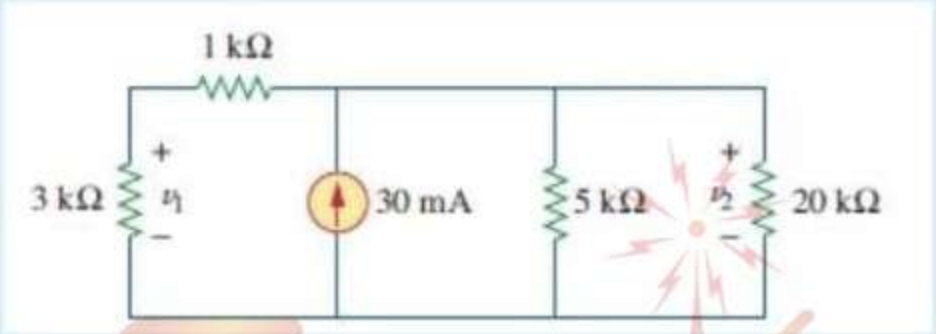
Spark POLYTECHNIC

The voltage across a 500 W load that produces 5 A is :

Select one:

- A. 100 V
- B. 1kV
- C. 200V
- D. 50V

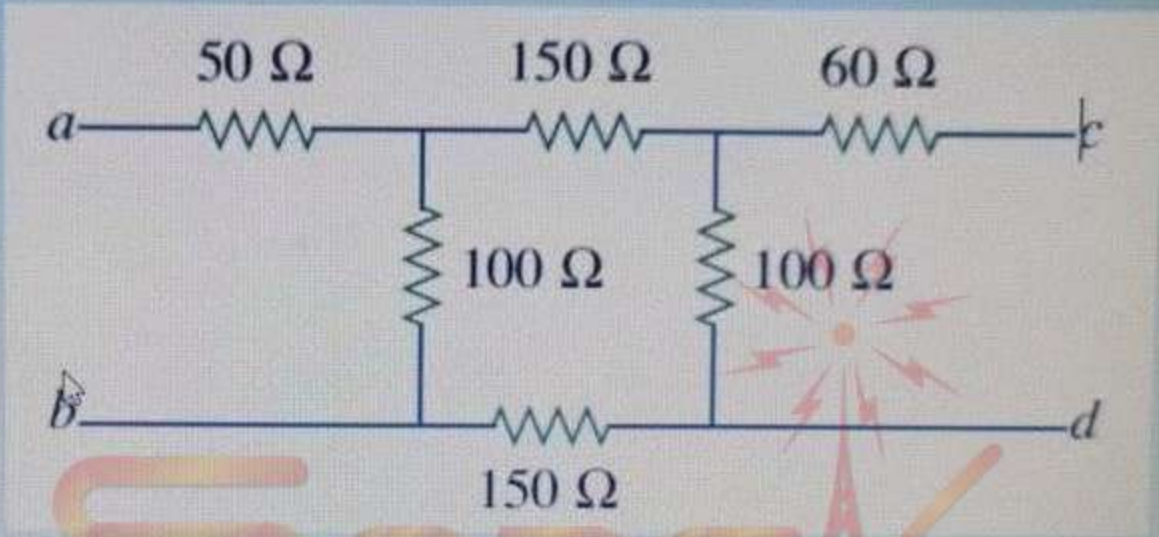
Refer to the figure shown, the power dissipated in 3k Ω is:



Select one:

- A. 17.5W
- B. 675mW
- C. 160mW
- D. 607 kW

: Refer to figure ; the value of the equivalent resistance at terminals cd



Spark
POLYTECHNIC

اختر احد الخيارات

A. 175 Ω

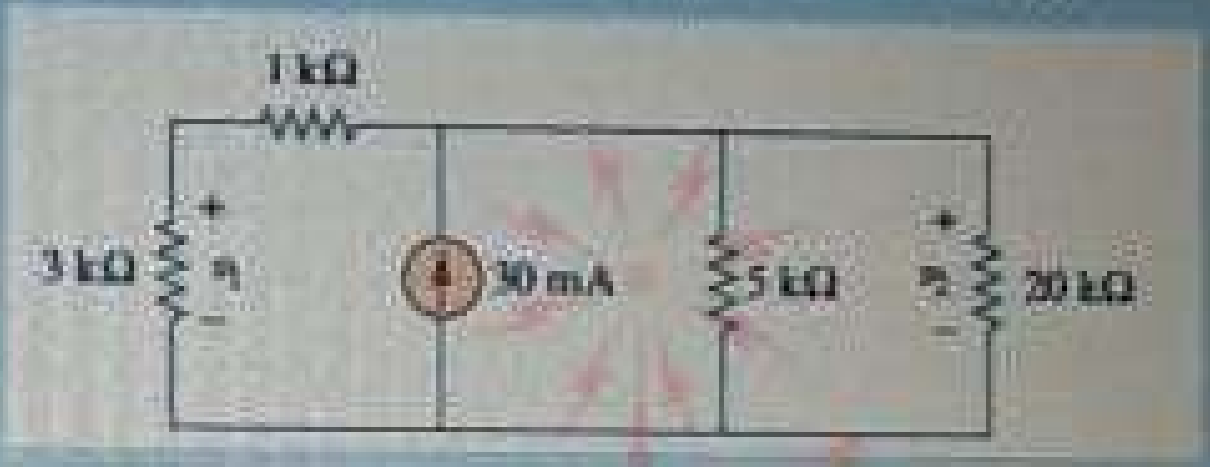
B. 160Ω

C. 140Ω

D. 260 Ω

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Refer to the figure shown, the power supplied by the current source is:



Select one:

- A. 12W
- B. 1.8 W
- C. 175 mW
- D. 160mW

The conductance of a 20 Ω resistor in milli siemens is:

Select one:

- A. 0.01mS
- B. 10 mS
- C. 50mS
- D. 0.1mS

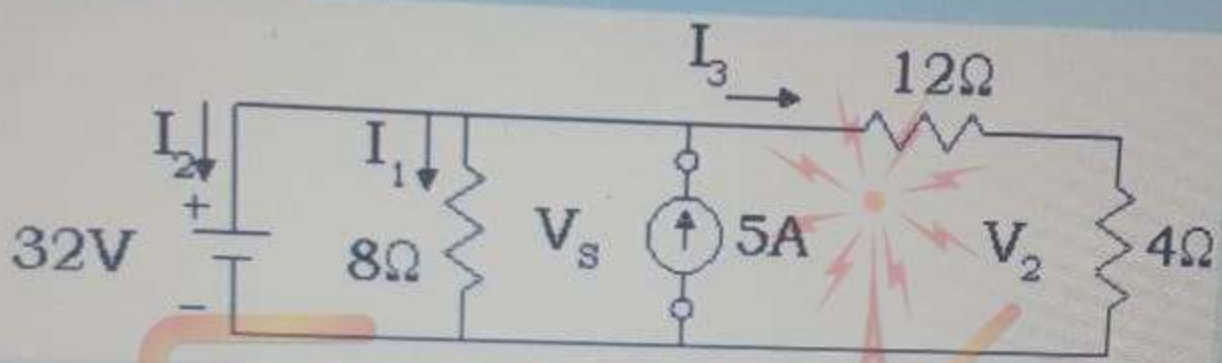
The maximum current that a 2W, 80 kΩ resistor can safely conduct is:

Select one:

- A. 5 mA
- B. 25 μA
- C. 160 kA
- D. 40 kA

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Refer to the figure shown, the current I_3 is:



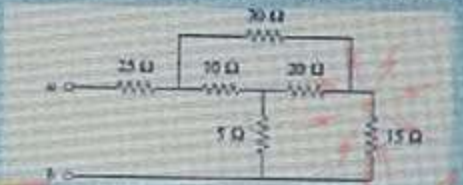
Select one:

- A. 2A
- B. 5A
- C. 3A
- D. 4A

Spark POLYTECHNIC

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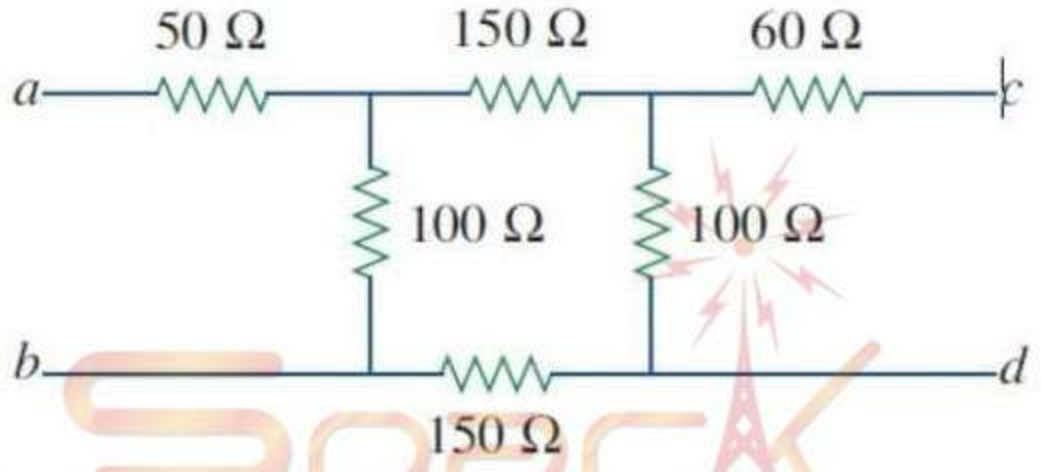
Refer to figure shown; the value of the equivalent resistance between a-b is:



Select one:

- A. 62.5 Ω
- B. 45.5 Ω
- C. 36.25 Ω
- D. 75 Ω

Refer to figure ; the value of the equivalent resistance at terminals ab is:

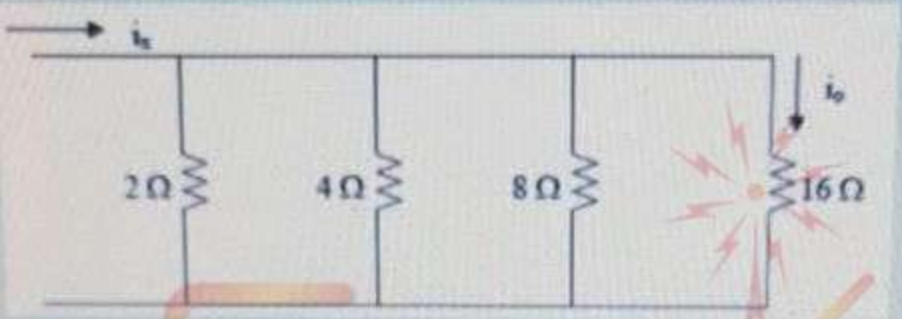


Select one:

- A. 160 Ω
- B. 130 Ω
- C. 175Ω
- D. 260 Ω

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: For the circuit in the figure, i_0 =2 A. Calculate ix



اختر أحد الخيارات

A. 120

B. 15

C. 60

D. 30A

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The current through a branch in a linear network is 5 A when the input source voltage is 10 V. If the voltage is reduced to 1 V and the polarity is reversed, the current through the branch is:

Select one:

- A. 2 A
- B. 0.2 A
- C. -0.5 A
- D. -2 A

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A current of 3.2 A flows through a conductor. Calculate how much charge passes through any cross-section of the conductor in 20 seconds.

اختر أحد الخيارات

A. 128

B. 64 C

C. 16

D. 32

Spark POLYTECHNIC

Protected with free version of Watermarkly. Full version doesn't put this mark.

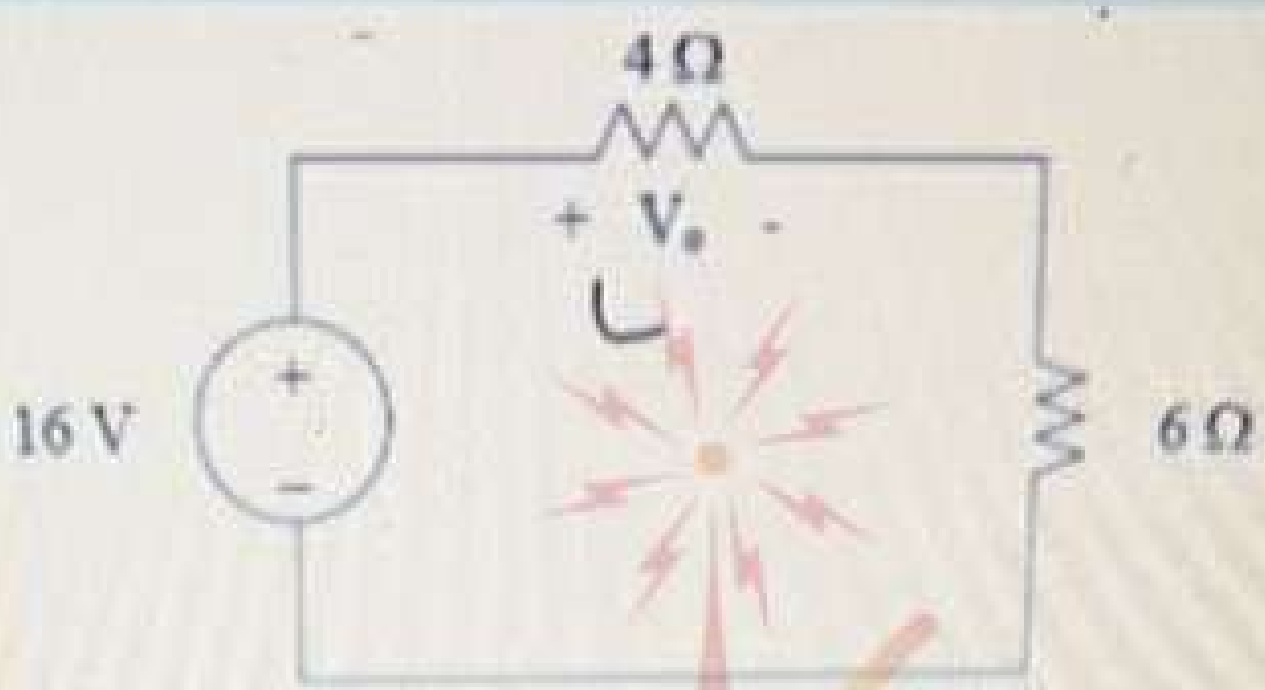
The current through a branch in a linear network is 6 A when the input source voltage is 10 V. If the voltage is reduced to 1 V and the polarity is reversed, the current through the branch is:

Select one:

- A. 0.2 A
- B. -0.6 A
- C. 2 A
- D. -2 A

Protected with free version of Watermarkly. Full version doesn't put this mark.

Calculate V_0 in the circuit of Figure.



Select one:

- A. 6.4 V
- B. 12.8
- C. 16
- D. 32

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The prefix pico stands for:

Select one:

- A. 10^{-12}
- B. 10^{-15}
- C. 10^{-6}
- D. 10^{-9}

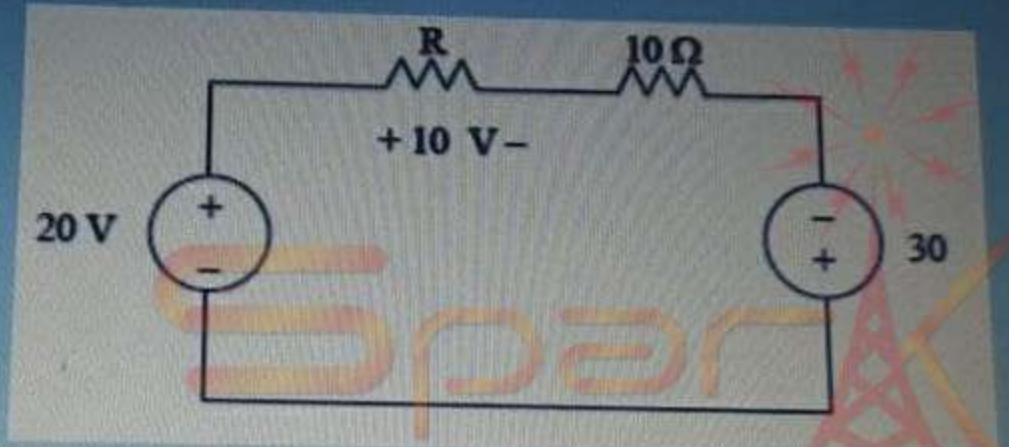
The current through a branch in a linear network is 5 A when the input source voltage is 10 V. If the voltage is reduced to 1 V and the polarity is reversed, the current through the branch is:

Select one:

- A. -0.5 A
- B. 0.2 A
- C. 2 A
- D. -2 A

Spark POLYTECHNIC

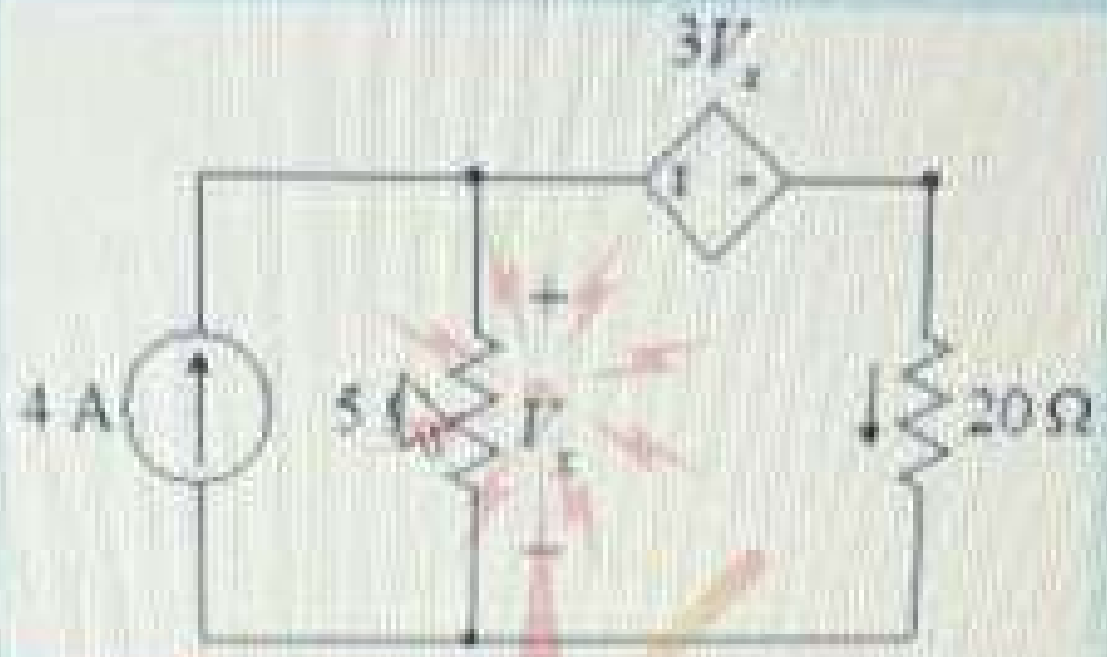
Find R for the circuit in Figure.



Answer:

Protected with free version of Watermarkly. Full version doesn't put this mark.

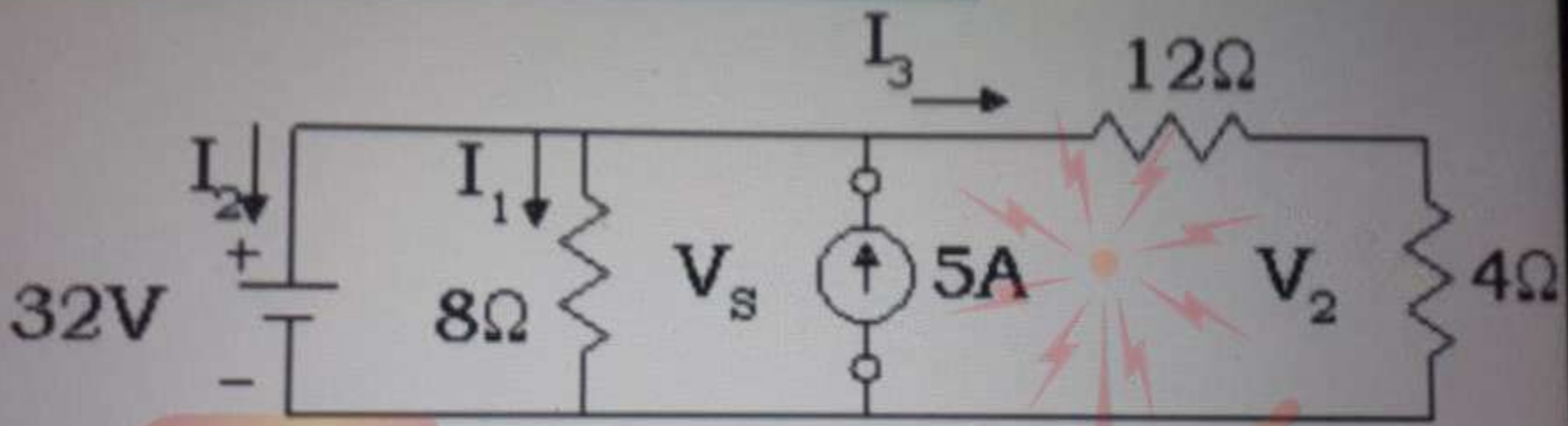
In the circuit shown, The current in 20 Ω resistor is:



Select one

- A. 200 mA
- B. 2A
- C. 1A
- D. 2.5A

Refer to the figure shown, the current I_3 is:



Select one:

- A. 3A
- B. 5A
- C. 4A
- D. 2A

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: For the circuit in the figure, i_0 =2 A. Calculate ix



اختر احد الخيارات

A. 120

B. 15

C. 60

D. 30A

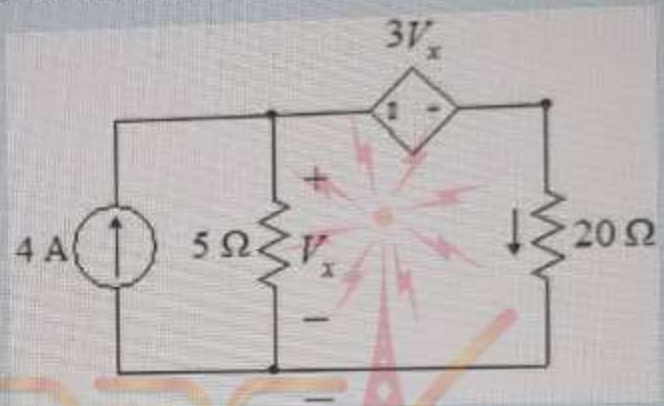
Calculate V_0 in the circuit of Figure.



Select one:

- A. 6.4 V
- B. 12.8
- C. 16
- D. 32

In the circuit shown, The current in 20 Ω resistor is:



Select one:

- A. 2.8A
- B. 1A
- C. 200 mA
- D. 2A

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A resistor absorbs an instantaneous power of $30\cos^{2}t$ mW when connected to a voltage source v=10Cos t V. Find *R*.

Select one:

- A. 33 Ω
- B. 330 Ω
- C. 3.33 kΩ
- D. 330 mΩ

Spark POLYTECHNIC

A current of 3.2 A flows through a conductor. Calculate how much charge passes through any cross-section of the conductor in 20 seconds.

اختر أحد الخيارات

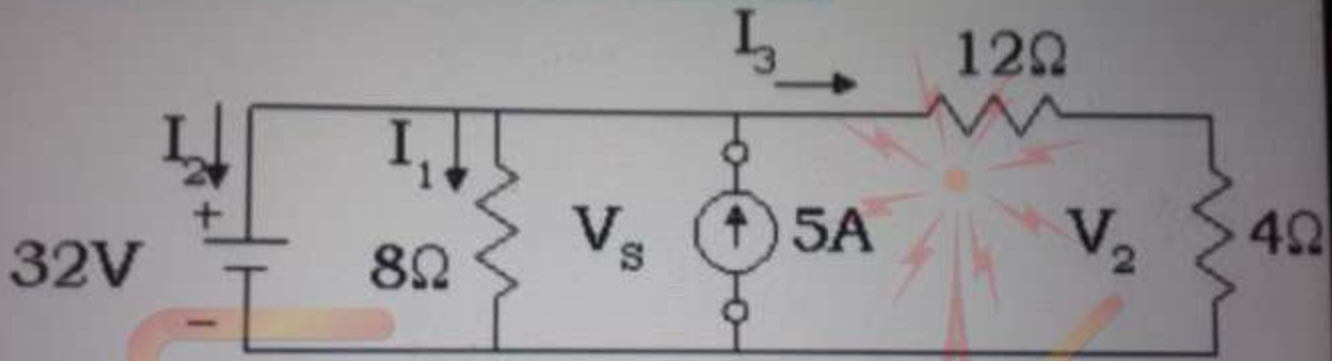
A. 128

B. 64 C

C. 16

D. 32

Refer to the figure shown, the current I_3 is:



Select one:

- A. 3A
- B. 5A
- C. 4A
- D. 2A

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The prefix pico stands for:

Select one:

- A. 10^{-12}
- B. 10^{-15}
- C. 10^{-6}
- D. 10^{-9}

The voltage across a 500 W load that produces 5 A is :

Select one:

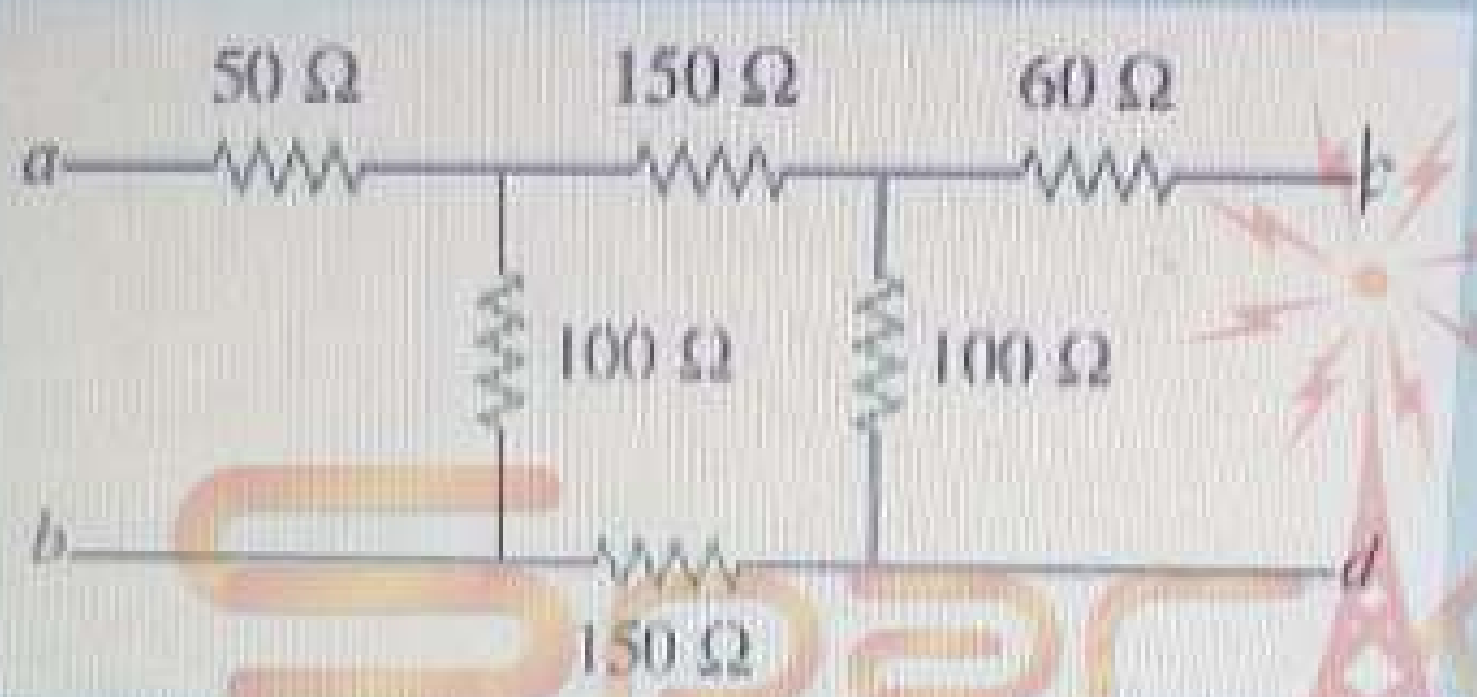
- A. 100 V
- B. 1kV
- C. 200V
- D. 50V

Internal resistance of ideal voltage source is:

Select one:

- A. finite
- B. zero
- C. None of the answers
- D. Infinite

Refer to figure ; the value of the equivalent resistance at terminals a

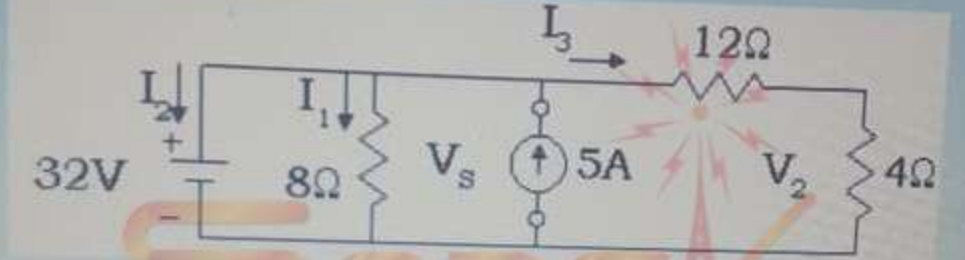


Select one:

- A. 160 Ω
- B. 260 Ω
- C. 175Ω
- D. 130 Ω

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Refer to the figure shown, the current I_3 is:



Select one

- A. 2A
- B. 5A
- C. 3A
- D. 4A

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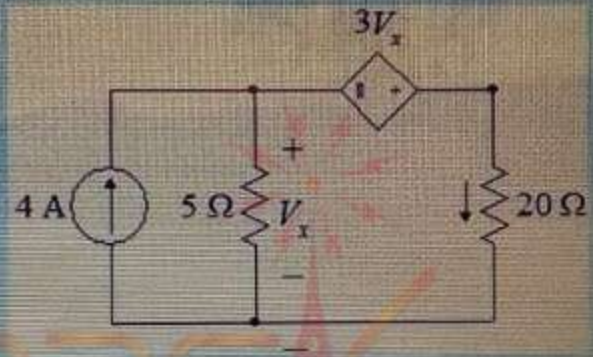
The current through a branch in a linear network is 5 A when the input source voltage is 10 V. If the voltage is reduced to 1 V and the polarity is reversed, the current through the branch is:

Select one:

- A. 2 A
- B. 0.2 A
- C. -0.5 A
- D. -2 A

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In the circuit shown, The current in 20 Ω resistor is:

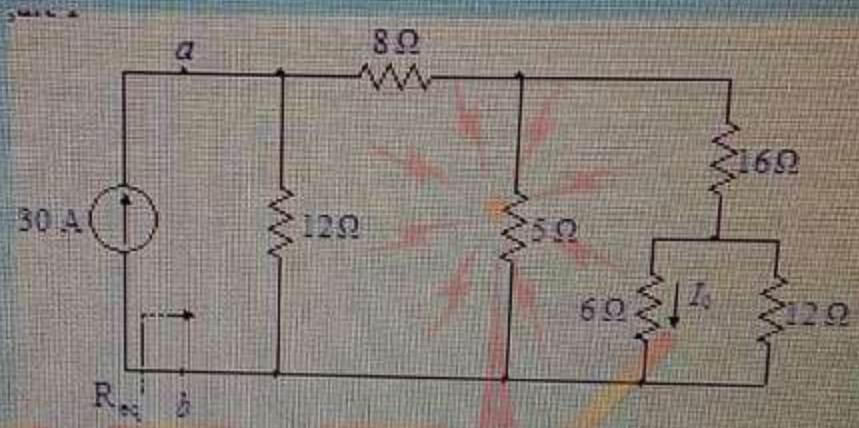


Select one:

- A. 200 mA
- B. 2.8A
- C. 2A
- D. 1A

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The voltage across 8Ω resistor is:



Select one:

- A. 240V
- B. 120V
- C. 60 V
- D. 72V

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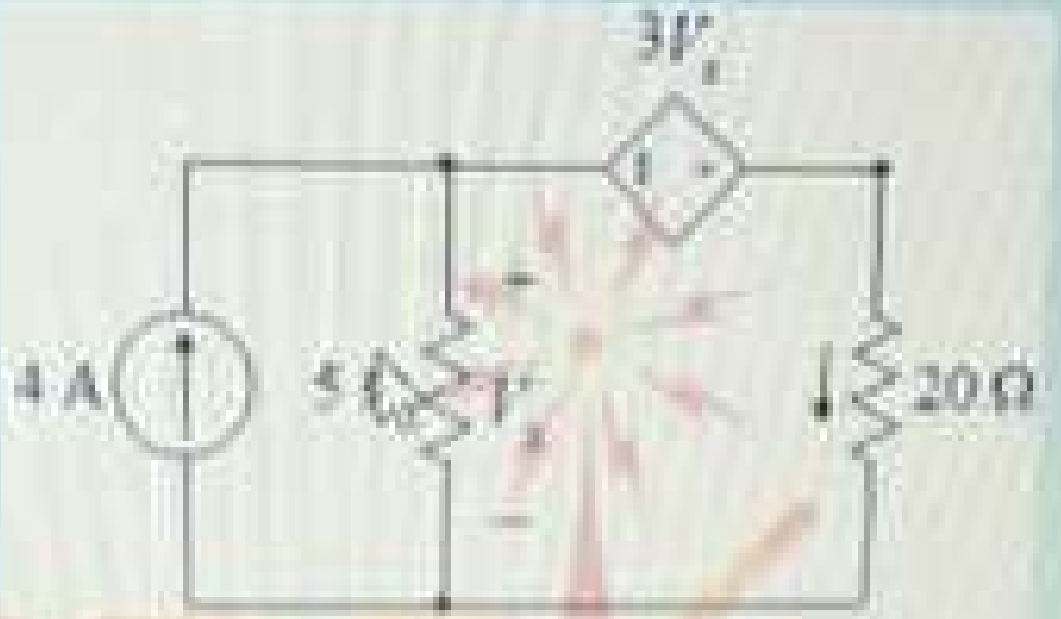
Refer to the figure shown, the power supplied by the current source is:



Select one:

- A. 12W
- B. 1.8W
- C. 175 mW
- D. 180mW

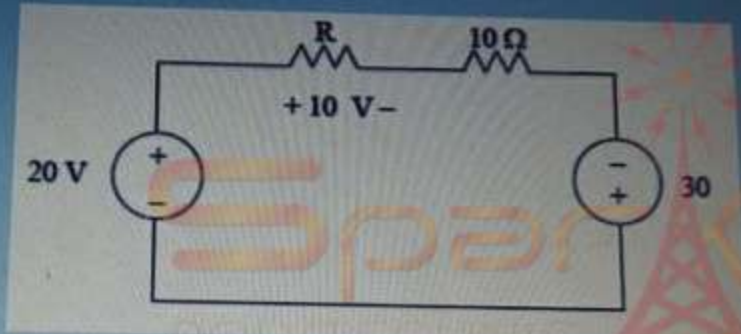
In the circuit shown, The current in 20 Ω resistor is:



Select one:

- A. 200 mA
- B. 2A
- C. 1A
- D. 2.8A

Find R for the circuit in Figure.



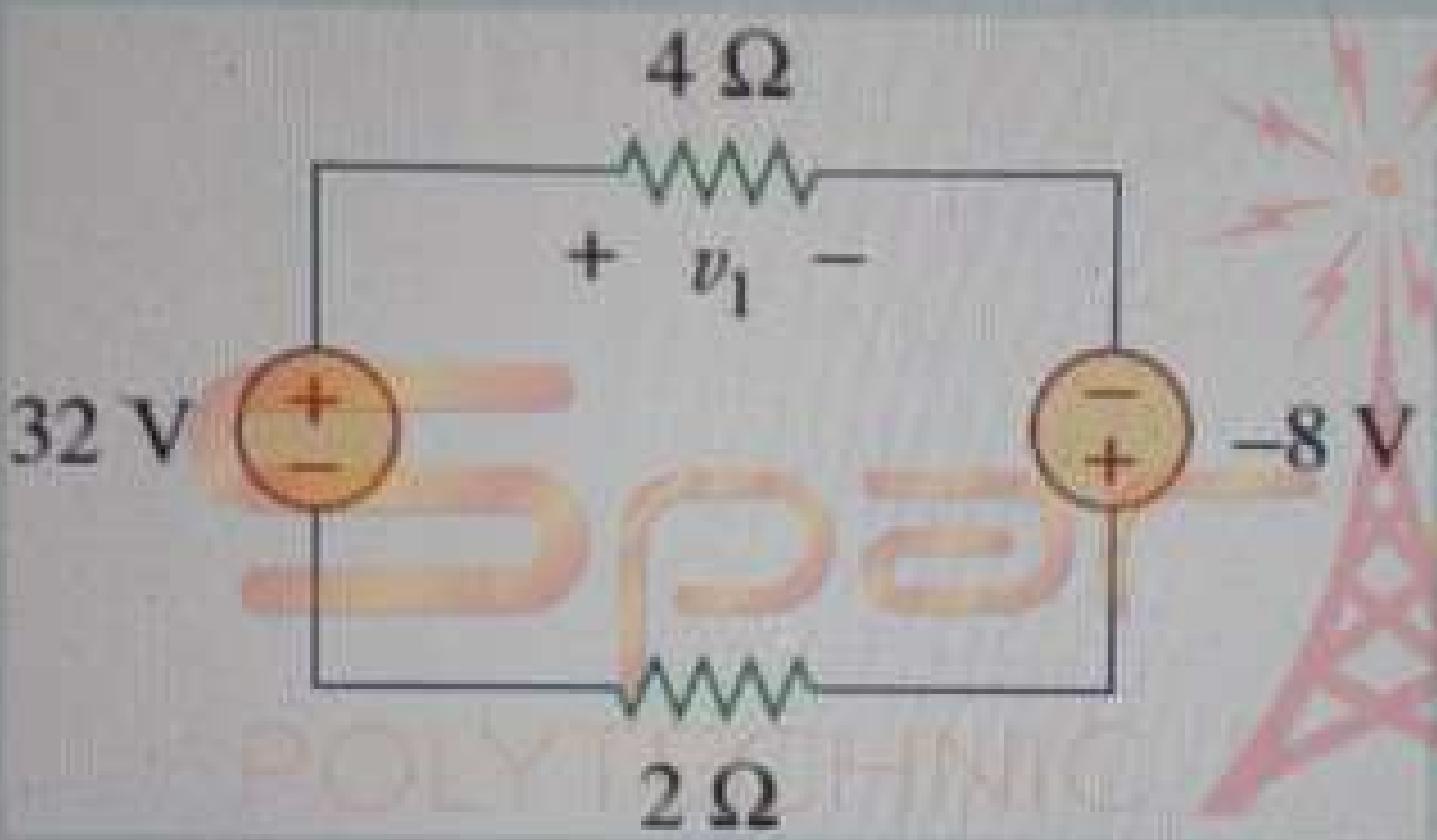
Answer:

A resistor absorbs an instantaneous power of $30\cos^2 t$ mW when connected to a voltage source v=45 Cos t V. Find *R*.

Select one:

- A. 67.5 kΩ
- B. 675 Ω
- C. 667 mΩ
- D. 667 Ω

For the circuit in figure the voltage v_1 is:



Answer:

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The maximum current that a 2W, 80 kΩ resistor can safely conduct is:

Select one:

- A. 5 mA
- B. 25 µA
- C. 160 kA
- D. 40 kA

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The superposition principle applies to power calculation.

إختر واحداً:

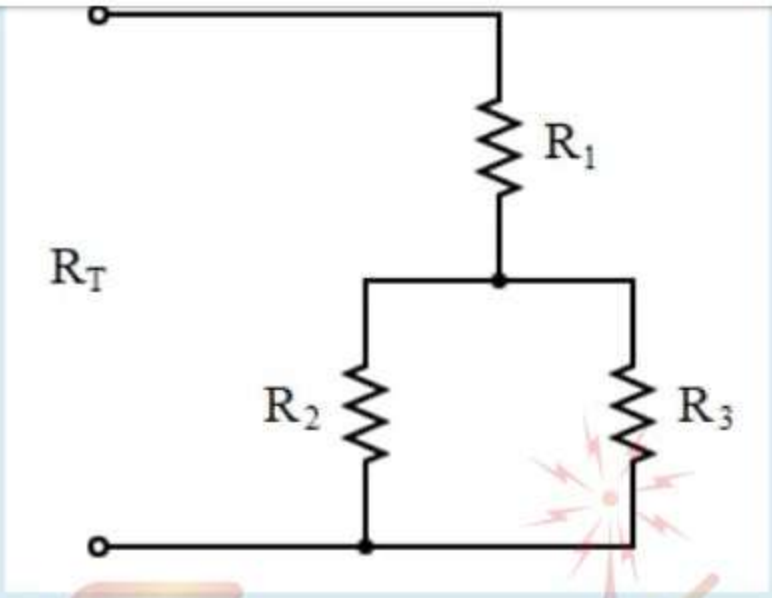
- صح
- خطأ

Spar POLYTECHNIC

.Circuit elements that absorb power have

إختر أحد الخيارات:

- .a. A positive value of P
- b. None of all
- .c. Zero value of P
- .d. A negative value of P



إختر أحد الخيارات:

a. 11.43 Ω

b. 6.66 Ω

c. 5.33 Ω

d. None of all

A charge entering a terminal is given by $q = .5t^2$ mC. Calculate the current at t = 0.5 S

إختر أحد الخيارات:

a. None of all

b. 5 mA

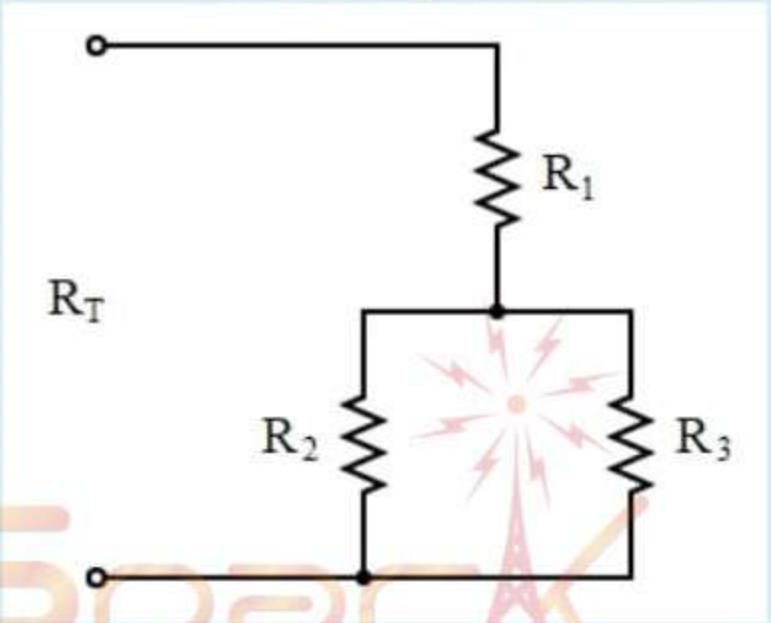
c. 2.5 mA

d. 5 A

Spark POLYTECHNIC

Protected with free version of Watermarkly. Full version doesn't put this mark.

For the given network find RT if R1 =10 Ω; .R2 = 2 Ω; and R3 = 5 Ω



POLYTECHNIC

إختر أحد الخيارات:

a. 11.43 Ω

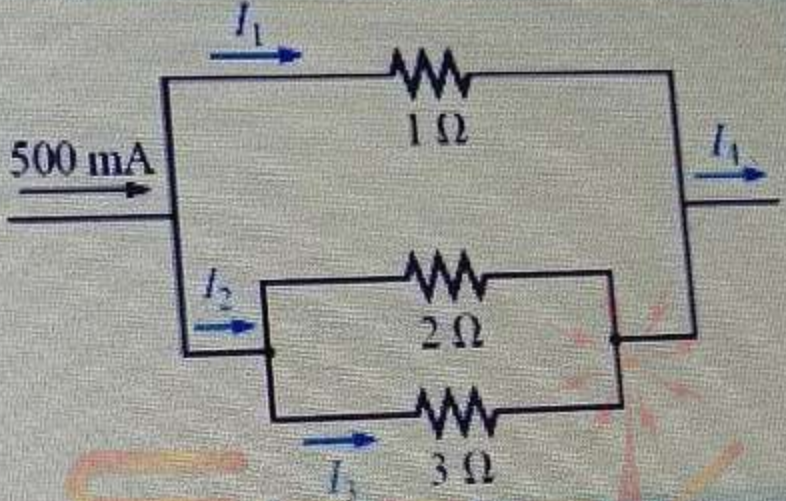
b. 6.66 Ω

c. 5.33 Ω

d. None of all

Protected with free version of Watermarkly. Full version doesn't put this mark.

In the given network, find I_2 :

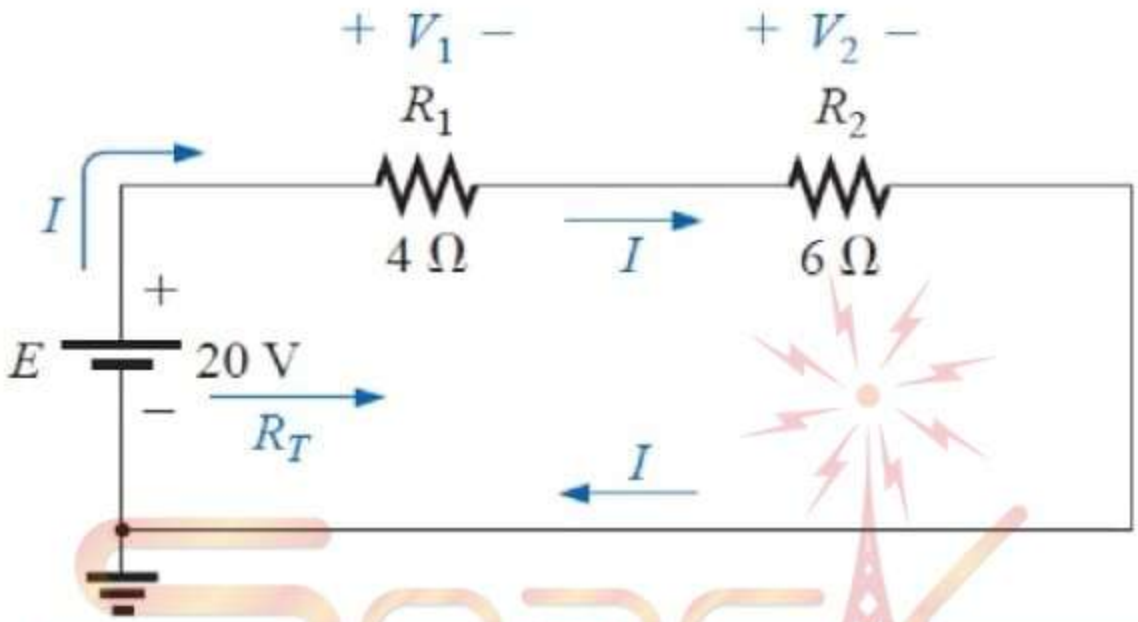


Select one:

- a. None of all
- b. 91 mA
- c. 273 mA
- d. 227 mA

Produced with free version of Watermarkly. Full version doesn't put this mark.

: For the circuit shown, find V_1



Spark
POLYTECHNIC

إختر أحد الخيارات:

a. 12 V

b. 3.33 V

c. 8 V

d. None of all

A Norton equivalent circuit with IN=3A and RN = 10Ω is equivalent to Thevenin circuits with:

إختر أحد الخيارات:

a. Vth = 15 V, Rth = 5 Ω

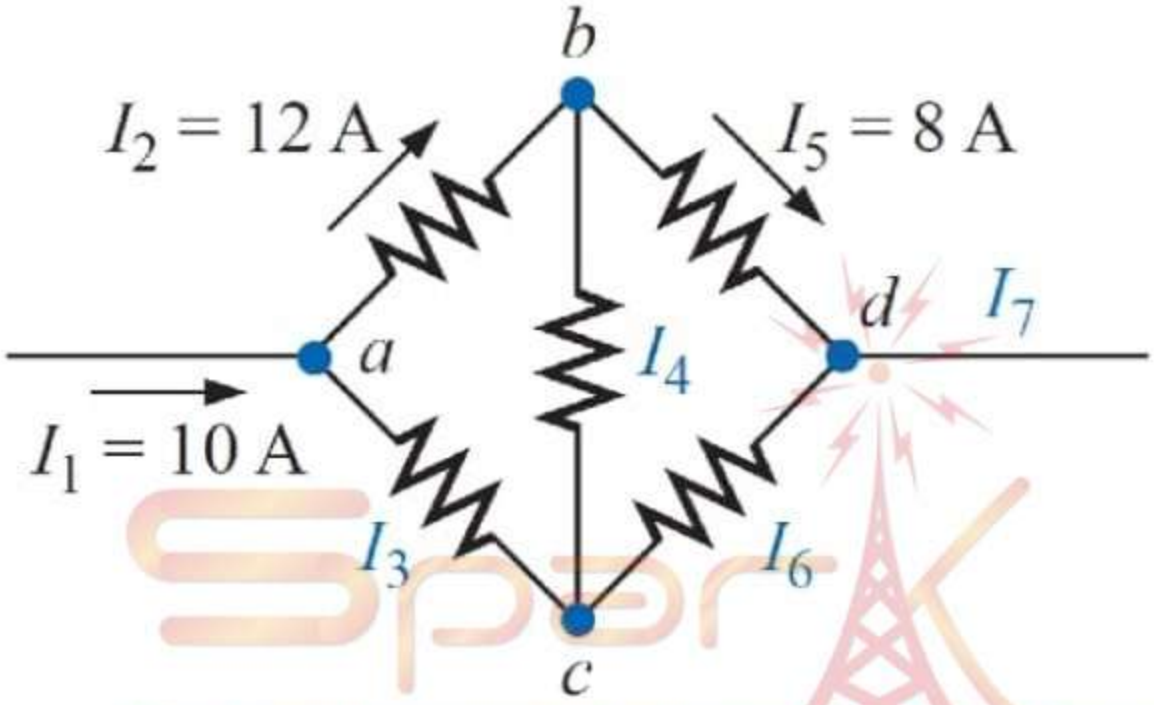
b. Vth = 3 V, Rth = 4 Ω

c. Vth =30 V, Rth = 10 Ω

d. Vth = 3 V, Rth = 10 Ω

Proceed with free version of Watermarkly. Full version doesn't put this mark.

Find the magnitude of the current I6 for the given network:



Spark POLYTECHNIC

إختر أحد الخيارات:

a. 2 A

b. 10 A

c. None of all

d. 4 A

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The voltage across a 5 kW load that produces 25 A is

Select one:

- a. 2kV
- b. 250V
- c. 500V
- d. 200 V

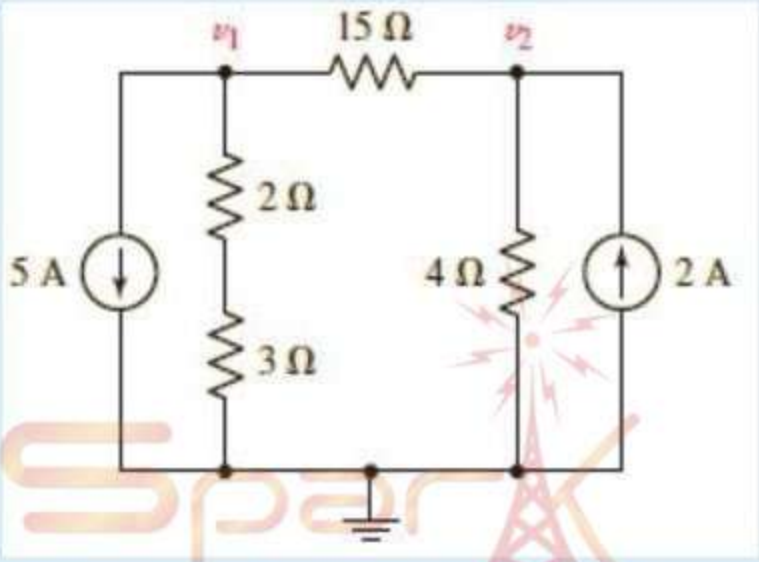
Protected with free version of Watermarkly. Full version doesn't put this mark.

The conductance of a 100Ω resistor in milli siemens is:

Select one:

- A. 100 mS
- B. 0.1mS
- C. 0.01mS
- D. 10mS

In the circuit shown, The power delivered by 2 A current source is:



Spark
POLYTECHNIC

Select one:

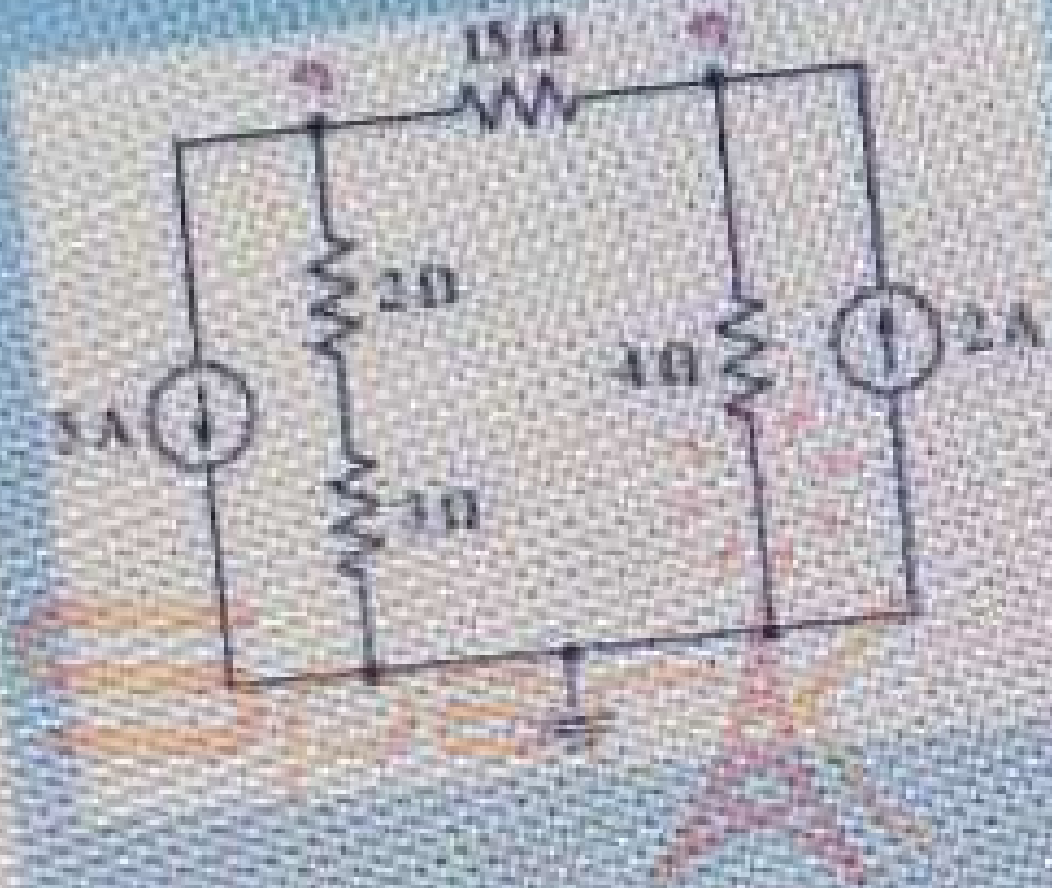
- A. 5W
- B. -44.8W
- C. 4W
- D. 25W

A load draws 30 A when connected to a 240V source.How long does it take to consume 180 kJ?

Select one:

- A. 5s
- B. 25s
- C. 2.5 minutes
- D. 250 ms

In the circuit shown, The power delivered by 2 A current source is:



Select one:

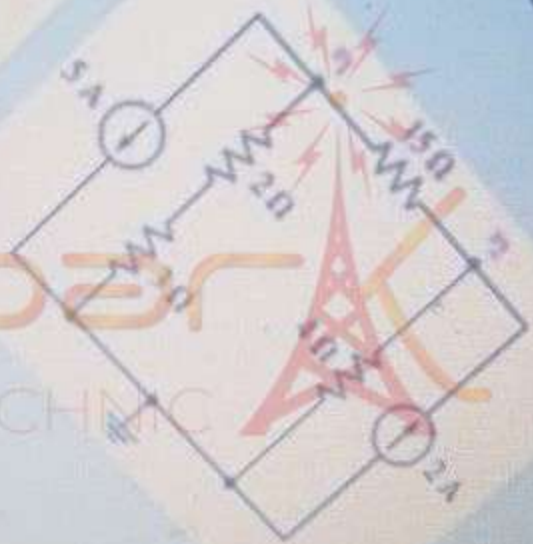
A. 5W

B. -11.8W

C. 4W

D. 25W

In the circuit shown, The va
is:



Select one:

A. 18.125V

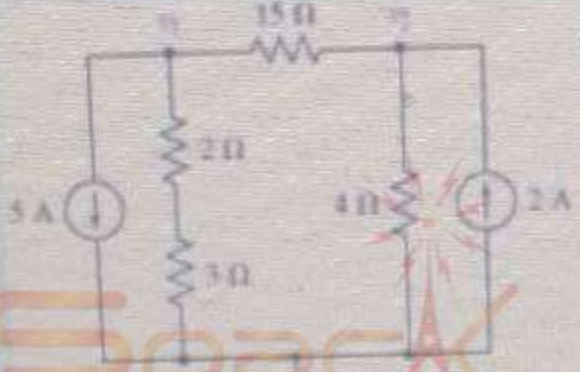
B. 36.5V

C. -4.8V

D. 365 mV

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In the circuit shown, The value of V_1 is:



Select one:

A. -18.125V

B. 36.5V

C. -4.8V

D. 365 mV

Protected with free version of Watermarkly. Full version doesn't put this mark.

Refer to the figure shown, the voltage V_2 is:



Select one:

A. 15V

B. 75V

C. 60V

D. 50V

Proceed with free version of Watermarkly. Full version doesn't put this mark.

A load draws 30 A when connected to a 240V source.How long does it take to consume 180 kJ?

Select one:

a. 25s

b. 2.5 minutes

c. 250ms

d. 5s

Spark POLYTECHNIC

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A resistor absorbs an instantaneous power of $30 \cos^2 t$ mW when connected to a voltage source v=10Cos t V. Find *R*.

Select one:

- a. 667 Ω
- b. 675 Ω
- c. 6 7.5 kΩ
- d. 667 mΩ

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Refer to the figure below. The value of the equivalent resistance at terminals AB is



The parallel internal resistance of ideal current source is:

Select one:

- a. Infinite
- b. zero
- c. finite
- d. None of the answers

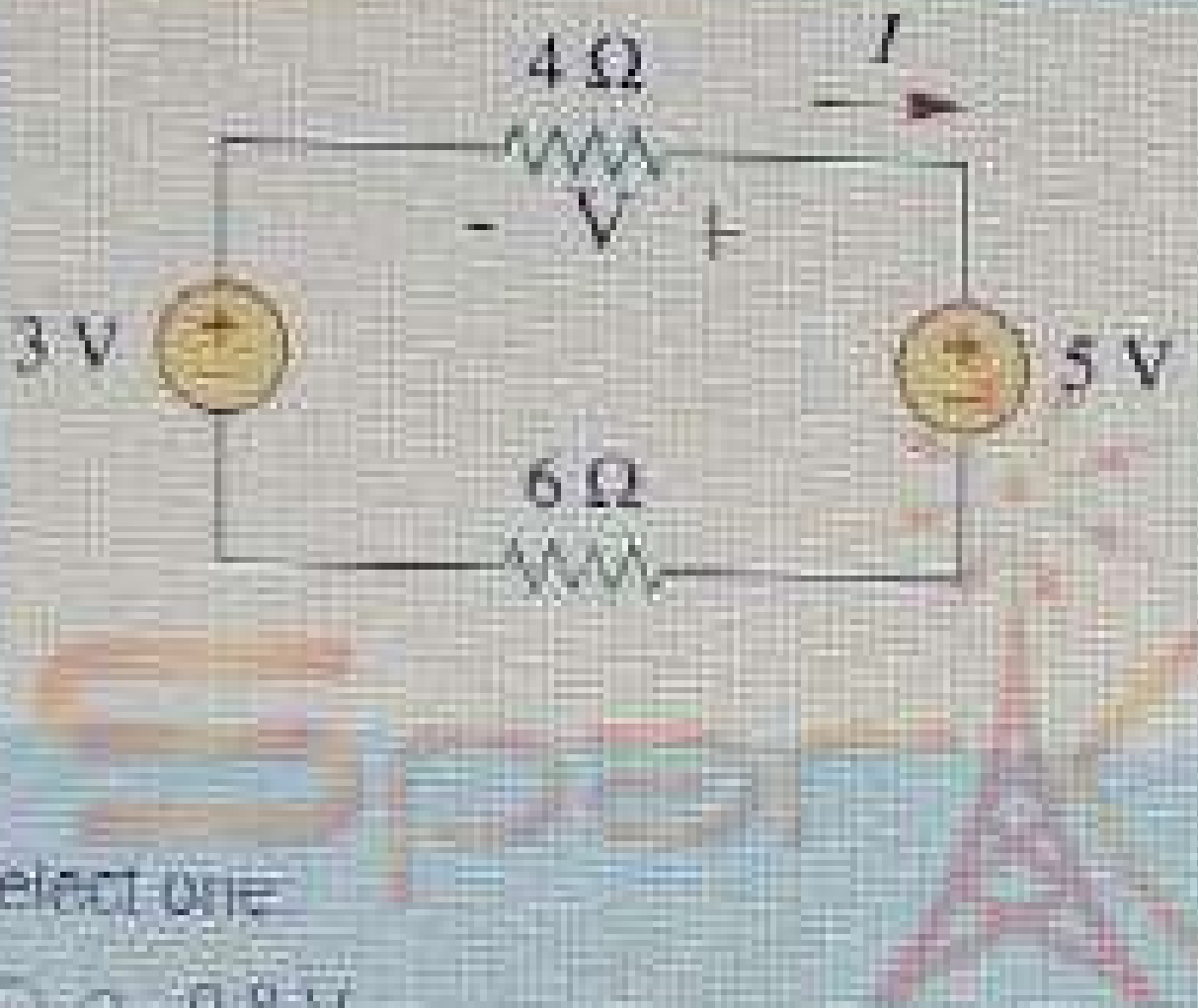
Spark POLYTECHNIC

The conductance of a 100Ω resistor in milli siemens is:

Select one:

- a. 100 mS
- b. 10mS
- c. 0.01mS
- d. 0.1mS

The voltage across 4 Ω resistor in the circuit is



Select one:

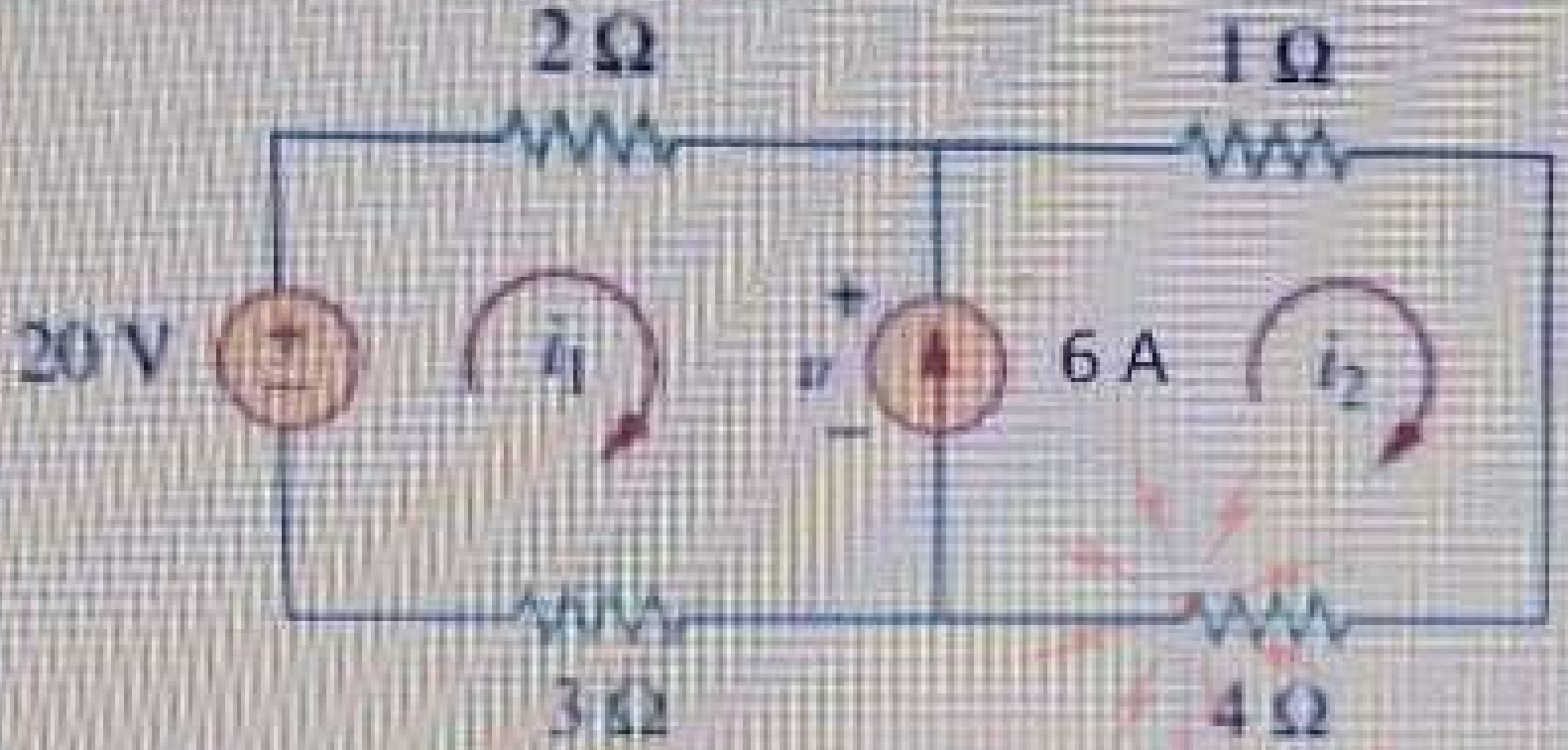
a. -0.8 V

b. 0.8 V

c. -1.2 V

d. 1.2 V

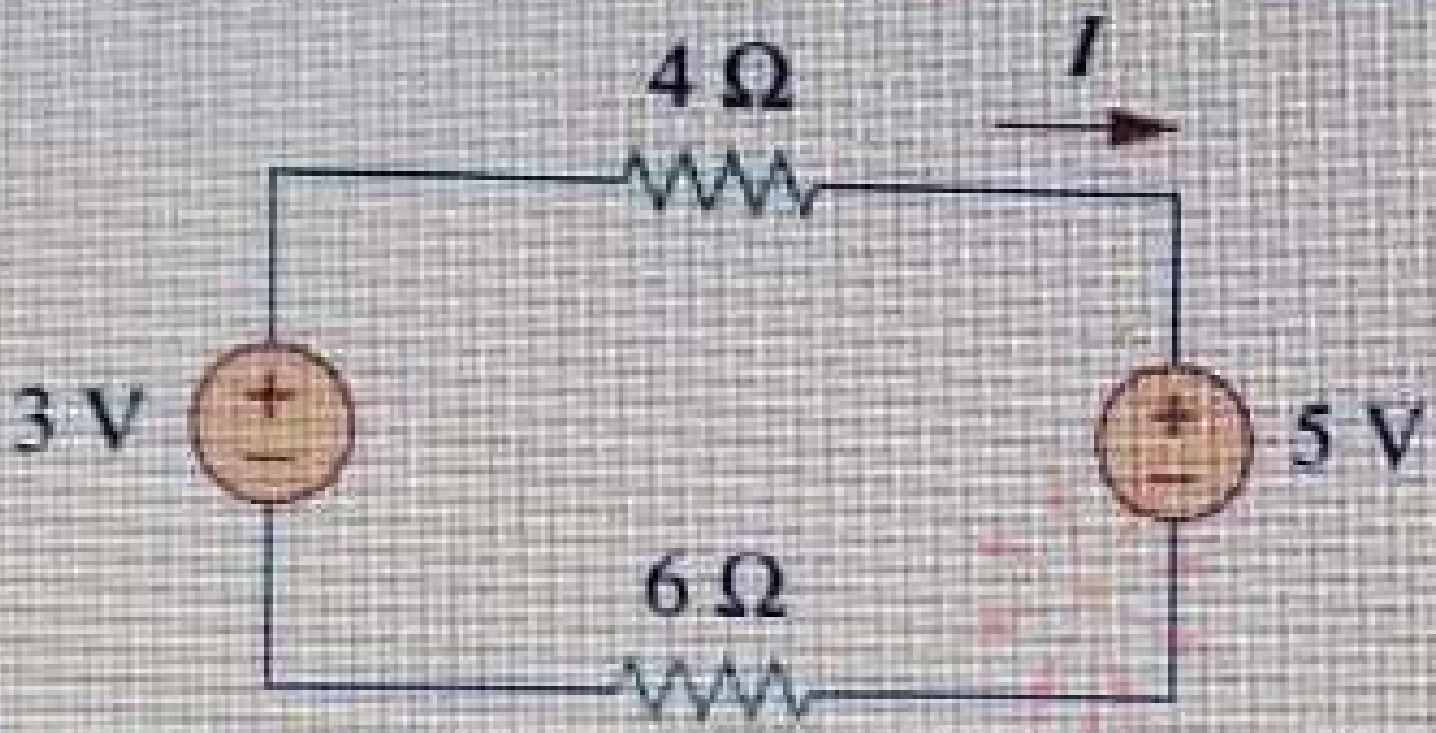
In the circuit of the Figure, current *i1* is :



Select one:

- a. 3 A
- b. -1 A
- c. 2 A
- d. 4 A

The current I in the circuit is:



Select one:

a. -0.2 A

b. 0.2 A

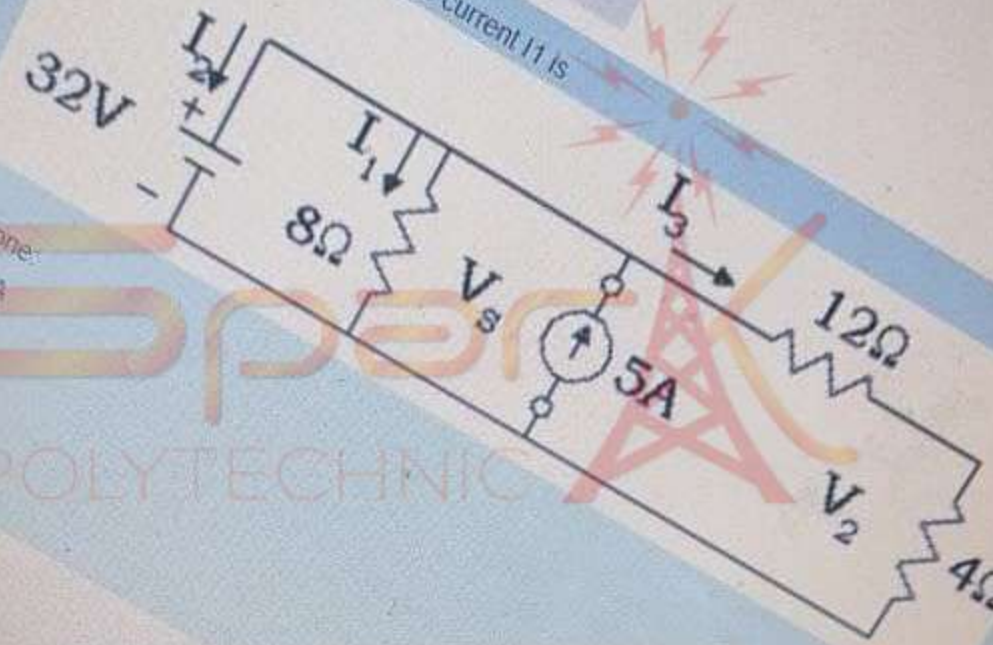
c. 0.8 A

d. -0.8 A

CUITS 1-1

1-1 ▶ General ▶ الامتحان النصفي

Refer to the figure below. the current I1 is



Select one:

- a. 4A
- b. 5A
- c. 3A
- d. 2A

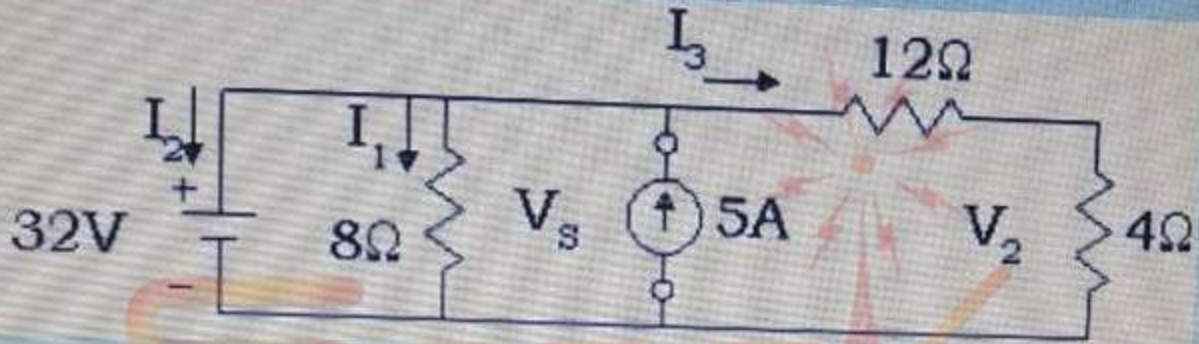
Marked out of 2.00

Flag question

POLYTECHNIC

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Refer to the figure below, the current I1 is

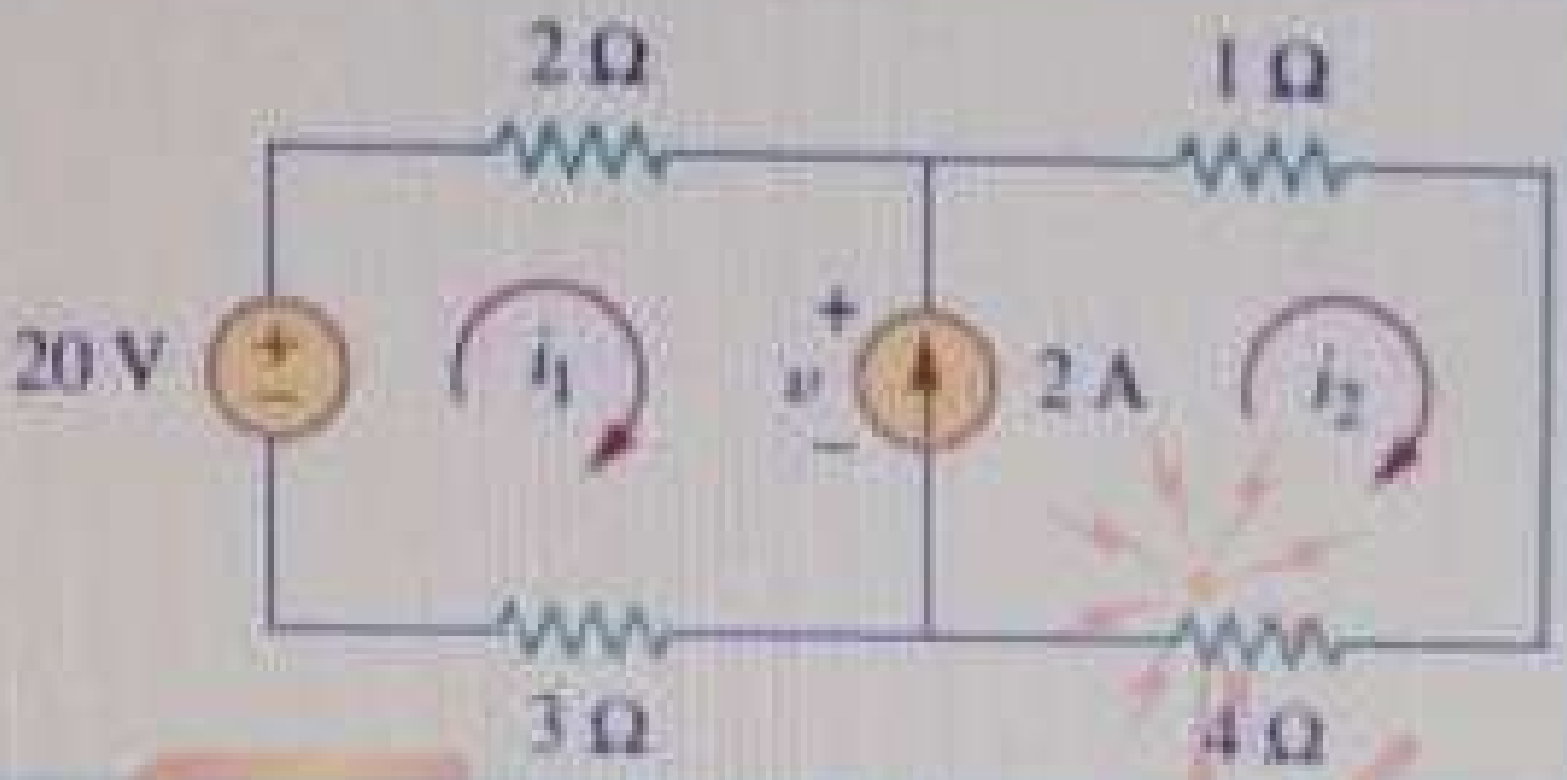


Select one:

- a. 4 A
- b. 2 A
- c. 5 A
- d. 3 A

Protected with free version of Watermarkly. Full version doesn't put this mark.

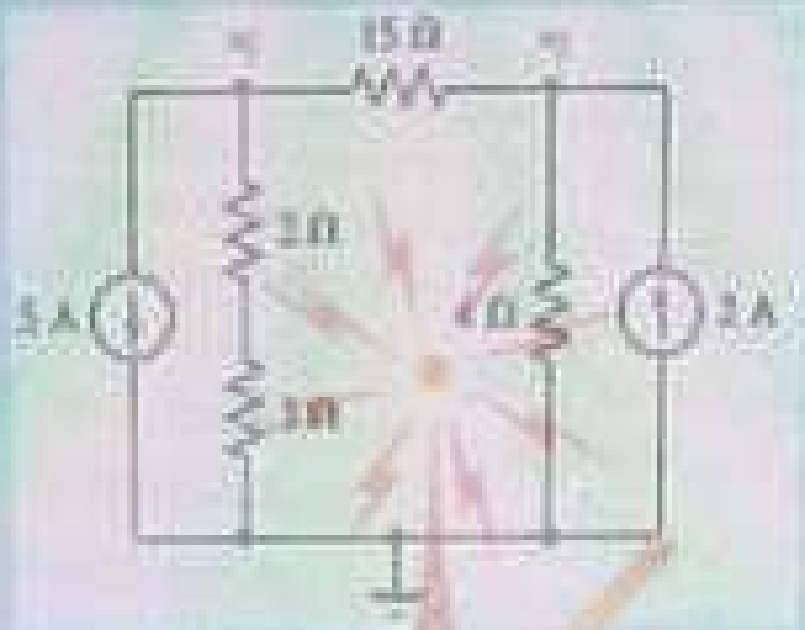
In the circuit of the Figure, current $i2$ is :



Select one:

- a. 3A
- b. 1 A
- c. 4 A
- d. 2 A

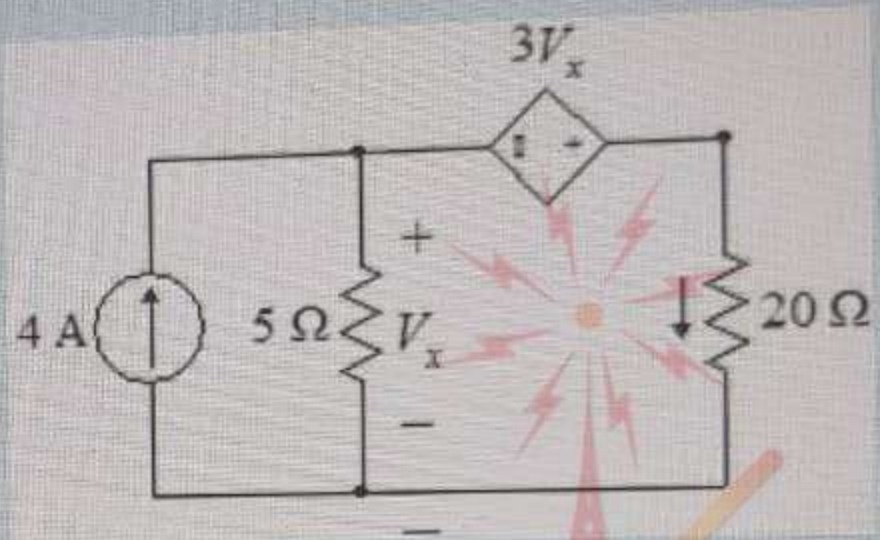
In the circuit shown, The power absorbed by 4 Ω resisler is:



Select one:

- A. 6.48W
- B. 10.48W
- C. 5.23W
- D. 1.583W

In the circuit shown, The current in 20 Ω resistor is:

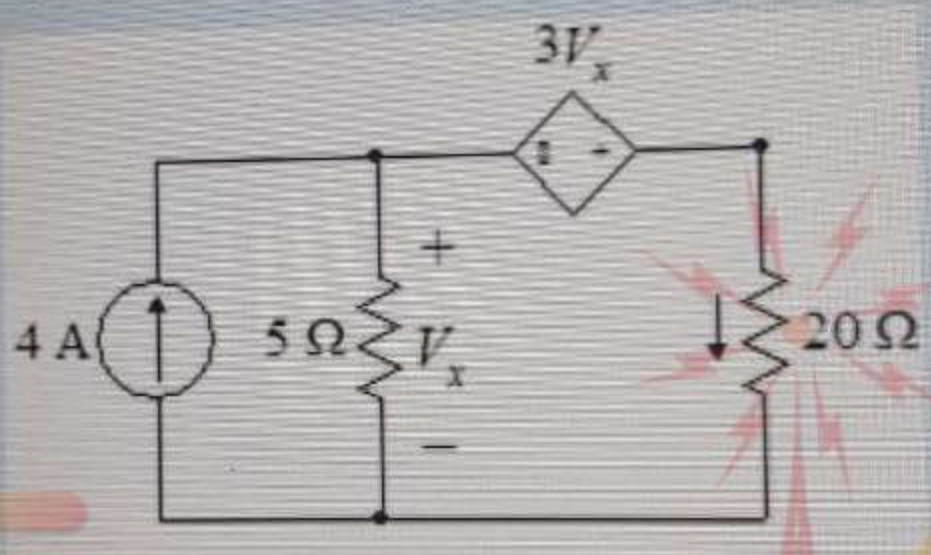


Select one:

- A. 2.8A
- B. 1A
- C. 200 mA
- D. 2A

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In the circuit shown, The value of V_x is:

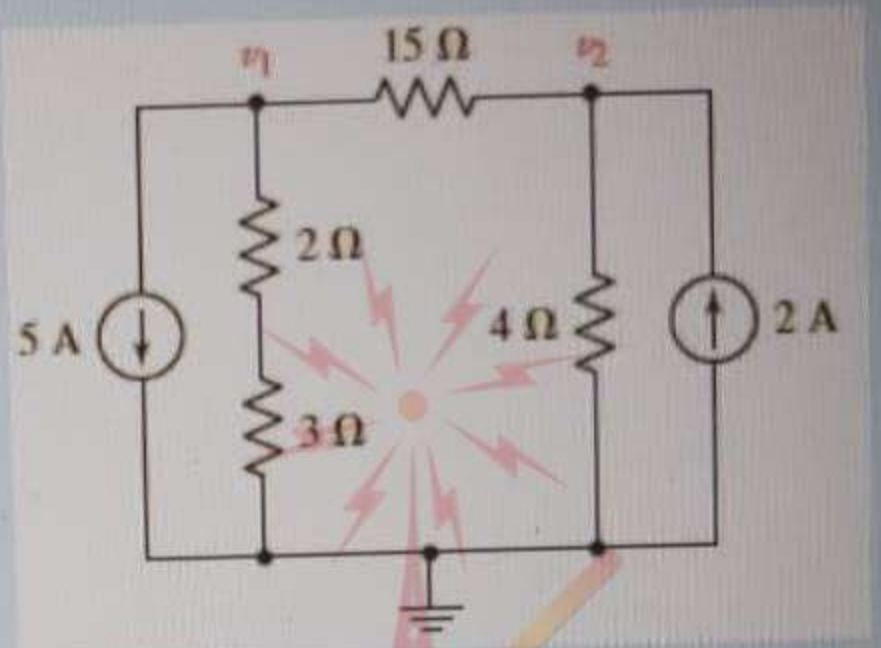


Select one:

- A. 100V
- B. 20V
- C. 5V
- D. 10V

Protected with free version of Watermarkly. Full version doesn't put this mark.

In the circuit shown, The power absorbed by 4 Ω resister is:

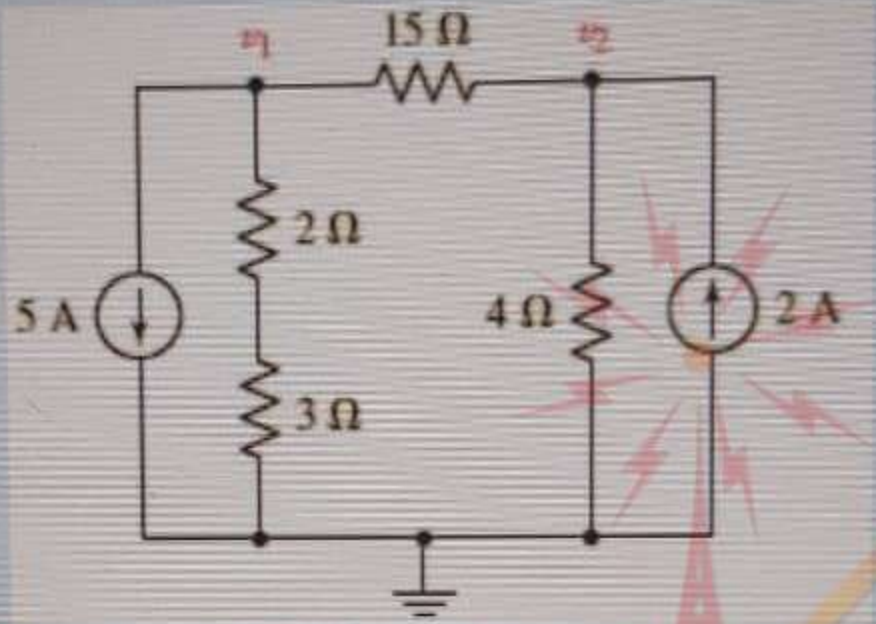


Select one:

- A. 5.23W
- B. 1.563W
- C. 10.46W
- D. 6.46W

Protected with free version of Watermarkly. Full version doesn't put this mark.

In the circuit shown, The value of V_2 is

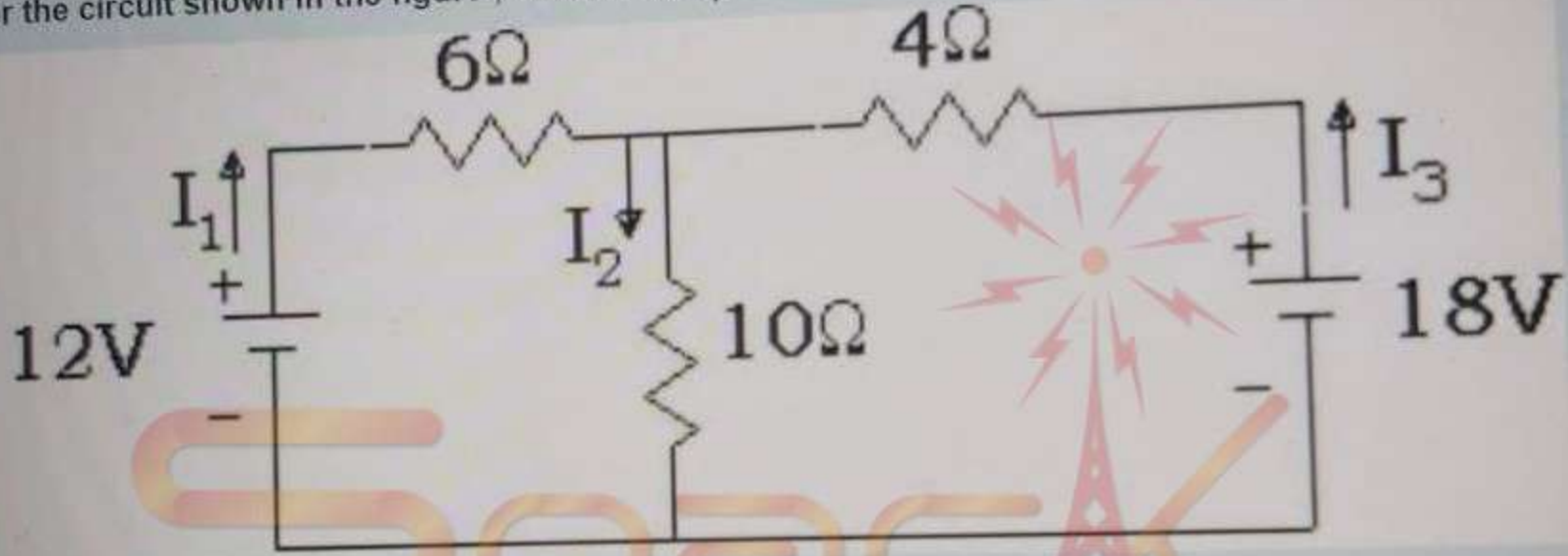


Select one:

- A. 5V
- B. -500mV
- C. 200mV
- D. 2.5V

Protected with free version of Watermarkly. Full version doesn't put this mark.

For the circuit shown in the figure , the current I_1 is:

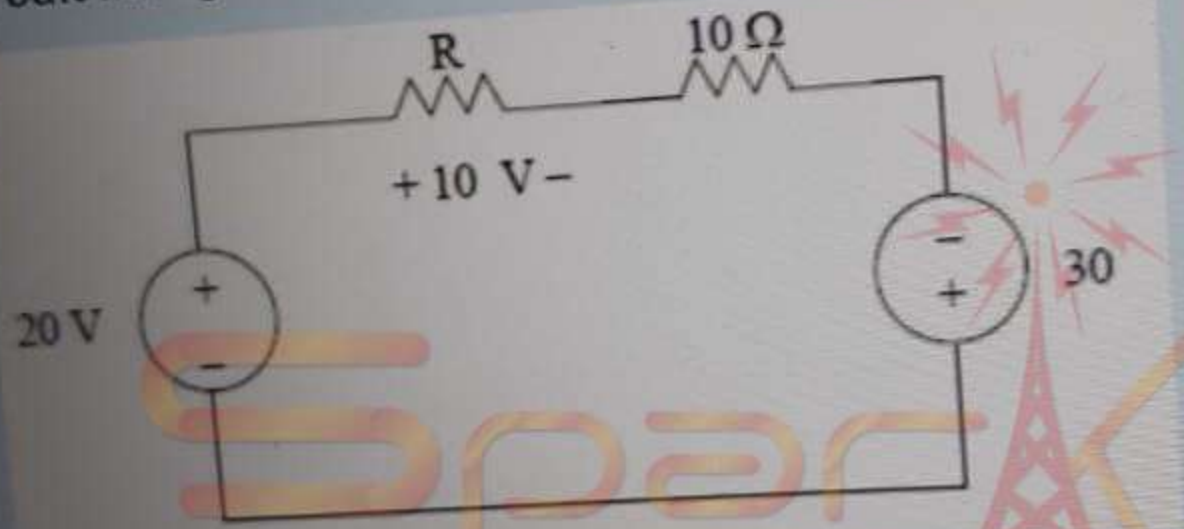


Select one:

- A. -1.62A
- B. -0.1A
- C. 1.62A
- D. 1.42A

Protected with free version of Watermarkly. Full version doesn't put this mark.

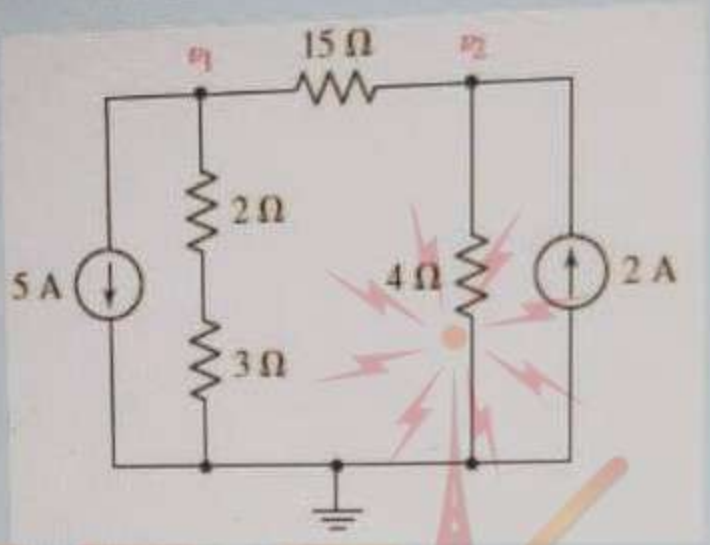
for the circuit in Figure.



swer:

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In the circuit shown, The value of the current (i) in 2 Ω resistor is:

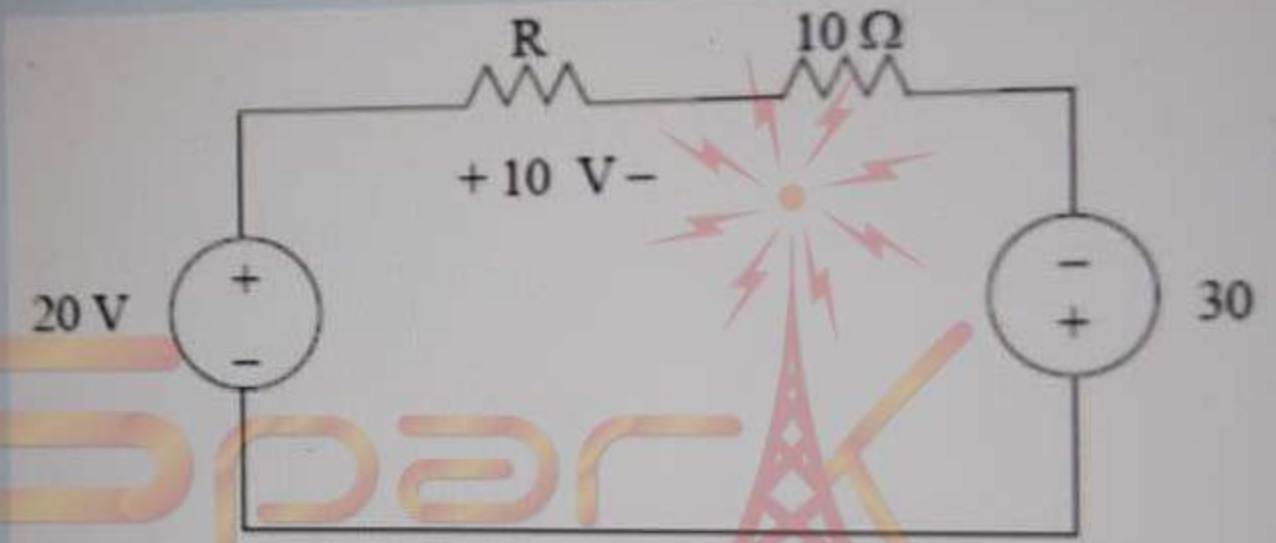


Select one:

- A. 1.48A
- B. 0.74A
- C. -0.7A
- D. 3.625A

Protected with free version of Watermarkly. Full version doesn't put this mark.

Find R for the circuit in Figure.



Answer:

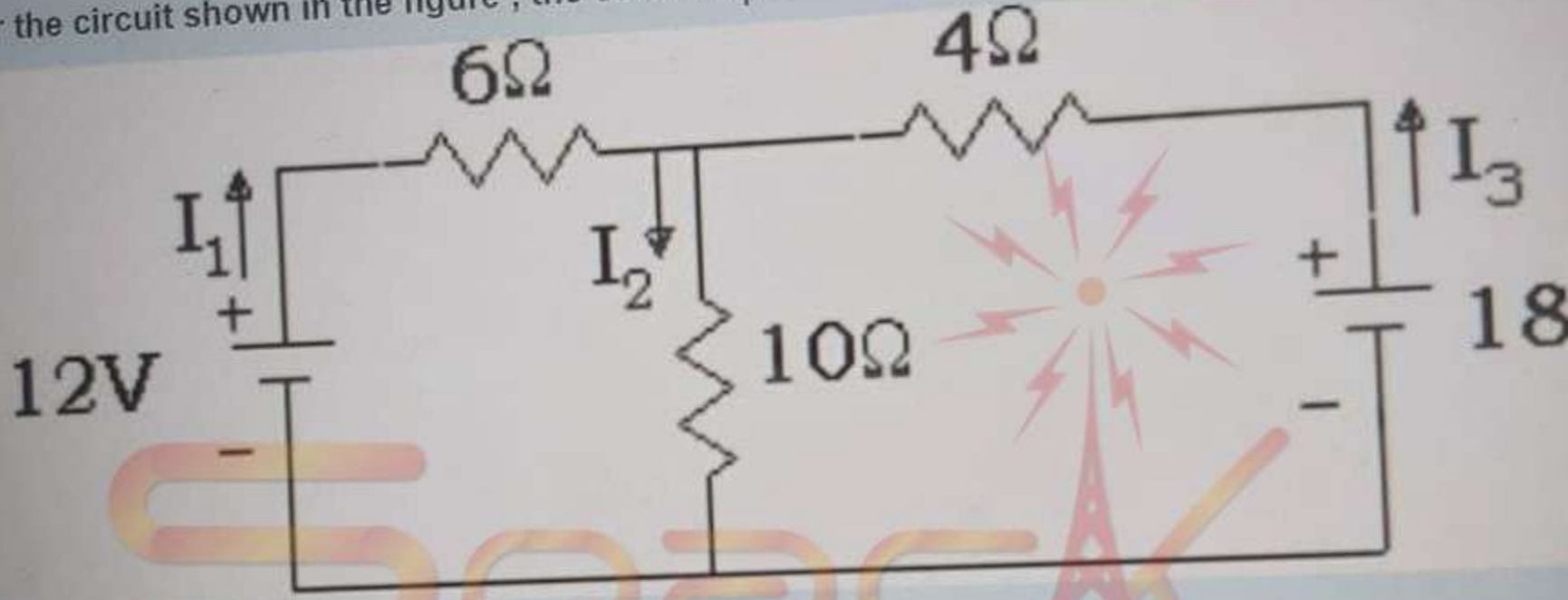
Protected with free version of Watermarkly. Full version doesn't put this mark.

A resistor absorbs an instantaneous power of 30 $\cos^2 t$ mW when connected to a voltage source v=10Cos t V
Find *R*.

Select one:

- A. 330 mΩ
- B. 33 Ω
- C. 330 Ω
- D. 3.33 kΩ

For the circuit shown in the figure , the current I_1 is:



Select one:

- A. -1.62A
- B. -0.1A
- C. 1.62A
- D. 1.42A

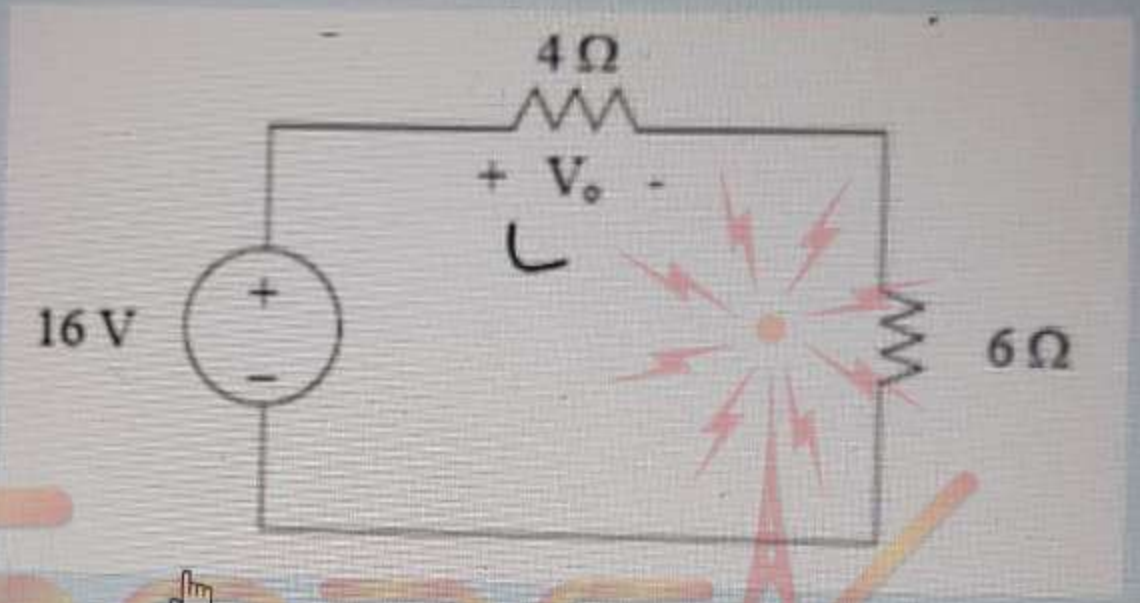
Protected with free version of Watermarkly. Full version doesn't put this mark.

A resistor absorbs an instantaneous power of 30 $\cos^2 t$ mW when connected to a voltage source v=10Cos t V. Find *R*.

Select one:

- A. 330 mΩ
- B. 33 Ω
- C. 330 Ω
- D. 3.33 kΩ

Calculate V_0 in the circuit of Figure.



Select one:

- A. 16
- B. 6.4 V
- C. 3.2
- D. 12.8

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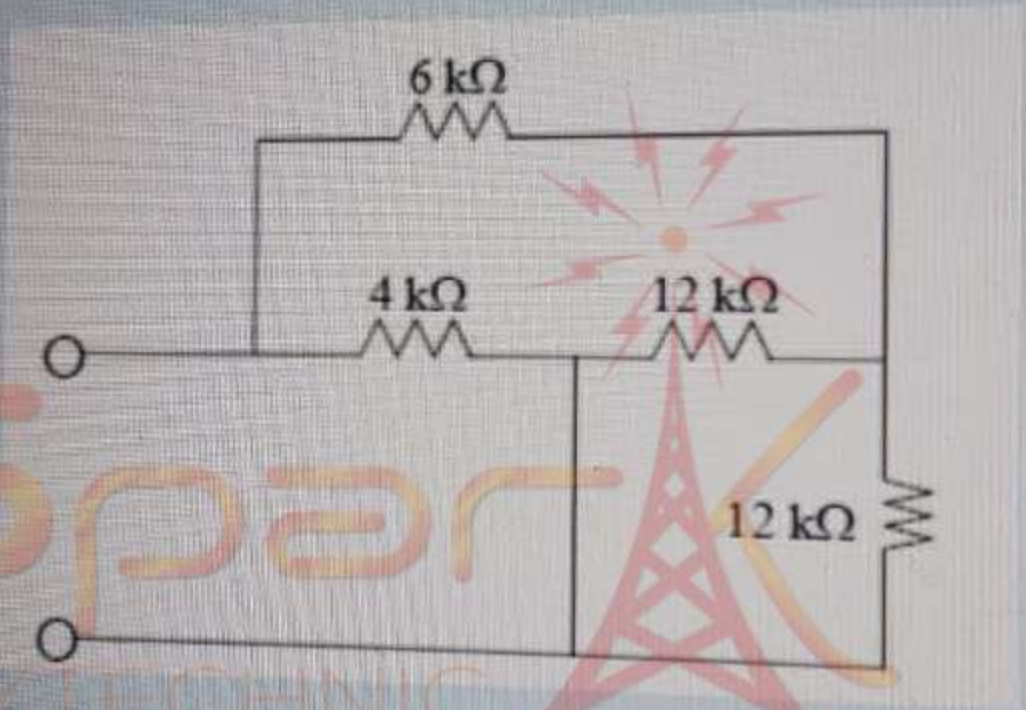
The conductance of a 10 Ω resistor in milli siemens is:

Select one:

- A. 10 mS
- B. 0.01mS
- C. 100mS
- D. 0.1mS

Protected with free version of Watermarkly. Full version doesn't put this mark.

Evaluate R_{eq} in kΩ for the circuit shown in Figure:



Answer:

Protected with free version of Watermarkly. Full version doesn't put this mark.

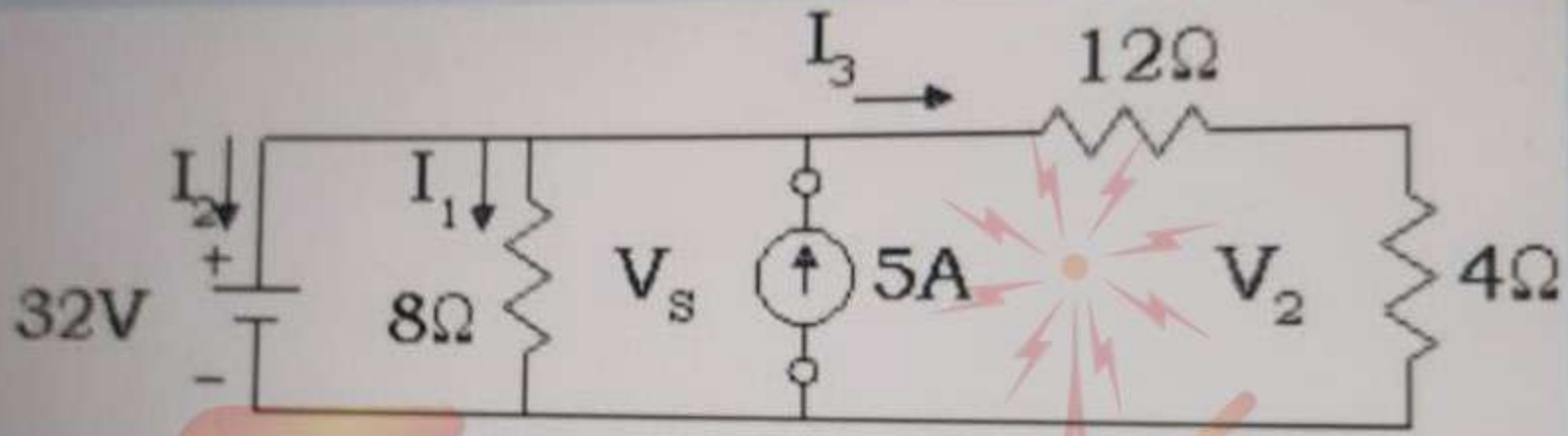
The maximum current that a 20 W, 80 kΩ resistor can safely conduct is:

Select one:

- A. 40 A
- B. 1600 A
- C. 34.6 µA
- D. 17.3 mA

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Refer to the figure shown, the current I_3 is:



Select one:

- A. 5A
- B. 3A
- C. 4A
- D. 2A

Spark POLYTECHNIC

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Refer to the figure shown, the voltage V_1 is:

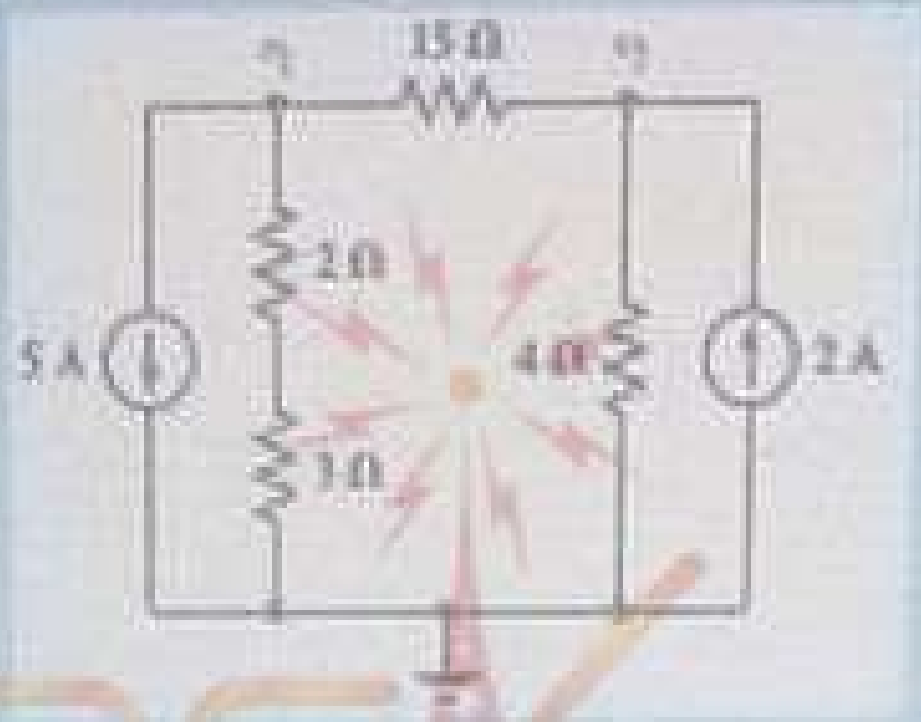


Select one:

- A. 15V
- B. 50V
- C. 60V
- D. 45V

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In the circuit shown,The value of V_2 is



Select one:

- A. 2.5V
- B. -500mV
- C. 5V
- D. 200mV

Spark Polytechnic

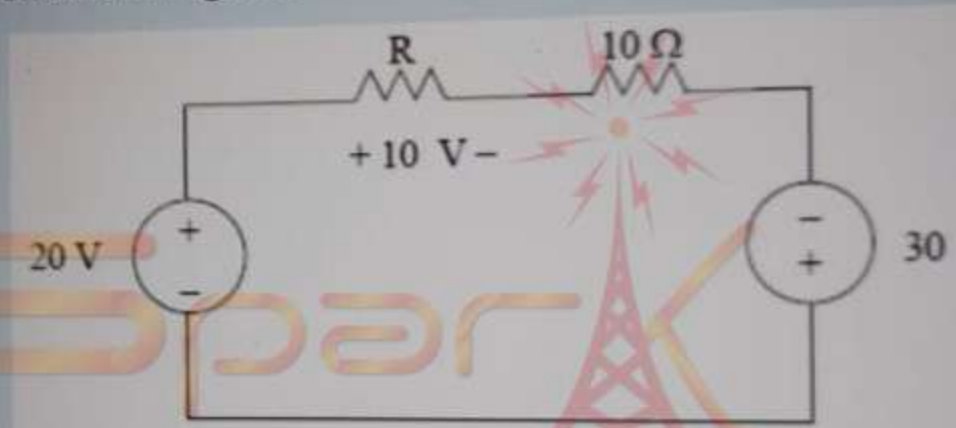
The voltage across a 5 kW load that produces 25 A is :

Select one:

- A. 200 V
- B. 500V
- C. 250V
- D. 2kV

Spark
HOLY TECHNICAL

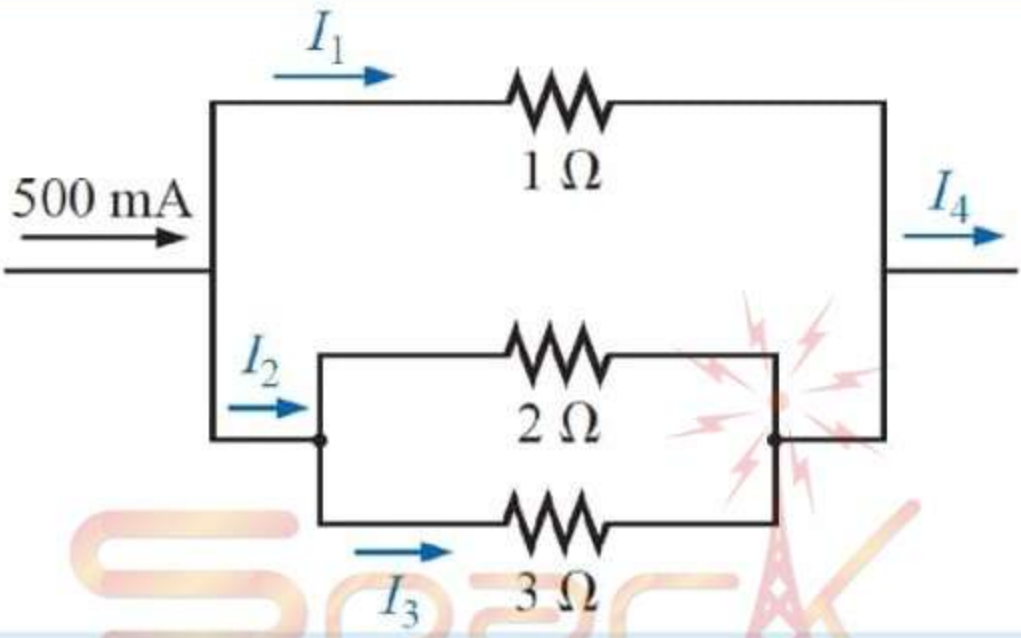
Find R for the circuit in Figure.



Answer:

Protected with free version of Watermarkly. Full version doesn't put this mark.

: In the given network, find I4



POLYTECHNIC

إختر أحد الخيارات:

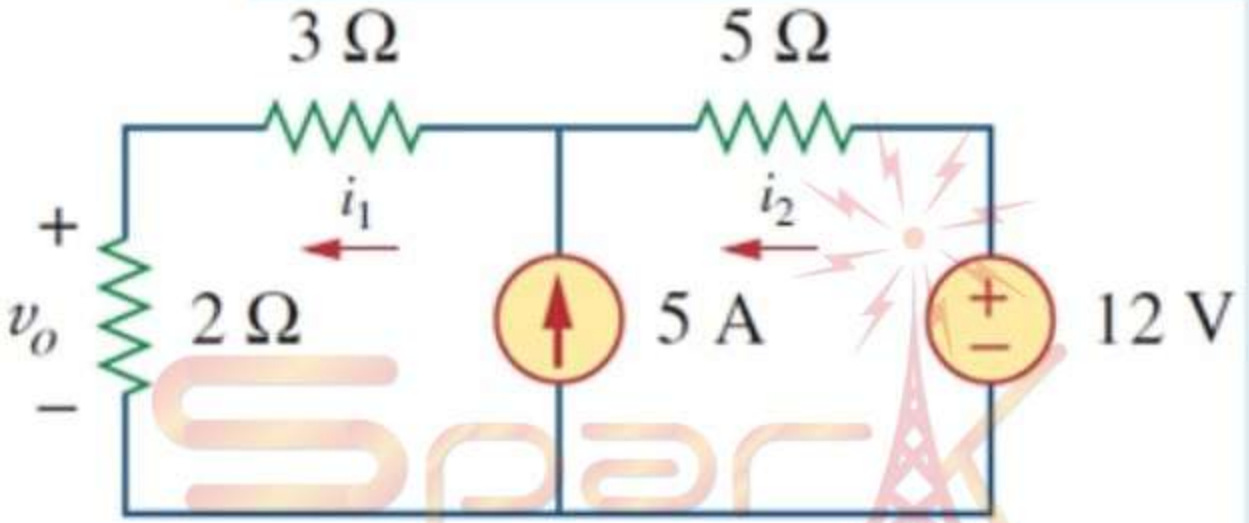
a. 227 mA

b. None of all

c. 0.5 A

d. 273 mA

Using the superposition theorem, find the power absorbed by 5 Ω resistor in the :circuit of the figure



Spark POLYTECHNIC

إختر أحد الخيارات:

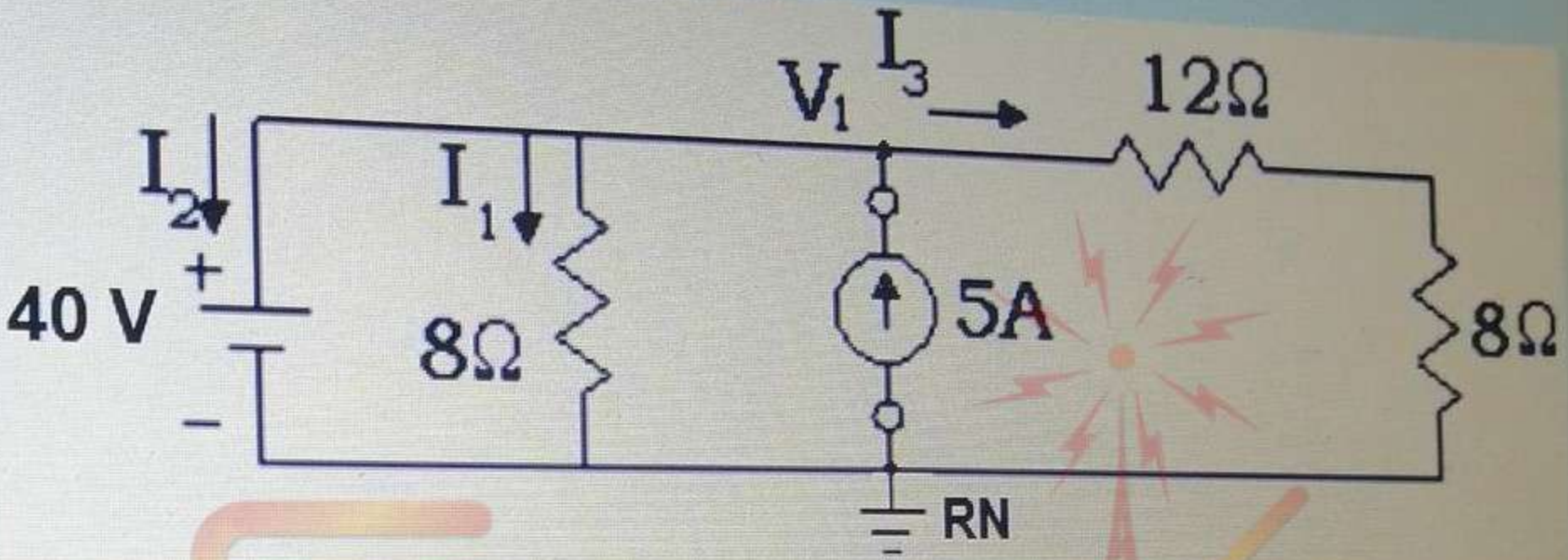
a. 456 mW

b. 46 W

c. 8.45 W

d. 25 mW

Refer to the figure, the voltage V_1 is:



Select one:

- a. 40 V
- b. 50 V
- c. 20 V
- d. 100 V

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Circuit elements that absorb power have.

Select one:

- a. A positive value of P.
- b. A negative value of P.
- c. None of all
- d. Zero value of P.

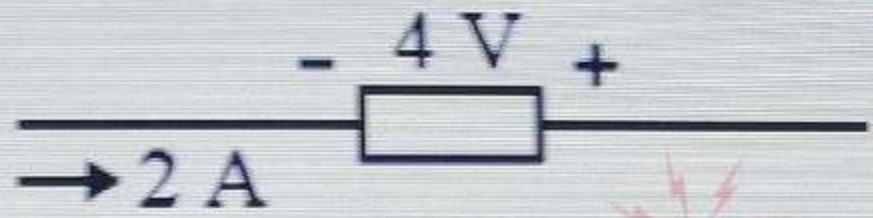
Protected with free version of Watermarkly. Full version doesn't put this mark.

The superposition principle applies to power calculation.

Select one:

- True
- False

Find the power generated by the device.



Select one:

- a. 8 W
- b. - 8 W
- c. None of all
- d. - 4 W

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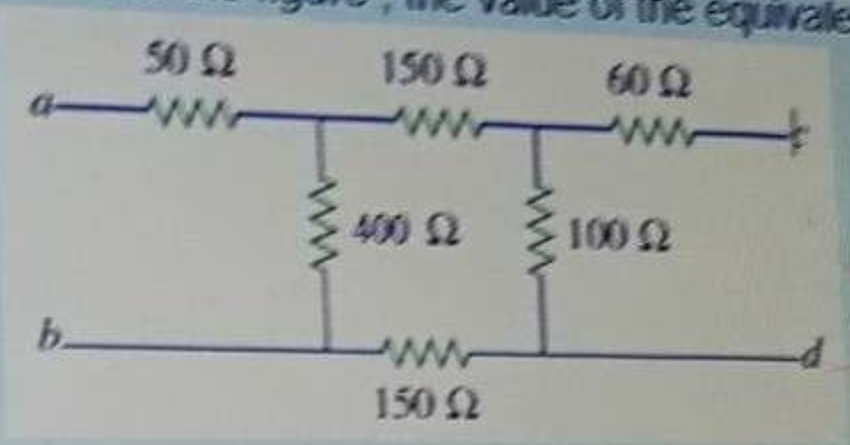
A Norton equivalent circuit with I_N=3A and R_N = 10Ω is equivalent to Thevenin circuits with:

Select one:

- a. V_{th} = 15 V, R_{th} = 5 Ω
- b. V_{th} =30 V, R_{th} = 10 Ω
- c. V_{th} = 3 V, R_{th} = 10 Ω
- d. V_{th} = 3 V, R_{th} = 4 Ω

Protected with free version of Watermarkly. Full version doesn't put this mark.

Refer to the figure ; the value of the equivalent resistance at terminals ab is:

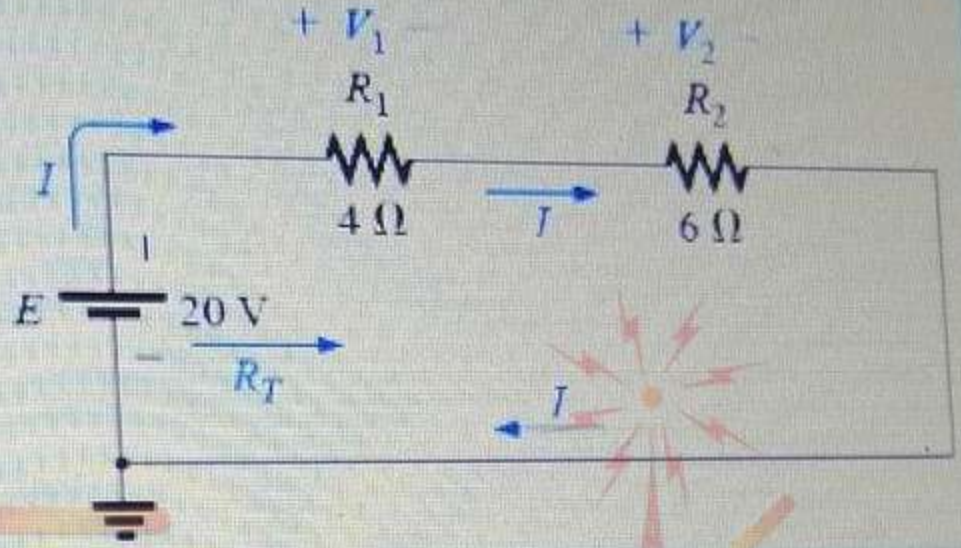


Select one:

- a. 275 Ω
- b. 310 Ω
- c. 250 Ω
- d. 360 Ω

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For the circuit shown, find V_2:

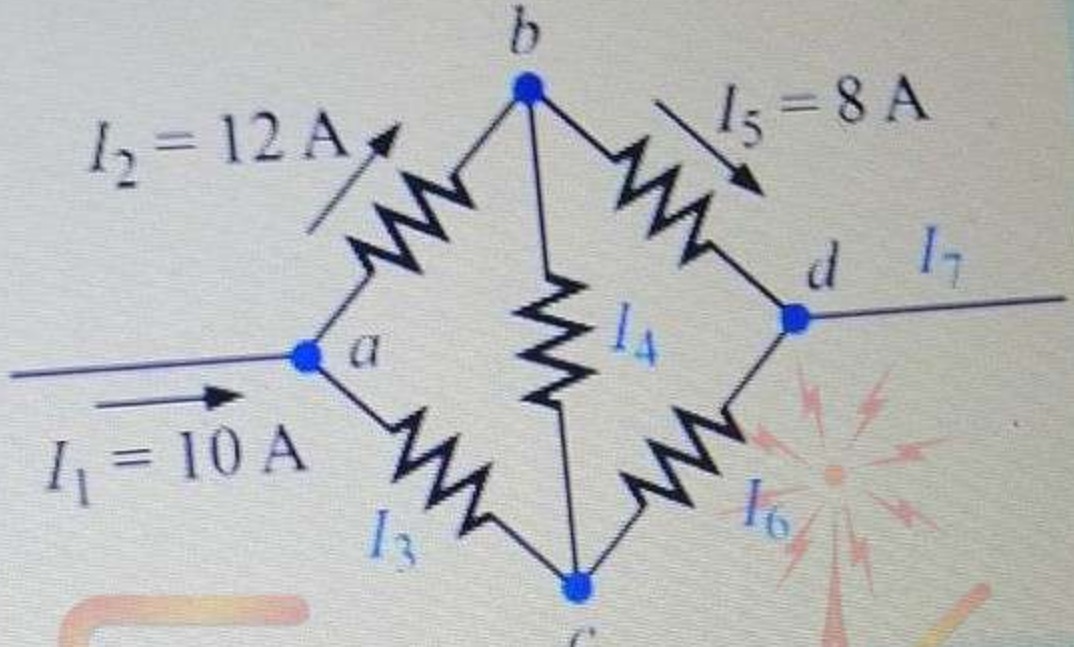


Select one:

- a. None of all
- b. 12 V
- c. 3.33 V
- d. 8 V

Spark

Find the magnitude of the current I_3 for the given network:

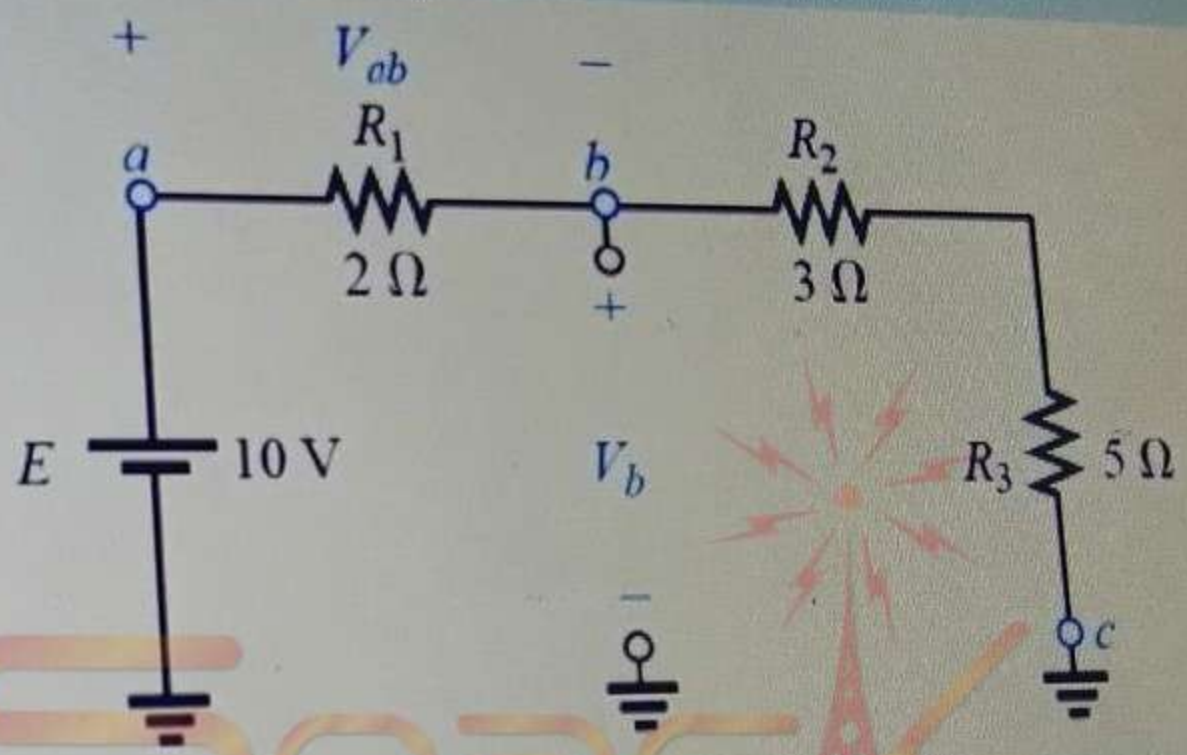


Select one:

- a. 4 A
- b. 10 A
- c. 2 A
- d. None of all

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For the given network, find V_b:

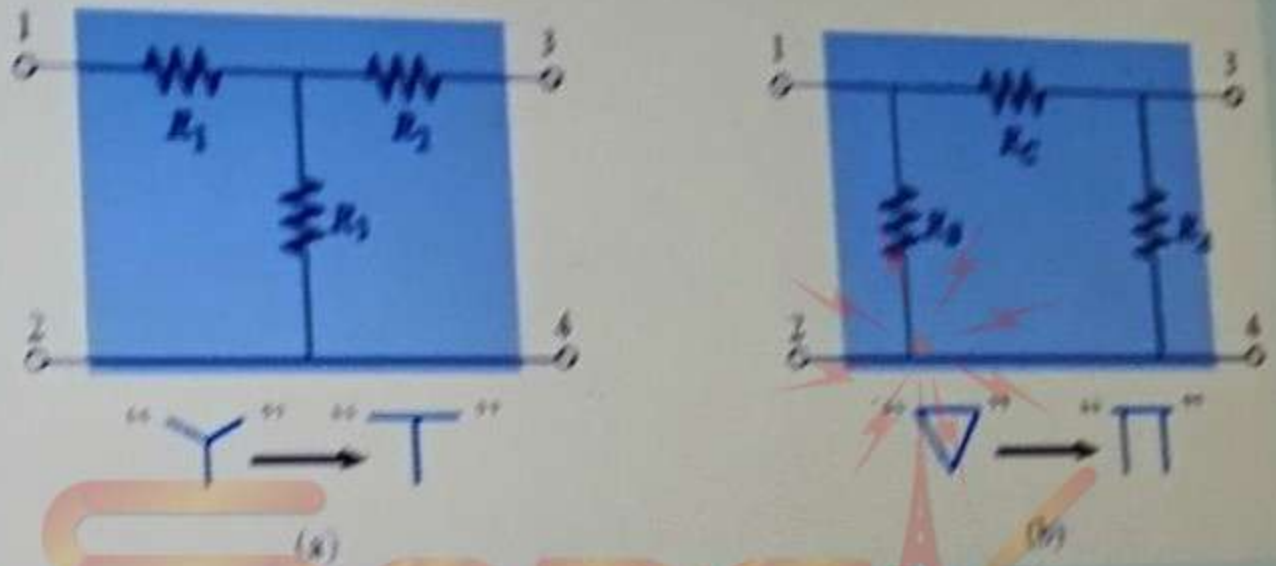


Select one:

- a. - 2 V
- b. 2 V
- c. None of all
- d. 8 V

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If the networks a and b are equivalent, find R_1 if $R_a = 10\,\Omega$; $R_b = 20\,\Omega$; $R_c = 5\,\Omega$.



Select one:

- a. 2.86 Ω
- b. None of all
- c. 1.43 Ω
- d. 5.71 Ω

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EXAM Electrical Circuits 1 (01-12-2020) 16:00 -16:30

A load is connected to a network. At the terminals to which the load is connected, $R_{th} = 10\Omega$ and $V_{th}=20V$. The maximum possible power supplied to the load is:

Select one:

- a. 80 W
- b. 160 W
- c. 1 W
- d. 10 W

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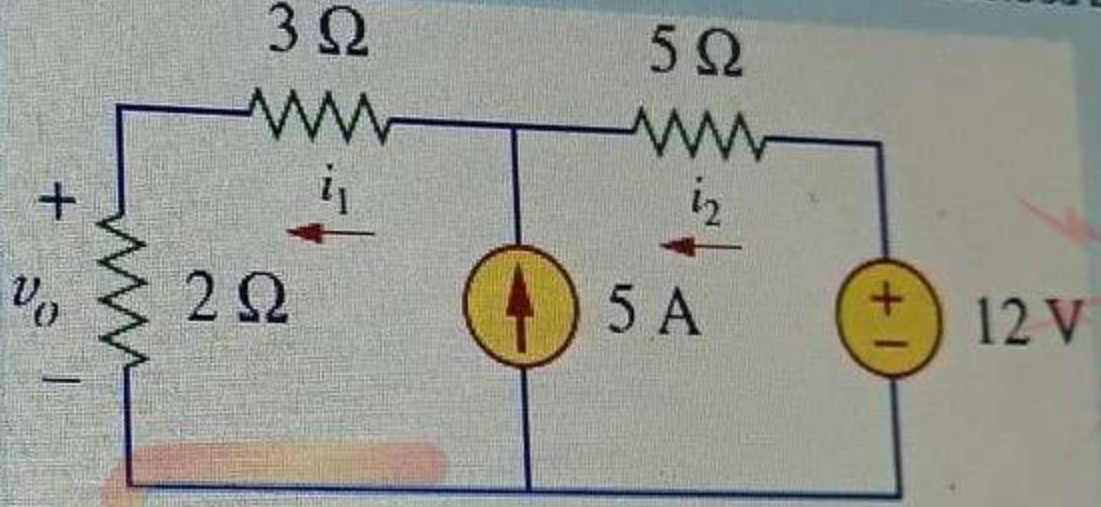
How much energy does a 10 – kW load consume in two days:

Select one:

- a. 480 W.h
- b. None of all
- c. 480 J
- d. 480 kW.h

Protected with free version of Watermarkly. Full version doesn't put this mark.

Using the superposition theorem, find the power absorbed by 5 Ω resistor in the circuit of the figure:

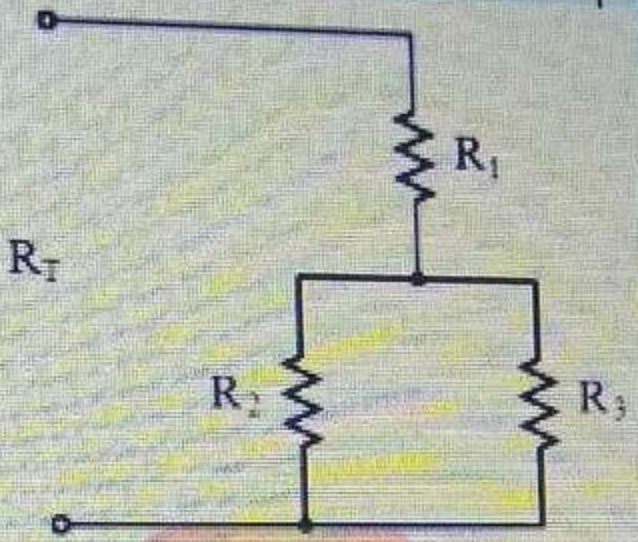


Select one:

- a. 25 mW
- b. 8.45 W
- c. 456 mW
- d. 46 W

Protected with free version of Watermarkly. Full version doesn't put this mark.

For the given network find R_T if $R_1 = 5\ \Omega$; $R_2 = 4\ \Omega$; and $R_3 = 10\ \Omega$.



Select one:

- a. 5.33 Ω
- b. 6.66 Ω
- c. 11.43 Ω
- d. 7.86 Ω

Protected with free version of Watermarkly. Full version doesn't put this mark.

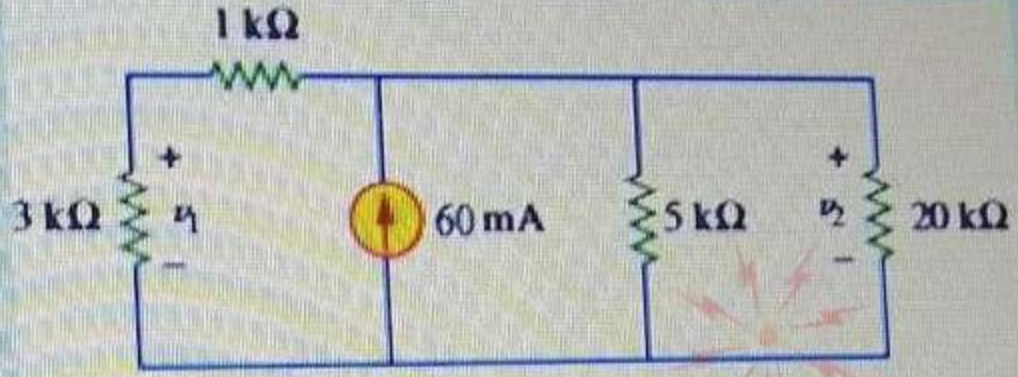
The current through a branch in a linear network is 16 A when the input source voltage is 100 V. If the voltage is reduced to 50 V and the polarity is reversed, the current through the branch is:

Select one:

- a. - 4 A
- b. 12 A
- c. 6.25 A
- d. - 8 A

Spark
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Refer to the figure , the voltage V_1 is:

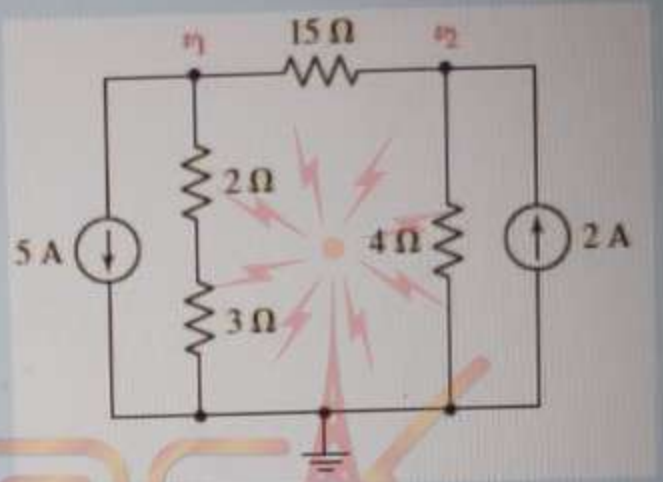


Select one:

- a. 100 V
- b. 120 V
- c. 30 V
- d. 90 V

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In the circuit shown, The power absorbed by 4 Ω resister is:

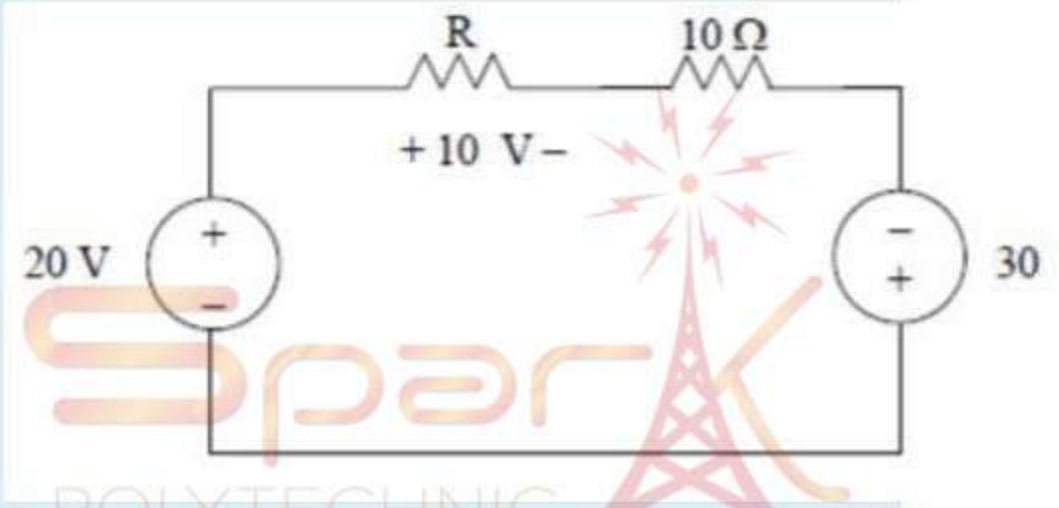


Select one:

- A. 5.23W
- B. 1.563W
- C. 10.46W
- D. 6.46W

Protected with free version of Watermarkly. Full version doesn't put this mark.

Find R for the circuit in Figure.



Spark POLYTECHNIC

Answer:

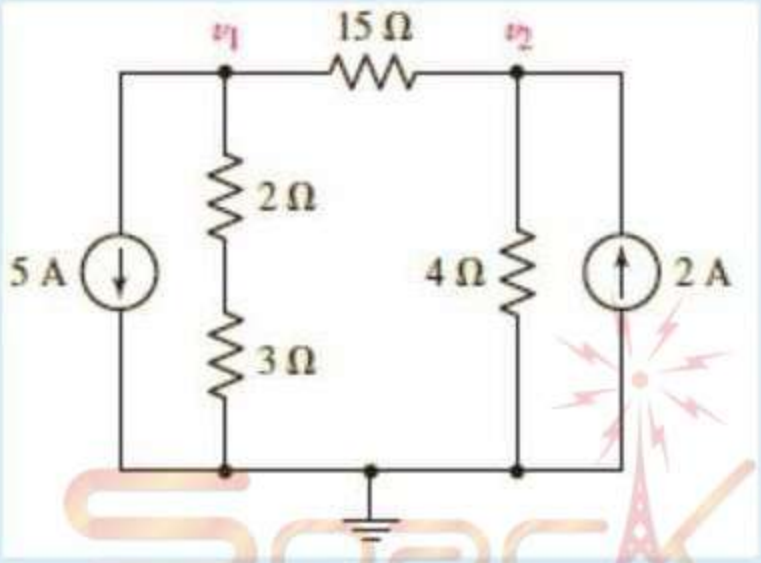
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The prefix Tera stands for

Select one:

- A. 10^{12}
- B. 10^{-15}
- C. 10^{-9}
- D. 10^{-9}

In the circuit shown, The value of the current (i) in 2 Ω resistor is:

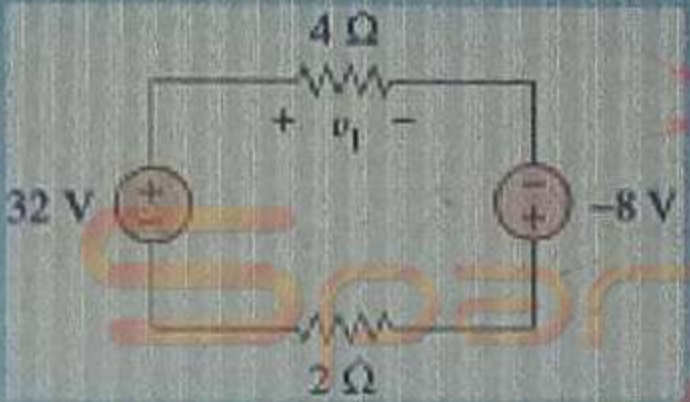


Spark POLYTECHNIC

Select one:

- A. -0.7A
- B. 0.74A
- C. 3.625A
- D. 1.48A

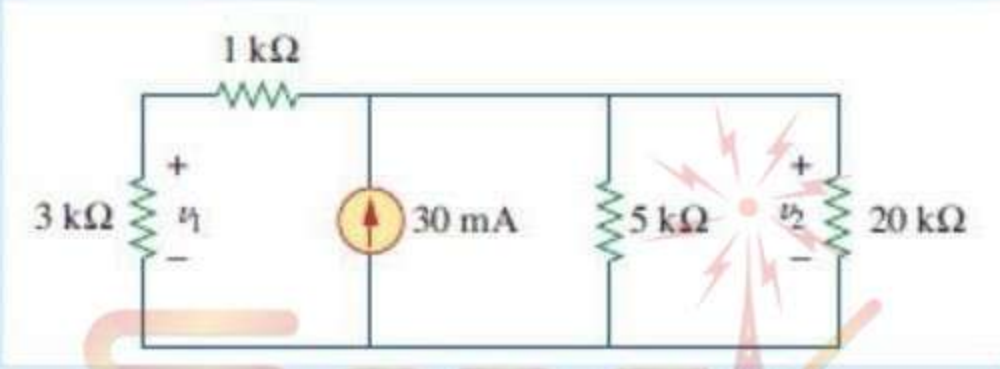
For the circuit in figure the voltage v_1 is:



Answer:

Protected with free version of Watermarkly. Full version doesn't put this mark.

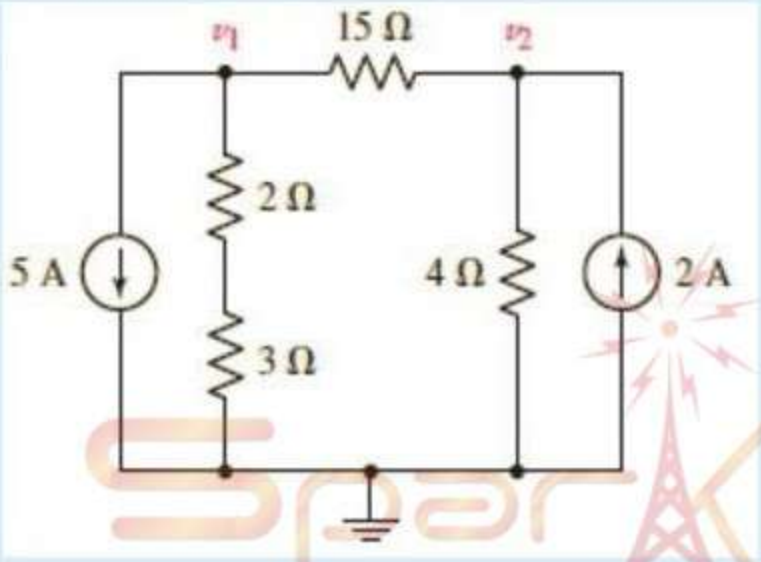
Refer to the figure shown, the voltage V_2 is:



Select one:

- A. 15V
- B. 75V
- C. 60V
- D. 50V

In the circuit shown, The value of V_1 is:



Select one:

- A. - 18.125V
- B. 36.5V
- C. -4.8V
- D. 365 mV

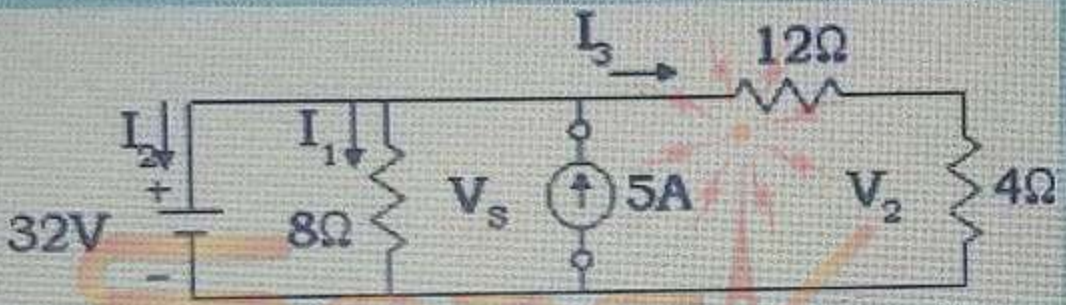
The conductance of a 10 kΩ resistor in milli siemens is:

Select one:

- A. 100mS
- B. 0.1mS
- C. 10 mS
- D. 0.01mS

Spark POLYTECHNIC

Refer to the figure shown, the current I_1 is :



Select one:

- A. 4A
- B. 2A
- C. 5A
- D. 3A

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The current through a branch in a linear network is 4A when the input source voltage is 10 V. If the voltage is reduced to 1 V and the polarity is reversed, the current through the branch is:

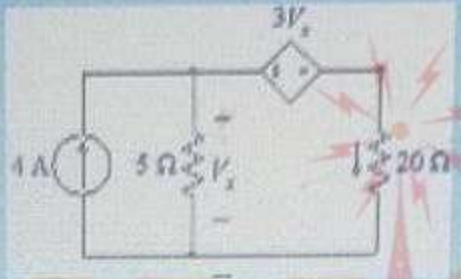
Select one:

- A. -0.4 A
- B. 2 A
- C. -2 A
- D. 0.2 A

Spark POLYTECHNIC

Protected with free version of Watermarkly. Full version doesn't put this mark.

In the circuit shown, The current in 20 Ω resistor is:

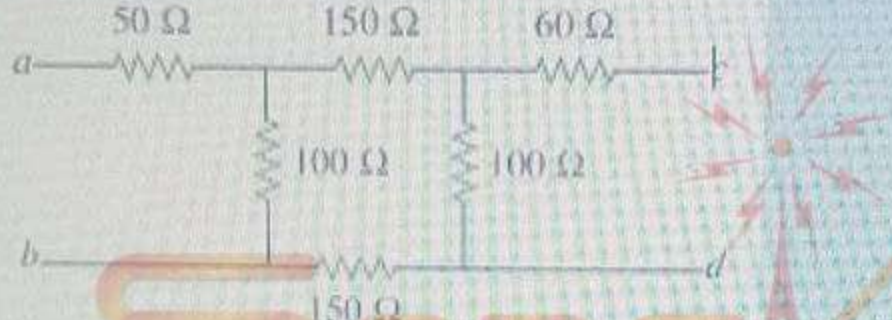


Select one:

- A. 200 mA
- B. 2.8A
- C. 1A
- D. 2A

Spark POLYTECHNIC
Protected with free version of Watermarkly. Full version doesn't put this mark.

Refer to figure ; the value of the equivalent resistance at terminals ab is:

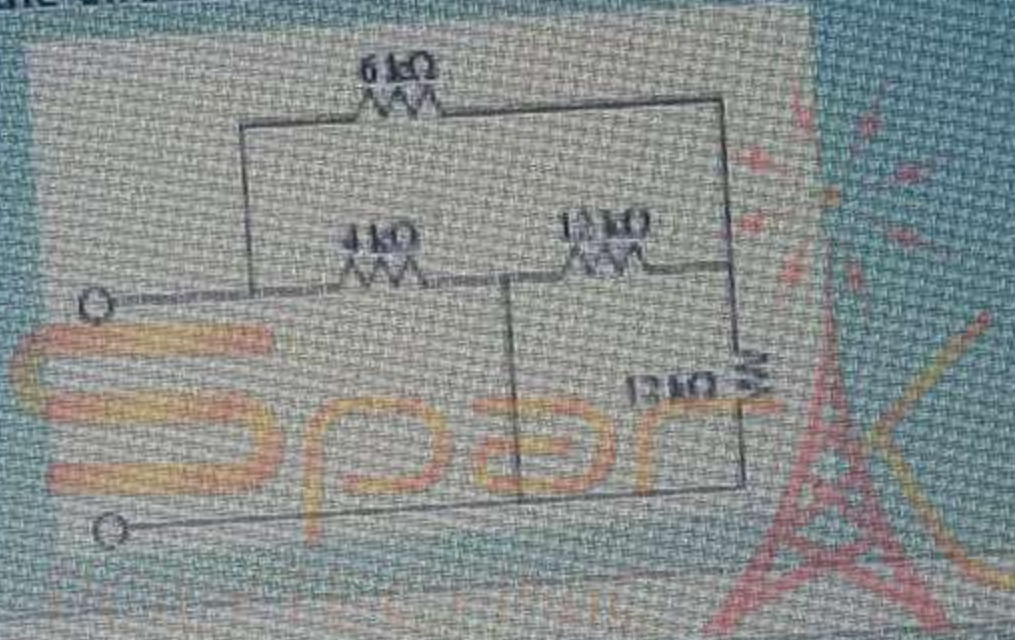


Select one:

- A. 160 Ω
- B. 260 Ω
- C. 175Ω
- D. 130 Ω

Protected with free version of Watermarkly. Full version doesn't put this mark.

Evaluate R_{eq} in kΩ for the circuit shown in Figure:



Answer:

Protected with free version of Watermarkly. Full version doesn't put this mark.

The maximum current that a 10W, 80 kΩ resistor can safely conduct is:

Select one:

- A. 40 kA
- B. 10 mA
- C. 160 A
- D. 10 μA

The conductance of a 10 kΩ resistor in milli siemens is:

Select one:

- A. 100mS
- B. 0.1mS
- C. 0.01mS
- D. 10 mS

Protected with free version of Watermarkly. Full version doesn't put this mark.

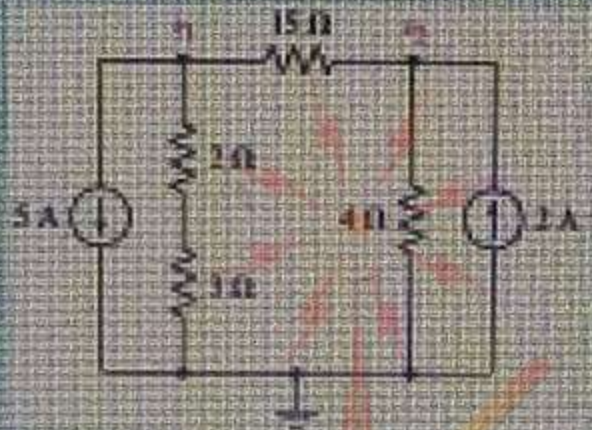
The voltage across a 50 kW load that produces 25 A is :

Select one:

- A. 200 V
- B. 250V
- C. 2kV
- D. 500V



In the circuit shown, The value of the current (i) in 2 Ω resistor is:

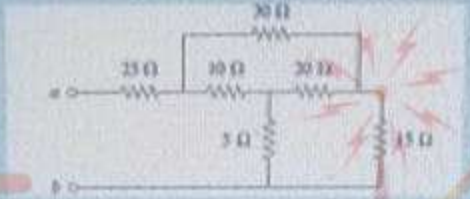


Select one:

- A. -0.7A
- B. 1.48A
- C. 0.74A
- D. 3.625A

Protected with free version of Watermarkly. Full version doesn't put this mark.

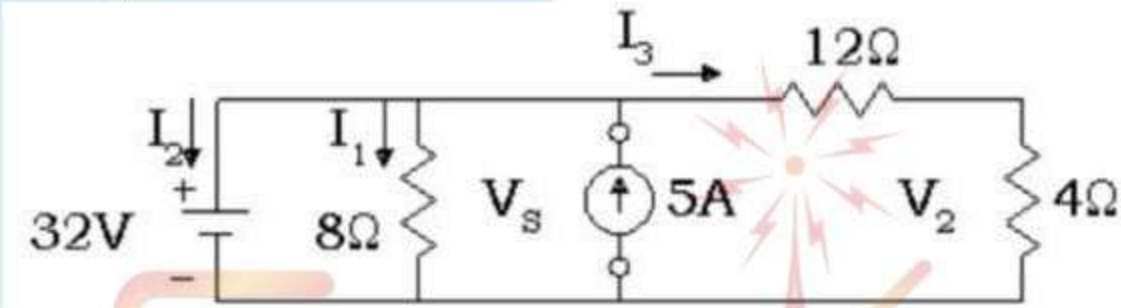
Refer to figure shown; the value of the equivalent resistance between a-b is:



Select one:

- A. 62.5 Ω
- B. 45.5 Ω
- C. 36.25 Ω
- D. 75 Ω

Refer to the figure shown, the current I_3 is:

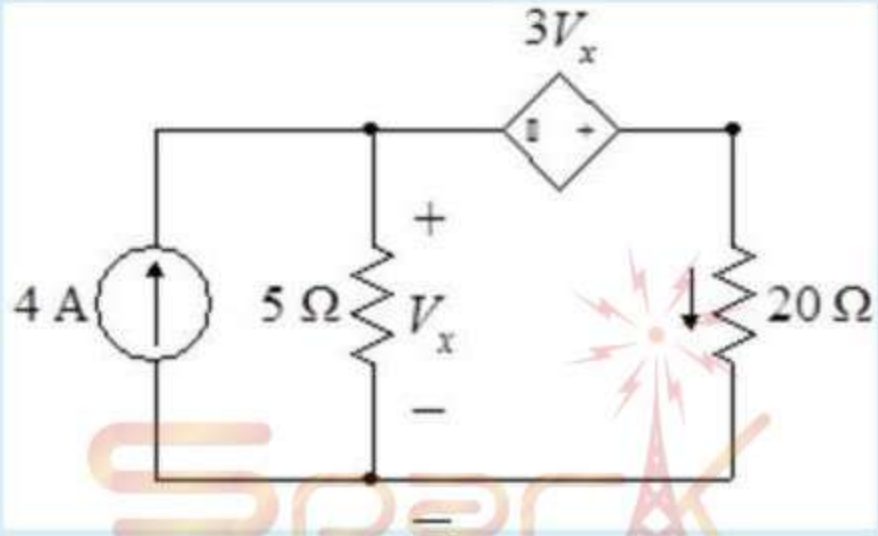


Select one:

- A. 4A
- B. 5A
- C. 2A
- D. 3A

Spark POLYTECHNIC

In the circuit shown, The delivered power by the 4 A current source is:



Select one:

- A. 40 W
- B. 16 W
- C. 20 W
- D. 6 W

A circuit element draws 15 A when connected to a 240-V line. How long does it take to consume 180 kJ?

Select one:

- A. 5 ms
- B. 50 s
- C. 2.67s
- D. 0.5 ms

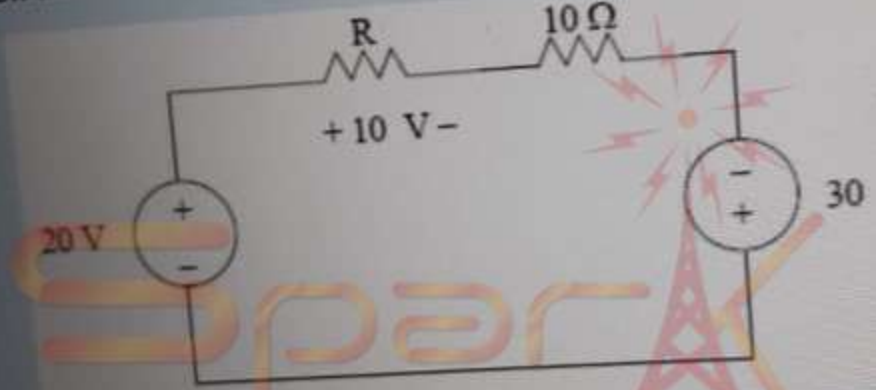
Internal resistance of ideal voltage source is:

Select one:

- A. finite
- B. zero
- C. None of the answers
- D. Infinite

Spark POLYTECHNIC

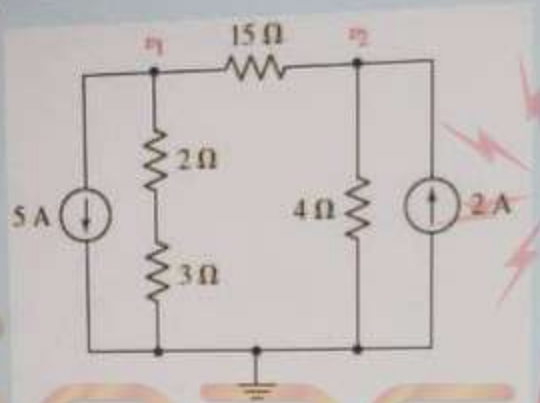
for the circuit in Figure.



swer:

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In the circuit shown, The value of the current (i) in 2 Ω resistor is:



Select one:

- A. 1.48A
- B. 0.74A
- C. -0.7A
- D. 3.625A

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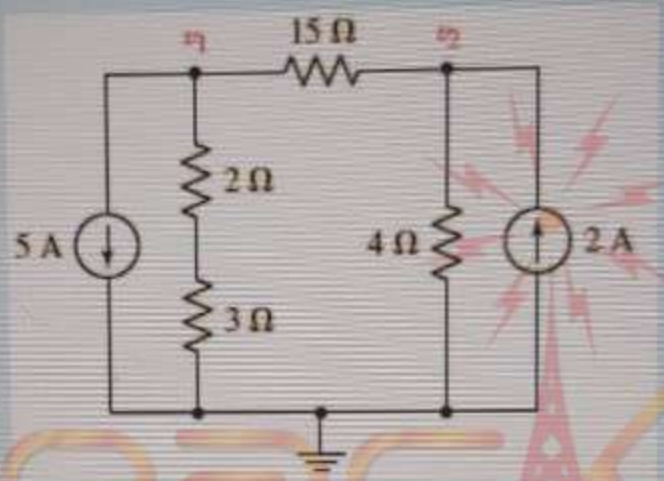
The maximum current that a 20 W, 80 kΩ resistor can safely conduct is:

Select one:

- A. 40 A
- B. 1600 A
- C. 34.6 μA
- D. 17.3 mA

Spark POLYTECHNIC

In the circuit shown, The value of V_2 is

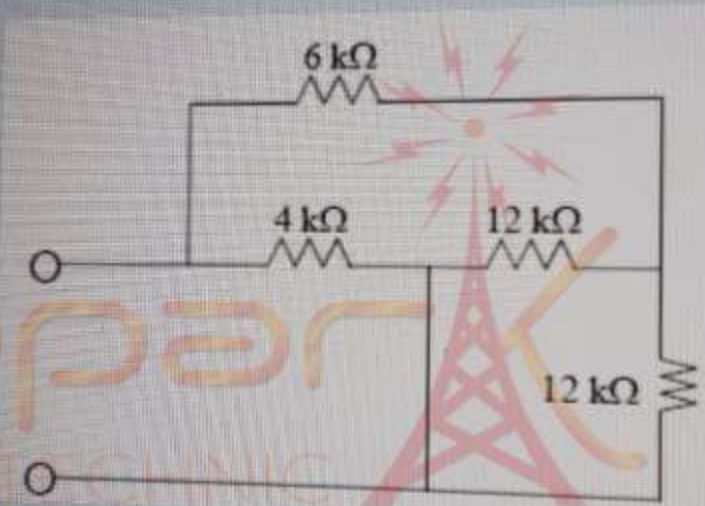


Select one:

- A. 5V
- B. -500mV
- C. 200mV
- D. 2.5V

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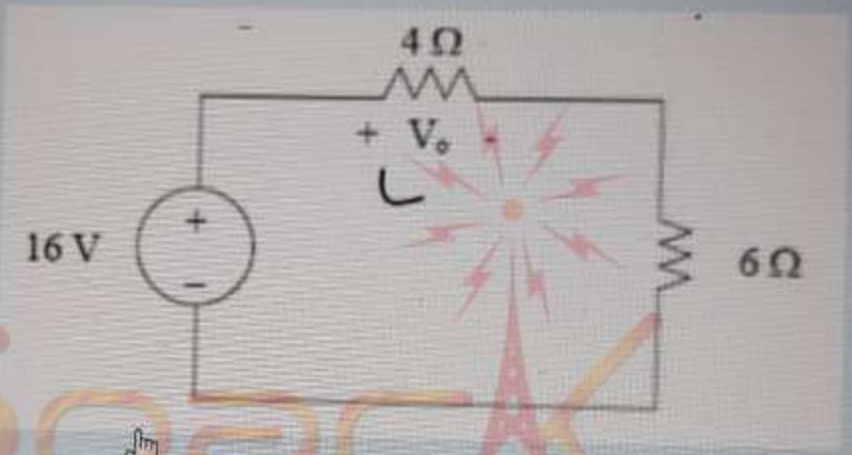
Evaluate R_{eq} in kΩ for the circuit shown in Figure:



Answer:

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Calculate V_0 in the circuit of Figure.



Select one:

- A. 16
- B. 6.4 V
- C. 3.2
- D. 12.8

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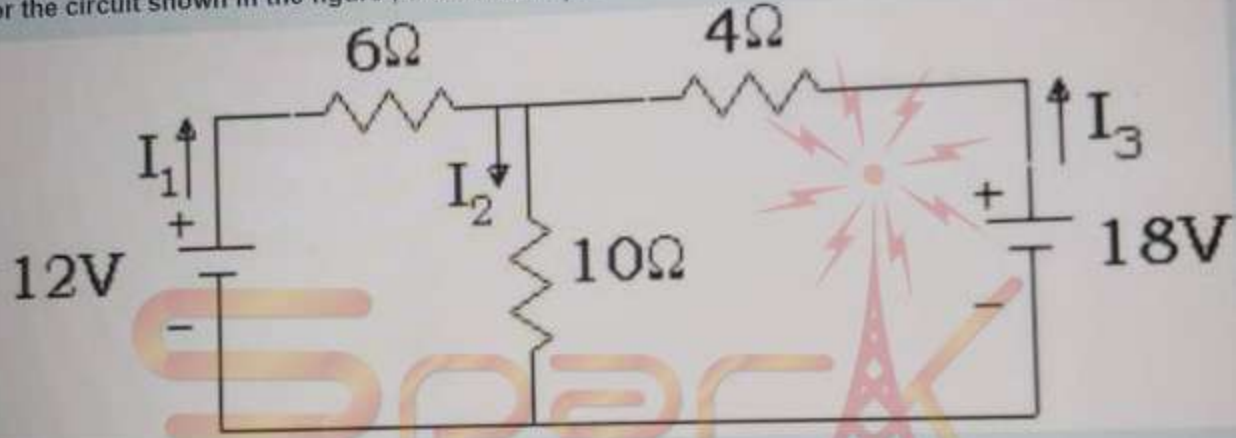
A resistor absorbs an instantaneous power of 30 $\cos^2 t$ mW when connected to a voltage source v=10Cos t V
Find *R*.

Select one:

- A. 330 mΩ
- B. 33 Ω
- C. 330 Ω
- D. 3.33 kΩ

Spark POLYTECHNIC

For the circuit shown in the figure , the current I_1 is:

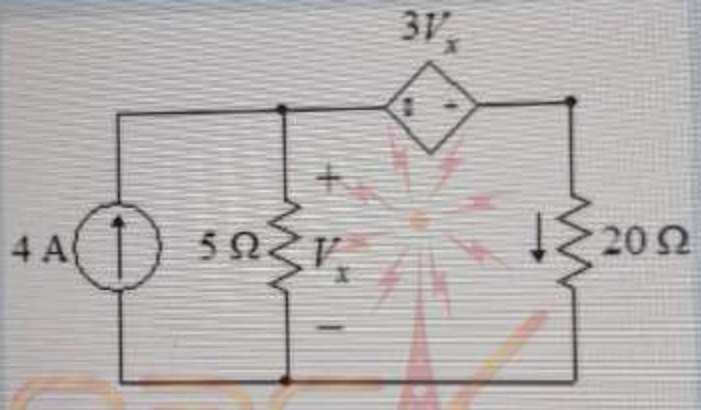


Select one:

- A. -1.62A
- B. -0.1A
- C. 1.62A
- D. 1.42A

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In the circuit shown, The value of V_x is:

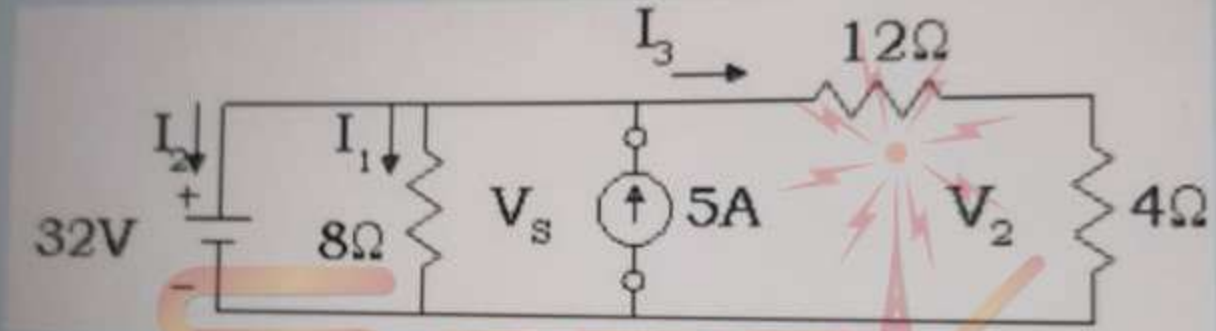


Select one:

- A. 100V
- B. 20V
- C. 5V
- D. 10V

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Refer to the figure shown, the current I_3 is:



Select one:

- A. 5A
- B. 3A
- C. 4A
- D. 2A

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The conductance of a 10 Ω resistor in milli siemens is:

Select one:

- A. 10 mS
- B. 0.01mS
- C. 100mS
- D. 0.1mS

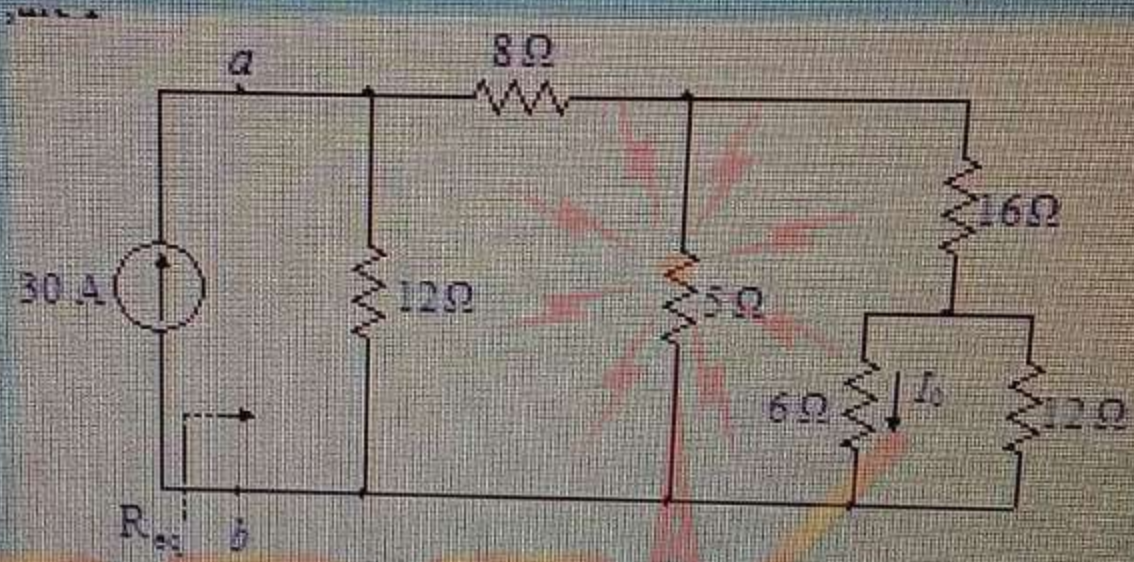
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A resistor absorbs an instantaneous power of $30\cos^2 t$ mW when connected to a voltage source v=10Cos t V. Find *R*.

Select one:

- A. 330 mΩ
- B. 33 Ω
- C. 330 Ω
- D. 3.33 kΩ

The voltage across 8Ω resistor is:

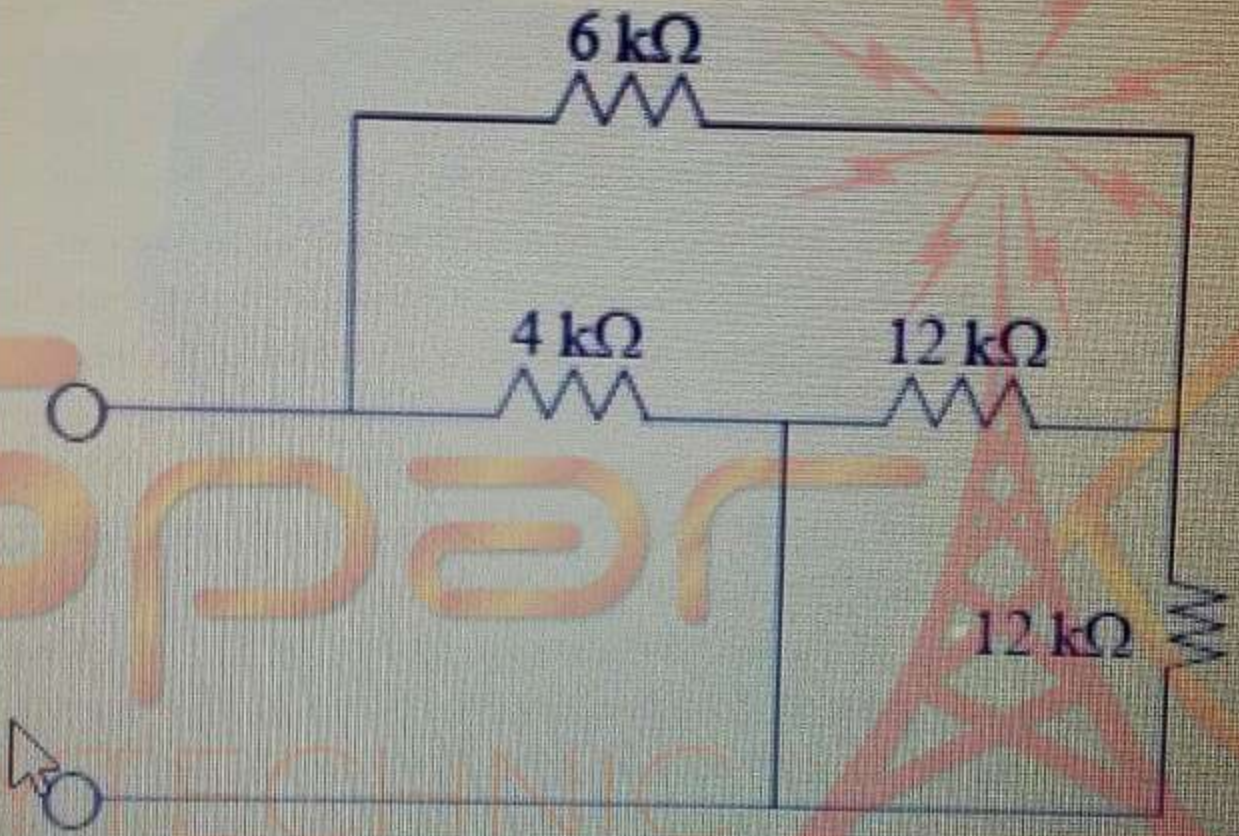


Select one:

- A. 240V
- B. 120V
- C. 60 V
- D. 72V

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Evaluate R_{eq} in kΩ for the circuit shown in Figure:



Answer

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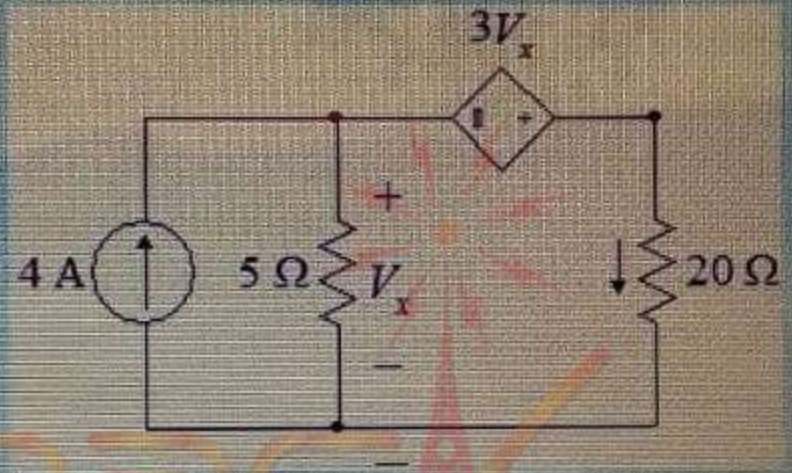
A resistor absorbs an instantaneous power of $30\cos^2 t$ mW when connected to a voltage source v=45 Cos t V. Find *R*.

Select one:

- A. 67.5 kΩ
- B. 675 Ω
- C. 667 mΩ
- D. 667 Ω

Spark POLYTECHNIC

In the circuit shown, The current in 20 Ω resistor is:

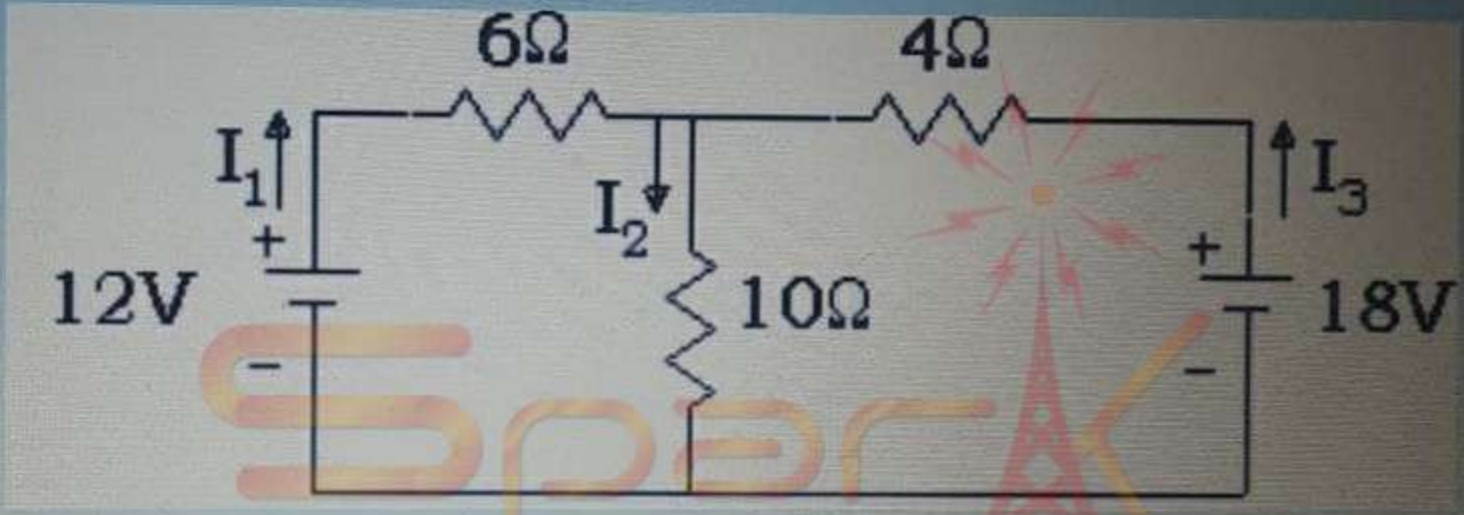


Select one:

- A. 200 mA
- B. 2.8A
- C. 2A
- D. 1A

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For the circuit shown in the figure , the current I_1 is:

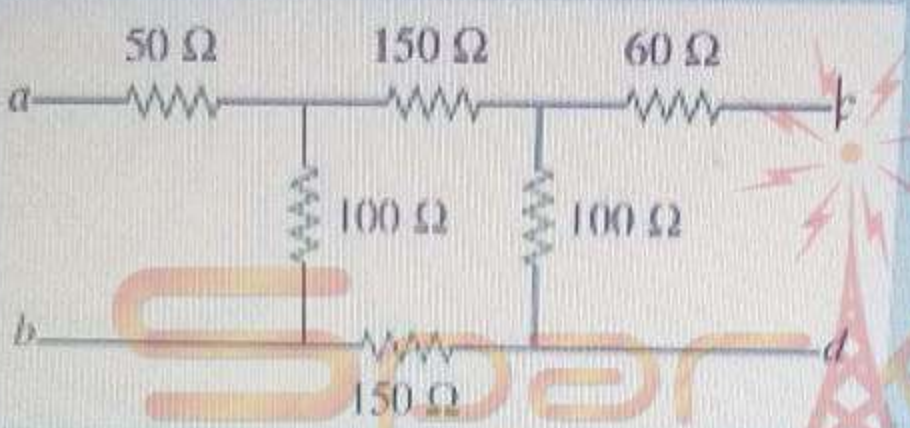


Select one:

- A. 1.62A
- B. 1.42A
- C. -0.1A
- D. -1.62A

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Refer to figure ; the value of the equivalent resistance at terminals a

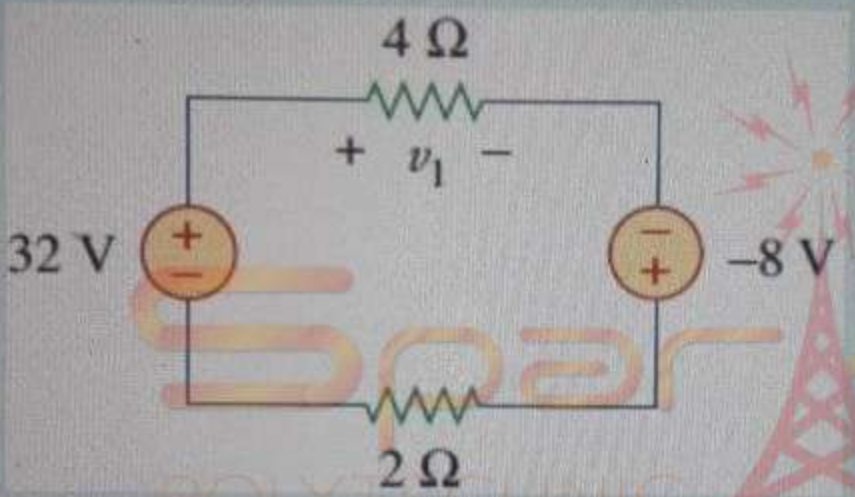


Select one:

- A. 160 Ω
- B. 260 Ω
- C. 175Ω
- D. 130 Ω

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For the circuit in figure the voltage v_1 is:



Answer:

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